

Approved VCS Methodology VM0011

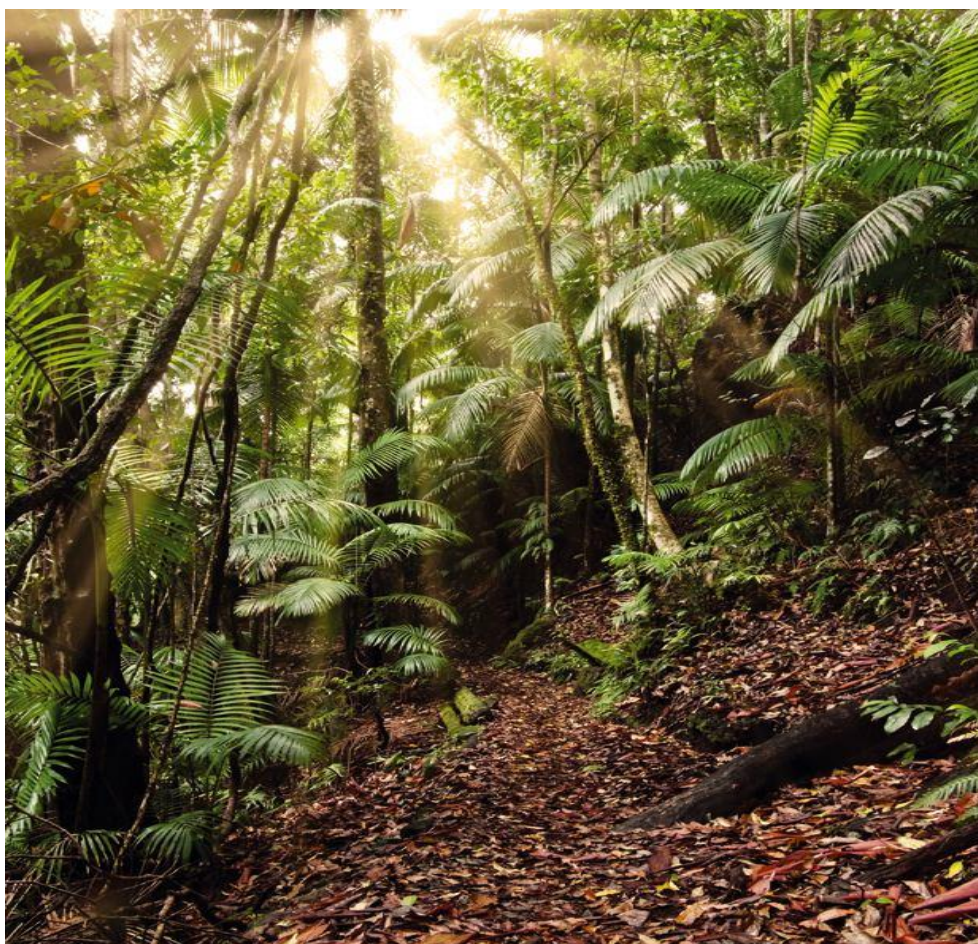
Version 1.0
Sectoral Scope 14

Methodology for Improved Forest
Management – Logged to Protected
Forest:

Calculating GHG Benefits from
Preventing Planned Degradation

Scope

This Methodology provides a procedure to determine the net greenhouse gas (GHG) emission reductions associated with an Improved Forest Management - Logged to Protected Forest (IFM-LtPF) activity where the baseline activity is selective logging.



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Executive Summary

The Methodology provides a procedure to determine the net anthropogenic greenhouse gas (GHG) emission reductions associated with an Improved Forest Management - Logged to Protected Forest (IFM-LtPF) activity which prevents the degradation of a forest through the cessation of a baseline activity of selective logging.

The Methodology is written to be compliant with VCS 2007.1. The Methodology specifically applies to previously logged or intact tropical forests, where the baseline activity can be clearly demonstrated and substantiated to be selective logging.

The key components of the Methodology are:

- (1) Applicability criteria and decision pathway for applying this Methodology based on the availability of data sources
- (2) Justification for a baseline activity of selective logging, additionality and definition of project boundary
- (3) Accounting for carbon changes of the baseline activity of selective logging
- (4) Accounting for emissions due to implementation of the baseline and IFM-LtPF project activity
- (5) Leakage assessment and management
- (6) Uncertainty analysis
- (7) Monitoring methodology

The application of this Methodology is based on the justification that selective logging is the most likely baseline activity above all other possible land use alternatives.

The Methodology calculates the net anthropogenic emission reductions by firstly estimating emissions due to forest degradation and the implementation of the baseline activity, selective logging. From this, GHG emissions associated with the implementation of the IFM-LtPF project activity, plus any emissions due to leakage of the baseline activity occurring outside the Project Area (as a result of the IFM-LtPF project activity), are subtracted from the baseline emissions to provide the net anthropogenic emission reductions of the IFM-LtPF project.

The baseline scenario, selective logging, as defined in the Methodology, involves the annual removal of merchantable trees with a minimum diameter at breast height as defined by the relevant authority in the host country. According to the data available, the Methodology provides guidance on determining the quantity and type of wood product that would be removed, as well as the nature of its fate as a result of the baseline scenario. In summary, emissions due to carbon lost from forest degradation from selective logging are attributable to (i) emissions from the oxidation of both short-term and long-term harvested wood products that are removed from the Project Area, (ii) emissions from the decay of the trimmings and branches, as well as any residual stand damage accumulating in the dead wood pool, and (iii) emissions from carbon that is forfeited due to forest growth that did not occur (foregone) as a result of the selective logging. These emissions are offset by (iv) regrowth in already harvested areas following logging.

The monitoring methodology provides guidance for the monitoring of the parameters employed to calculate carbon changes due to forest degradation, as well as emissions due to implementation of project and baseline activities. Following the implementation of the monitoring methodology, the results are applied to the above-mentioned accounting components (3) to (6) to revise the net anthropogenic GHG emission reductions for the subsequent reporting period.

The application of the Methodology is intended to not only mitigate GHGs, but to trigger a long term sustainable shift in the status of the forest in the covered region, hence aiding in the conservation of biodiversity. In addition, local communities will have new avenues for revenues that replace and exceed revenues to be derived from the removal of their forest.

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1 Introduction

The Methodology herein is applicable to a project activity that reduces greenhouse gas (GHG) emissions by preventing the degradation of a forest, either intact¹ or previously logged, that has been legally authorised by local, national or sub-national regulatory bodies to be harvested.

A common practice of IFM in tropical forests is selective logging. In selective logging operations (which is the baseline scenario of the Methodology), trees that meet the requirements as specified by the relevant authority in the host country of (i) minimum diameter at breast height, (ii) species, (iii) accessibility, and (iv) tree quality, are considered merchantable.

1.1 Methodology Applicability Criteria

The Methodology is applicable under the criteria and conditions presented in Table 1-1:

Table 1-1. Methodology applicability criteria

Criteria	Description
Project Type	Improved Forest Management - Logged to Protected Forest; with no removals (e.g. harvesting, planned biomass burning) occurring in the Project Area upon implementation of the actual project (with the exception of felling sample trees for validating or deriving project-specific parameters presented in Section 7.2.4).
Condition of the Forest	Intact forest or previously logged forest (also known as forest degraded due to logging) Land within the Project Area must have qualified as forest at least 10 years before the project start date.
Type of Forest	Tropical forests including evergreen tropical rainforests, moist deciduous forests, tropical dry forests and tropical upland forests (see Appendix A for definition), except peat swamp forests.
Forest Product Type	Harvested wood products i.e., sawlog, pulplog and commercially harvested fuelwood (See Appendices A and B.9).
Driver of Degradation	Legally sanctioned logging (timber and commercially harvested fuelwood) undertaken in accordance with the relevant laws, regulations and codes of practice of the country in which the Methodology is being applied.
Baseline Activities to be Displaced	Legally sanctioned selective logging for specific forest product types presented above.
Project Area	Must be designated, sanctioned or approved by the relevant authority in the host country for the selective logging

¹ Defined herein as forests that have not yet been logged.

Criteria	Description
Carbon Pools	<u>Carbon Pools considered:</u> - Aboveground biomass (AGB) of all trees as defined by the relevant authority in the host country - Harvested wood products (HWPs) based on domestic production not domestic consumption - Deadwood (DW). <u>Carbon Pools not considered:</u> - Aboveground biomass (non-trees) - Belowground biomass - Soil - Litter.

1.2 Methodology Overview

1.2.1 Net Anthropogenic GHG Emission Reductions

The Methodology estimates the net anthropogenic GHG emission reductions associated with an IFM-LtPF project ($C'_{IFM-LtPF,t}$) that replaces a baseline of selective logging in a tropical forest. It does this by estimating the annual total carbon emissions associated with the baseline scenario ($C'_{baseline,t}$) and by subtracting the emissions associated with project activity ($C'_{actual,t}$) and leakage ($C'_{leakage,t}$), in year, t , years elapsed since the start of the IFM-LtPF project activity. The parameters $C'_{baseline,t}$, $C'_{actual,t}$ and $C'_{leakage,t}$ are determined in Sections 3, 4 and 5, respectively.

The net anthropogenic GHG emission reductions in tonnes of carbon dioxide equivalents (tCO₂-e) of the IFM-LtPF project activity will be² calculated as follows:

Equation 1-1
$$C'_{IFM-LtPF,t} = C'_{baseline,t} - C'_{actual,t} - C'_{leakage,t}$$

Parameter	Description	Unit
$C'_{IFM-LtPF,t}$	Annual net anthropogenic GHG emission reductions in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$C'_{baseline,t}$	Annual total carbon emissions associated with the baseline scenario in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

² Throughout, four main terms are used: "must be", "should be", "will be" and "can be". The term "must be" represents an action to be taken, according to a mandatory condition specified in the Standards or Guidelines; the term "should be" represents an action to be taken, according to a best practice condition; the term "will be" represents an action to be taken, as developed specifically in this Methodology; whilst the term "can be" is used to indicate possibility.

Parameter	Description	Unit
$C'_{actual,t}$	Annual total carbon emissions associated with the project activity in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$C'_{leakage,t}$	Annual total carbon emissions associated with leakage in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

The term "ex ante" is defined as the estimation of the net GHG emission reductions, before the start of the IFM-LtPF project activity, for the project crediting period. The term "ex post" is defined as the calculation of the actual net GHG emission reductions for the years elapsed since the start of the IFM-LtPF project activity. See Appendix A of the Methodology for a list of definitions of other terms.

For an ex ante estimation of annual net anthropogenic GHG emission reductions, $C'_{IFM-LtPF,t}$, equations in Sections 3 and 4 for baseline accounting and actual project activity emissions, respectively, must be applied. The ex ante estimation for leakage assessment will be zero. The Project Proponent must provide estimates of the values of those parameters that are not available before the start of the monitoring activities and must retain a conservative approach in making these estimates.

1.2.2 Data Sources Required

Due to the nature of IFM-LtPF projects, the emission reductions will be largely dependent on emissions arising from the baseline scenario ($C'_{baseline,t}$), which, in turn, is a function of emissions from the implementation of the baseline activity (i.e. selective logging) ($C'_{emissions,t}$) and primarily, the emissions arising from forest degradation due to the baseline activity ($C'_{degradation,t}$). The method to calculate $C'_{degradation,t}$ depends on the available data for the Project Area.

This IFM-LtPF Methodology includes: (1) protecting currently logged or degraded forests from further logging and degradation; and (2) protecting unlogged forests that would be logged in the absence of carbon finance. For both, the Project Proponent must provide documentary evidence to prove that the Project Area will be legally harvested if the IFM-LtPF project is not implemented in the area and must include these documents Project Description (PD). The following documentary evidences are relevant, but not limited to:

- (i) A legally authorised letter or a document that clearly demonstrates the area will be selectively logged in the absence of the IFM-LtPF project; and
- (ii) An approved Forest Inventory Report (FIR) or an equivalent document including a forest management plan or a timber harvesting plan. This document must also include map(s) of the geographical extent of the Project Area where selective logging has occurred or would occur; or
- (iii) Preparation of the FIR or an equivalent document (by the relevant authority in the host country) which specifies the geographical extent of the Project Area where selective logging would occur.

Figure 1-1 presents a schematic of the dual-pathway for data availability in determining a suitable method. The Existing Inventory Data pathway represents the case where a Forest Inventory Report (FIR) or an equivalent document, is available. Within this pathway, two methods exist for determining

emissions due to degradation of the Project Area. These are: (i) combined methods of wood density and biomass conversion and expansion factor (BCEF) or (ii) allometric equations. The choice of method is based on the level of detail in the existing inventory data.

The Measured Data pathway must be applied for the following cases:

- (i) where there is no Forest Inventory Report (FIR) or equivalent document available
- (ii) where the FIR is still in preparation by the relevant authority in the host country
- (iii) where the validation of existing data is not accepted (see Section 3.2.1.1).

In order to apply this pathway, upfront monitoring is required, and carbon emissions will be based on the data obtained from the measurements taken in the Permanent Sample Plots (PSPs) in the forest using allometric equations (see Section 7.1.2).

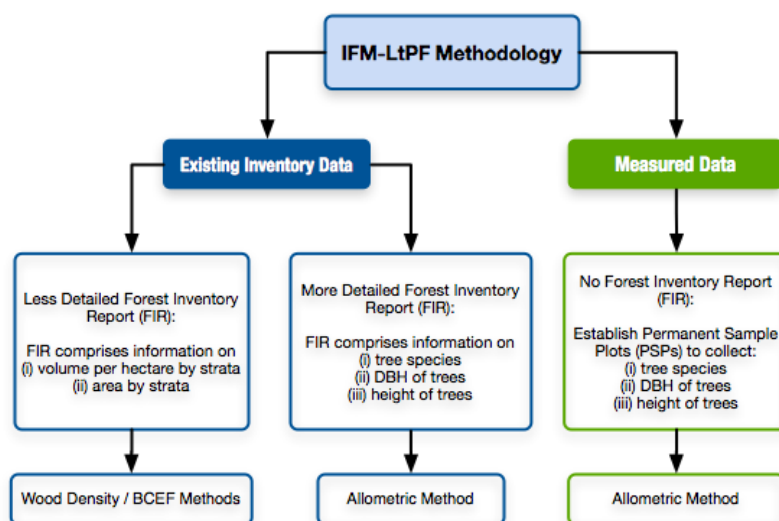


Figure 1-1. Schematic of data sources required for IFM-LtPF Methodology

1.2.3 Significance

To test for the significance of emission sources, the Project Proponent must use the most recent CDM Tool - "Tool for testing significance of GHG emissions in A/R CDM project activities" (see CDM-EB, 2007a).

1.2.4 VCs and the Non-Permanence Risk Withholding Buffer Percentage

A procedure for determining the non-permanence risk rating (i.e., the potential reversibility of sequestered/protected carbon) is described in Section 7.2.5. The outcome of the risk rating is the determination of the non-permanence buffer withholding percentage ($NP_{buffer}\%$) required to calculate the number of carbon credits to be deposited in the VCS buffer withholding pool. The latter is calculated as follows:

Equation 1-2

$$CC_{NP_{buffer},t} = NP_{buffer,t} \% \times C'_{degradation,t}$$

Parameter	Description	Unit
$CC_{NPbuffer,t}$	Annual carbon credits deposited in the VCS non-permanence buffer withholding account in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$NP_{buffer,t} \%$	Annual buffer withholding percentage required for the VCS buffer withholding pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	%
$C'_{degradation,t}$	Annual total carbon emissions associated with degradation as a result of the baseline activity in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

The number of VCUs that can be traded annually from the IFM-LtPF project activity is determined by subtracting the carbon credits deposited in the buffer account from the carbon credits determined post uncertainty deduction issued in year t .

Equation 1-3
$$VCU_t = CC_{IFM-LtPF,t} - CC_{NPbuffer,t}$$

Parameter	Description	Unit
VCU_t	Annual total tradeable Verified Carbon Units in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$CC_{IFM-LtPF,t}$	Annual carbon credits post uncertainty deduction in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$CC_{NPbuffer,t}$	Annual carbon credits deposited in the VCS non-permanence buffer withholding account in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

2 Project Baseline, Additionality and Boundary

2.1 Project Baseline Justification and Additionality

Since the Methodology is limited to a baseline scenario of selective logging, it must therefore be demonstrated that selective logging is the most likely baseline scenario above all other possible scenarios.

2.1.1 Selection of Baseline Amongst Alternative Scenarios

The procedure to identify the baseline scenario as selective logging for a VCS IFM-LtPF project activity, has been adapted from Steps 1 and 2 of the “Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities” (CDM-EB, 2007b, pp. 2-7).

Step 1: Identify credible alternative baseline scenarios to the proposed VCS IFM-LtPF project activity.

Sub-step 1a: Identify all realistic and credible alternative baseline scenarios to the proposed VCS IFM-LtPF project activity.

Realistic and credible baseline scenarios would be those that may have occurred in the Project Area in the absence of the VCS IFM-LtPF project activity. The scenarios must take into account relevant national and or sectoral policies and circumstances such as historical land use and practices, and economic trends and must at least include (CDM-EB, 2007b, p. 2):

- continuation of pre-project land use
- the protection of the land within the Project Area without being registered under the VCS as an IFM-LtPF project activity.

To identify realistic and credible baseline scenarios, data from the following sources must be applied:

- (i) land use records
- (ii) field surveys
- (iii) data and feedback from stakeholders.

To establish that all identified baseline scenarios are credible, the Project Proponent must ascertain the following:

- all land uses within the Project Area of the proposed IFM-LtPF project activity that are currently existing, or have existed but no longer exist, are deemed credible
- for all other baseline scenarios, credibility must be justified and justification must include elements of spatial planning information, if applicable, or legal requirements. In the absence of adequate justification, an assessment of economic feasibility of the proposed alternative baseline scenario must be provided.

Sub-step 1b: Identify realistic and credible alternative baseline scenarios with enforced mandatory legislation and regulations.

For the alternative baseline scenarios identified in Sub-step 1a, the Project Proponent must demonstrate that these alternatives comply with mandatory applicable legislation and regulations in the host country.

In the case where an identified alternative does not comply with mandatory applicable legislation and regulations, the Project Proponent must demonstrate that non-compliance of such legislation and regulations are the current practices in the host country, to the extent that the legislation and regulations are systematically not enforced and that non-compliance of such legislation and regulations is widespread.

Alternative baseline scenarios that do not comply with mandatory applicable legislation and regulations in the host country, and where widespread non-compliance cannot be demonstrated, must be removed from the list of realistic and credible baseline scenarios.

Selective logging must be present in the list from Sub-steps 1a and 1b for the Methodology to be applicable.

Step 2: Determine alternative baseline scenarios.

Sub-step 2a: Identify barriers that would prevent the implementation of at least one alternative baseline scenario.

For the alternative baseline scenarios identified in Sub-step 1b, the Project Proponent must identify any barriers that prevent the implementation of at least one of the alternatives. The barriers must not be specific to the Project Proponent but must apply to the activity itself, regardless of the entity that is developing the project.

A list of potential barriers can be found in CDM-EB (2007b), pp. 4-5.

Sub-step 2b: Eliminate baseline scenarios that are prevented by the identified barriers.

Determine which baseline scenarios identified in the Sub-step 1b are prevented by at least one of the barriers listed in Sub-step 2a. Eliminate these scenarios from the list (of alternative baseline scenarios) produced in Sub-step 1b.

Selective logging must be present in the resultant list for the Methodology to be applicable.

In applying Sub-steps 2a and 2b, the Project Proponent must provide transparent and documented evidence, and offer conservative interpretations of this documented evidence, as to how it demonstrates the existence and significance of the identified barriers. Anecdotal evidence can be included, but this alone is not sufficient proof of barriers (CDM-EB, 2007b, point 17, p. 6).

A list of the types of evidence acceptable can be found in CDM-EB (2007b), pp. 6-7.

Sub-step 2c: Determine the baseline scenario.

From Sub-step 2b, the refined list of alternative baseline scenarios that are in compliance with mandatory applicable legislation and regulations (or for which widespread non-compliance can

be demonstrated), and are not prevented by barriers, must contain selective logging as an alternative baseline for the current Methodology to be applied.

If selective logging is the only alternative scenario that is not prevented by any barrier, then this scenario is identified as the baseline activity.

If there are several alternative baseline scenarios in the list produced from Sub-step 2b (and this may include the proposed IFM-LtPF project activity undertaken without being registered as a VCS activity), then apply the Investment Analysis procedure (CDM-EB, 2007b, Step 3, pp. 7-11) to demonstrate that selective logging is the most economically and/or financially attractive baseline scenario amongst all the alternative baseline scenarios.

Selective logging must be demonstrated to be the most economically and/or financially attractive baseline scenario for the Methodology to be applicable.

2.1.2 Establishment of the Baseline Scenario: Selective Logging

Upon demonstrating that the most conservative baseline scenario is selective logging, and in order to establish this baseline, the Project Proponent must provide the following information:

- (i) documented history of the operator (e.g., operator shall have five to 10 years of management records to show normal historical practices)
- (ii) legal requirements for forest management and land use in the area; and
- (iii) proof that operator's environmental practices equal or exceed those commonly considered a minimum standard among similar landowners in the area.

If the Project Proponent is not a logging operator, but instead a new management entity with no history of logging practices in the project region and who takes over ownership of a property specifically to reduce forest emissions, then the baseline of selective logging is to be based on the projected management plans of the previous property owners. The established baseline must represent what would have most likely occurred in the absence of the IFM-LtPF project.

2.1.3 Additionality

For the specific case of demonstration and assessing additionality, the Project Proponent must use the latest version of the VCS tool for the Demonstration and Assessment of Additionality in VCS Agriculture Forestry and Other Land Use Project.

2.2 Project Boundaries

Project boundaries must be defined in terms of the following categories:

- Geographic boundary
- Temporal boundaries
- Carbon pools
- GHG sinks and sources.

2.2.1 Geographic Boundary

The Geographic Boundary includes the Project Area and the leakage area(s) and is described in the sections below. The land eligibility for the Project Area is discussed in Section 2.2.1.1.

2.2.1.1 Project Area and Land Eligibility

The Project Area is the forest area within the defined Geographic Boundary which would be degraded through selective logging under the baseline scenario. The term "forest area" must be appropriately defined in a way that is consistent with the thresholds used to define the term "forest" in the host country where the project activity will be implemented. Where the country has adopted a forest definition for the Kyoto Protocol, the minimum thresholds of the vegetation indicators used for defining "forest" will be used (see Appendix A). Otherwise, the definition used to define "forest" in the national GHG inventory will be used.

The Project Area must be designated, sanctioned or approved by the relevant authority in the host country.

The Project Proponent must provide sanctioned or approved document(s) which designates or specifies the geographical boundary of the Project Area. The Project Proponent must also provide the following information about the Project Area: name, local name, compartment number, harvesting blocks, identification number and area for each forest compartment in the Project Area, the geographic co-ordinates obtained from the Global Positioning System (GPS) or from the geo-referenced digital maps and, digital or GIS maps. In addition, the Project Proponent requires to provide digital map of the geographic boundary distinguishing the Project Area and the Leakage Area. In the case this Methodology is used for multiple discrete land parcels, each land parcel is treated as a separate Project Area and requires all the detail information/documentation as well as the digital maps as stated above.

If the Project Area has already been stratified, similar information must be provided for each stratum including stratified maps for the Project Area.

After the project has commenced, the Project Area will remain fixed throughout the crediting period.

2.2.1.1.1 Stratification of the Project Area

Stratification of the Project Area is required to divide the forest area into relatively homogeneous units or 'strata' in order to minimise variation among the strata. The stratification enhances the measuring precision and also minimises the cost of measurement. Where stratification of the Project Area has not been done, stratification is required before establishing PSPs for monitoring. Various criteria including forest parameters such as species, canopy density, tree height, age, etc., as well as physical parameters such as slope and drainage condition, are to be used to define the stratification. The most appropriate criteria for stratification are to be those which are directly related to the variables to be measured and monitored.

The forest stratification can be performed using forest type maps, topographic maps, interpretation of aerial photographs and satellite data with subsequent field verification. Remote sensing and GIS software is useful for spatial analysis and mapping for stratification.

The Project Area is a summation of the areas defining each stratum, j .

Equation 2-1

$$A_{project,t=0} = \sum_{j=1}^J A_{project,j,t=0}$$

Parameter	Description	Unit
$A_{project,t=0}$	Project Area where the IFM-LtPF project activity will be implemented; determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	ha
$A_{project,j,t=0}$	Project Area within each stratum j , (where $j=1,2,3 \dots J$ strata) where the IFM-LtPF project activity will be implemented; determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	ha

Both $A_{project,t=0}$ and $A_{project,j,t=0}$ are employed in both Sections 3.2 and 3.3 to calculate emissions from degradation due to selective logging.

In the case of existing stratification in the Project Area, the Project Proponent must review the strata using the latest available maps and data sources. If the existing stratification is not relevant for implementing the GHG project, the Project Proponent must update the strata, following the approach discussed above, and derive a Project Area map based on the revised stratification.

The Project Proponent must describe the approach used for stratifying the Project Area and also include GIS map(s) illustrating each stratum with specific detail. These maps must be revised if there is any update in the stratification in the event that there is a change in the carbon in the existing stratum.

2.2.1.2 Leakage area(s)

The leakage area(s) is defined as the area(s) to which the activity (i.e. selective logging) has shifted. This applies to lands owned and/or operated by the Project Proponent outside the Project Boundary, as well as to other areas within the host country (caused by market leakage effects), since both activity shifting and market leakage are considered in the IFM-LtPF Methodology. Section 5 addresses leakage accounting in more detail.

The Project Proponent must identify the strata in the leakage area similar to the Project Area through analysing the forest and topographical characteristics as described in Section 2.2.1.1.1. These areas must also be illustrated in the maps of the project.

2.2.2 Temporal Boundaries

2.2.2.1 Project crediting period

The project crediting period shall be determined in accordance with applicable VCS rules.

2.2.2.2 Monitoring and reporting periods

The monitoring period corresponds to the time taken between one monitoring event and the immediate next monitoring event for collecting measurements in the PSPs in the Project Area, and for reviewing non-monitored parameters. According to international industry practices for forest carbon assessment,

the maximum interval between one monitoring event and the immediate next, should not exceed five years (Pearson et al., 2005).

The results of project monitoring must be used to re-calculate the emissions associated with the project and this must be included in a report submitted for independent verification by an accredited third party.

2.2.2.3 Historical reference period

The historical reference period is a predetermined amount of time before the project start date, from which data can be taken in order to make ex ante estimation of natural disturbances and illegal harvesting and also to analyse leakage due to implementation of the project. A five-year timeframe for a historical reference period as close as possible to the project start date shall be used so as to limit uncertainty of the data collected.

2.2.3 Carbon Pools

Table 2-3 presents the justification, for inclusion or exclusion of carbon pools in the present IFM-LtPF Methodology in accordance with applicable VCS rules.

Table 2-3. Summary of carbon pools eligible under the Methodology

Carbon Pool	Status for IFM-LtPF Methodology	Justification
Aboveground Biomass (tree#)	Included	Anticipated to significantly increase under IFM-LtPF
Aboveground Biomass (non-tree)	Not included	Unlikely to decrease as a result of the project activities, or increase due to the baseline
Belowground Biomass	Not included	Unlikely to decrease as a result of the project activities, or increase due to the baseline
Deadwood	Included	Anticipated to significantly decrease under IFM-LtPF
Litter	Not included	Unlikely to decrease as a result of the project activities, or increase due to the baseline
Soil	Not included	Unlikely to decrease as a result of the project activities, or increase due to the baseline

Carbon Pool	Status for IFM-LtPF Methodology	Justification
Harvested Wood Products	Included	Anticipated to significantly decrease under IFM-LtPF

Trees are perennial plants and primarily constitute forest biomass. The FAO considers all living trees with a minimum diameter of 10 cm (over bark) when estimating the biomass in a tropical forest (Brown, 1997). This Methodology applies to all trees with a minimum diameter at breast height as specified by the relevant authority in the host country.

2.2.4 GHG Sources and Sinks

2.2.4.1 Consideration of GHGs

The six primary GHGs that contribute to climate change (WBCSD and WRI, 2004) and their global warming potentials (Forster et al., 2007) are listed in Table 2-4. Forster et al. (2007) present global warming potentials (GWPs) for three given time horizons: 20-year, 100-year and 500-year in the IPCC's Fourth Assessment Report (AR4) (IPCC, 2007).

The Project Developer shall use GWP of the 100-year time horizon as specified in the VCS rules. Table 2-4 presents the GWPs for the six primary GHGs for 100-year time horizon in accordance to the IPCC's Second Assessment Report. However, hydrofluorocarbons, perfluorocarbons and sulphur hexafluoride are synthetic industrial gases and therefore not included under Agriculture, Forestry and Other Land Use (AFOLU) methodologies.

Methane and nitrous oxide gases are not included for soil fluxes since soil is not included as a carbon pool in this Methodology. However, methane and nitrous oxide gases are included for calculating emissions from fossil fuel use in machinery, subject to significance (see Section 1.2.3). This significance accounting approach is in line with IPCC's definition of good practice (Penman et al., 2003) and the CDM "Tool for testing significance of GHG emissions in A/R CDM project activities" (CDM-EB, 2007a).

Carbon dioxide is included since it is intuitively present in both sources and sinks. Tables 2-5 and 2-6 present potential sources and sinks for CO₂, CH₄ and N₂O and whether or not a source/sink is included in the baseline and actual project implementation, respectively.

Table 2-4. The six primary greenhouse gases that contribute to climate change (WBCSD and WRI, 2004) and their associated global warming potentials for a 100-year time horizon

Greenhouse Gas	Chemical Formula	Global Warming Potential (100-year horizon)	Consideration
carbon dioxide*	CO ₂	1	included

Greenhouse Gas	Chemical Formula	Global Warming Potential (100-year horizon)	Consideration
methane	CH ₄	21	not included for soil fluxes; included for fossil fuel use in machinery but subject to significance. It is also included to calculate emissions from natural disturbances such as forest fires, subject to significance (see Section 1.2.3).
nitrous oxide	N ₂ O	310	not included for soil fluxes; included for fossil fuel use in machinery but subject to significance. It is also included to calculate emissions from natural disturbances such as forest fires, subject to significance (see Section 1.2.3).
hydrofluorocarbons	HFCs	see Forster et al. (2007) for list of HFCs	not included
perfluorocarbons	PFCs	see Forster et al. (2007) for list of PFCs	not included
sulphur hexafluoride	SF ₆	23,900	not included

* Where CH₄ and N₂O are determined to be insignificant (determined from Section 1.2.3) to the overall calculations, and HFCs, PFCs and SF₆ are not included, then CO₂ becomes the sole contributor. Throughout this Methodology, even if CO₂ becomes the sole contributor toward GHG emissions, it is expressed as CO₂-e.

Table 2-5. Sources and sinks for the three primary greenhouse gases (further to consideration stated in Table 2-3), carbon dioxide, methane and nitrous oxide for baseline consideration in the Project Area

Greenhouse Gas	Source (S) / Sink (CS)	Project Area Baseline Consideration and Justification
carbon dioxide	Forest degradation (S)	included; carbon stock determination
	Fossil fuel use in machinery (S)	included; since logging is the baseline activity
	Electricity consumption (S)	included
	Forest fires (S)	not included
	Commercially harvested fuelwood (S)	included
	Fuelwood gathered for domestic use (S)	not included
	Biomass burning in the course of land use conversion (S)	not included
	Embodied carbon in AGB (CS)	included, carbon stock determination
	Forest regrowth (CS)	included
	Harvested wood products (S and CS)	included
	Deadwood (S and CS)	included
methane	Pestilence (S)	not included; unlikely scenario
	Biomass burning in the course of land use conversion (S)	not included; unlikely scenario
	Fossil fuel use in machinery (S)	included but subject to significance; since logging is the most likely scenario
	Deadwood (S and CS)	not included
nitrous oxide	Biomass burning in the course of land use conversion (S)	not included; unlikely scenario
	Fossil fuel use in machinery (S)	included but subject to significance; since logging is the most likely scenario

Table 2-6. Sources for the three primary greenhouse gases (further to consideration stated in Table 2-3), carbon dioxide, methane and nitrous oxide for the actual project implementation

Greenhouse Gas	Source [^]	Actual Project Activity Consideration and Justification
carbon dioxide	Electricity consumption	included
	Flights	included
	Ground travel	included
	Aerial surveillance	included
	Natural disturbances such as forest fires	included
methane	Electricity consumption	included but subject to significance
	Flights	included but subject to significance
	Ground travel	included but subject to significance
	Aerial surveillance	included but subject to significance
	Natural disturbances such as forest fires	included but subject to significance
nitrous oxide	Electricity consumption	included but subject to significance
	Flights	included but subject to significance
	Ground travel	included but subject to significance
	Aerial surveillance	included but subject to significance
	Natural disturbances such as forest fires	included but subject to significance

[^] Note that for the actual project implementation, there are no sinks for GHG emissions and as such, the term 'sink (CS)' is not stated.

3 Baseline Accounting

3.1 Estimation of Emissions from Degradation

Section 3 of the Methodology provides the method for calculating emissions for the baseline activity. As the Methodology is developed for a baseline activity of legally sanctioned selective logging, the annual GHG emissions resulting will be due to degradation of the Project Area ($C'_{degradation,t}$), as well as annual emissions due to the selective logging operations ($C'_{emissions,t}$):

Equation 3-1
$$C'_{baseline,t} = C'_{degradation,t} + C'_{emissions,t}$$

Parameter	Description	Unit
$C'_{baseline,t}$	Annual total carbon emissions associated with the baseline scenario in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$C'_{degradation,t}$	Annual total carbon emissions associated with degradation as a result of the baseline activity in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$C'_{emissions,t}$	Annual total carbon emissions associated with the baseline activity of selective logging operations in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

For an IFM-LtPF project, determination of GHG emission reductions relies on comparing a baseline scenario of selective logging to a “with GHG project” scenario that assumes all the selective logging will cease. Emissions from degradation due to the baseline selective logging operations are determined by conducting a carbon mass balance over the entire Project Area.

In the baseline scenario described by this Methodology, selective logging involves the harvesting of merchantable trees larger than the minimum diameter at breast height as specified by the relevant authority in the host country. The selective logging operations have been treated as an annual event resulting in the removal of a specified volume of merchantable logs from the harvested merchantable trees in the Project Area. The annual harvest volume, i.e. the merchantable logs’ volume, is obtained from the inventoried growing stock of the forest by employing a sustainable and commonly used method in the host country. Here the term “growing stock” implies the total volume (over bark) of all the living trees as defined by the relevant authority in the host country. The growing stock data for the Project Area is either available from the Existing Inventory Data pathway or Measured Data pathway before the start date of the IFM-LtPF project.

At the site of the logging operations, the felled trees are converted to merchantable logs. The branches, bark and foliage are trimmed off the felled trees and left on the forest floor, becoming part of the deadwood (DW) pool and decaying slowly over time. Trees that are damaged during the logging operations are considered as residual stand damage, and are also included as carbon input to the DW pool. Carbon emissions from the DW pool into the atmosphere are accounted for by considering the net accumulation per year into the pool, multiplied by the rate of decay of the dry matter.

The subsequent processing of merchantable logs is dependent upon their species type, diameter class and end-product use. Some merchantable logs are taken to an on-site processing plant (i.e. a sawmill) resulting in sawn timber. With further value-added processing, these sawn timbers are converted into long-term harvested wood products (ltHWPs). Other merchantable logs are removed directly from the Project Area, for example commercially harvested fuelwood, and are converted into short-term harvested wood products (stHWPs). Long-term HWPs are defined as those with a half-life of over 30 years, while short-term HWPs are those with a half-life of approximately 2 years (IPCC, 2006, Vol 4, Table 12-2). It is possible for a Project Area to have a combination of both ltHWPs and stHWPs.

In the case of merchantable logs intended to be used in the manufacture of ltHWPs, the proportion of log converted to sawn timber is known as the lumber recovery factor. The remaining proportion of the merchantable log (residues that include sawdust and wood chips) is set aside within the project boundary and can either decay in the stockpile or be utilised as an energy source for processing. The emissions from ltHWPs are then accounted for by applying a (delayed) rate of oxidation of the ltHWP pool.

In the case of merchantable logs intended to be used in the manufacture of stHWPs, the stHWPs are further classified into commercially harvested fuelwood and paper products. For commercially harvested fuelwood, immediate oxidation occurs when the fuelwood is consumed. For paper products, emissions due to paper products are accounted for by applying a (delayed) rate of oxidation of the stHWP (to be converted into paper products only) pool.

Over the lifetime of logging operations, the previously harvested areas will experience regrowth. The carbon increase in the forest due to regrowth must be subtracted from the carbon lost due to degradation. Conversely, the logging operations remove trees every year which would otherwise experience growth. This loss in growth, defined as the amount of growth foregone, must be determined and added to the overall carbon lost due to degradation. The total carbon emissions associated with degradation as a result of the baseline activity accounts for all of these processes and is calculated as follows:

Equation 3-2

$$C'_{degradation,t} = \left[(C_{DW_{decay},t} + C_{ltHWP_{oxidation},t} + C_{stHWP_{oxidation},t} + C_{growth_foregone,t} - C_{regrowth,t}) \times \frac{44}{12} \right]$$

Parameter	Description	Unit
$C'_{degradation,t}$	Annual total carbon emissions associated with degradation as a result of the baseline activity in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$C_{DW_{decay},t}$	Annual carbon leaving the deadwood pool due to the decay of deadwood in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{ltHWP_{oxidation},t}$	Annual carbon due to the combined delayed oxidation of long-term harvested wood products and immediate oxidation of long-term harvested wood products residues in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{stHWP_{oxidation},t}$	Annual carbon due to immediate oxidation of short-term harvested wood products (commercially harvested fuelwood) and delayed oxidation of short-term harvested wood products (paper products) in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{growth_foregone,t}$	Annual carbon lost due to growth foregone in the aboveground biomass in the Project Area in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{regrowth,t}$	Annual carbon increase in the biomass due to regrowth following logging in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
44/12	The ratio of molecular weight of carbon dioxide to carbon, <i>see</i> Appendix C	tCO ₂ -e tC ⁻¹

Ex ante estimations of $C'_{degradation,t}$ will be made using data from Forest Inventory Report (FIR) or an equivalent document, or measured data using the sequence of equations in Section 3.2.

Figure 3-1 presents a schematic of the dual-pathway for data availability in determining a suitable method to apply. As the baseline refers to selective logging that has been sanctioned by the host country's authority, it is most likely that a Forest Inventory Report (FIR) or an equivalent document, is available: the Existing Inventory Data pathway.

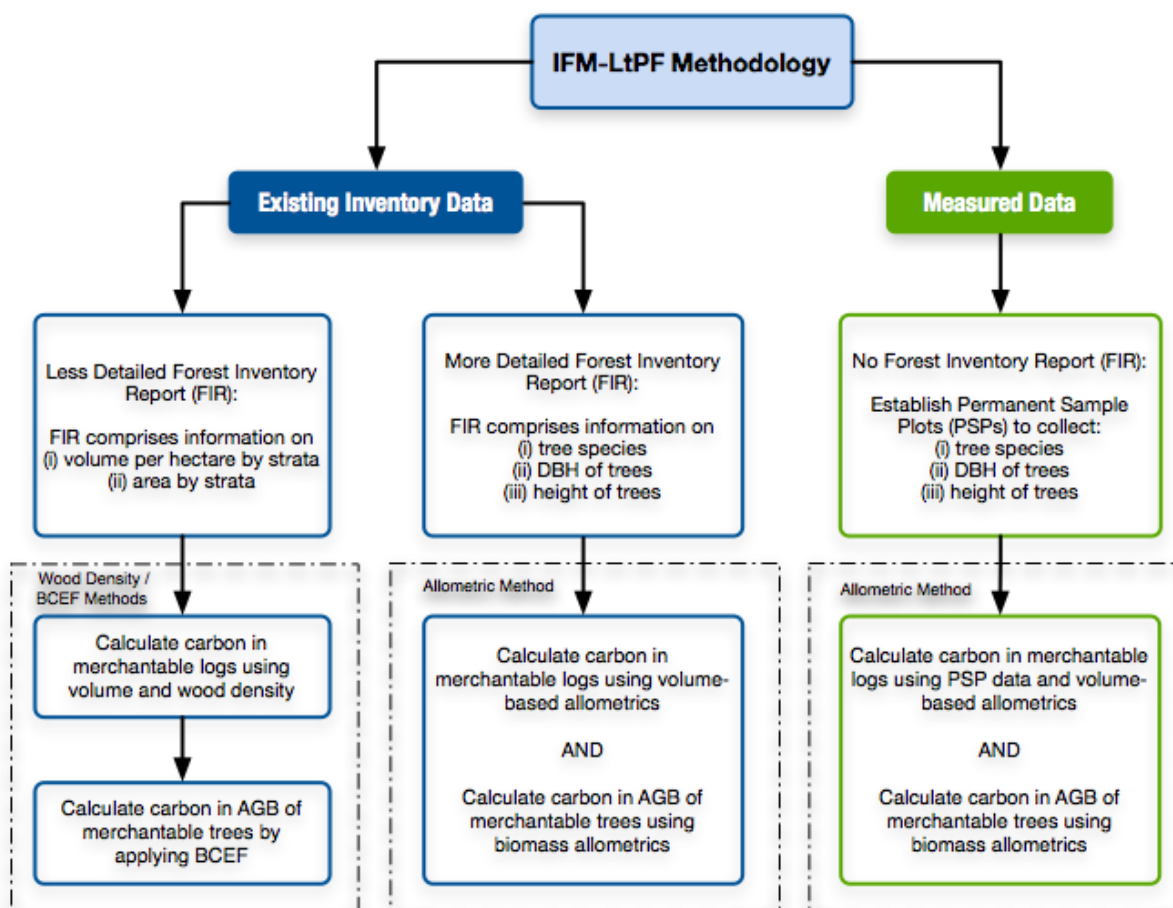


Figure 3-1. Schematic of accounting approach based on available data

There are two methods for determining the quantity of carbon in merchantable logs and the growing stock within the Existing Inventory Data pathway, depending on the detail of data in the FIR: Wood density plus the Biomass Conversion and Expansion Factor (BCEF) methods, or the use of the allometric method (see Section 3.2.1). The methods are selected based on the level of information provided in the FIR or equivalent document. For a less detailed FIR, where only stratified volume per hectare and area data is provided, the wood density plus BCEF methods must be applied. For a more detailed FIR, where information on tree species, diameter at breast height (DBH) and height of trees is provided, carbon is calculated using volume and biomass allometric equations. For the Existing Inventory Data pathway, prior to applying either the wood density and BCEF or allometric methods, the FIR must be validated conservatively according to Section 3.2.1.1.

In the case where there is legal approval for selective logging in the Project Area and the FIR or an equivalent document is under preparation by the relevant authority in the host country, there is only one pathway for determining the quantity of carbon in merchantable logs and the growing stock: the Measured Data pathway. This pathway is based on the data obtained from measurements in Permanent Sample Plots (PSPs) in the Project Area (see Section 7.1.2).

3.2 CALCULATION OF PRIMARY PARAMETERS IN THE PROJECT AREA

This section first presents the method to validate the existing inventory data. Then, the calculations to obtain the primary parameters: annual carbon in merchantable logs, $C_{merch,t}$, annual carbon in the AGB of the growing stock, $C_{AGB_gstock,t}$ and annual net harvest area, $A_{NHA_annual,t}$ are presented.

3.2.1 EXISTING INVENTORY DATA PATHWAY

3.2.1.1 Validation of existing forest inventory data

The Existing Inventory Data pathway applies where an existing legally approved FIR or an equivalent document, presents inventory data not more than five years old. According to Pearson et al. (2005), carbon in the aboveground biomass (AGB) is likely to change at a much faster rate than the carbon stock in the soil. It is thus appropriate that monitoring of the AGB in the forest be carried out at five-yearly intervals. This monitoring period also complies with the carbon verification and certification of the projects under the Clean Development Mechanism (CDM) policy (Pearson et al., 2005). Furthermore, the data in the FIR must have been obtained from an adequate number of representative sample plots in strata similar to the Project Area. The data in the FIR, or equivalent document, should only be employed after validation by the Project Proponent using the following procedure:

- Step 1: Stratify the Project Area by following the procedure described in Section 2.2.1.1.1 and check with the existing stratification to verify whether it is similar to the Project Area stratification or not.
- Step 2: If the existing stratification is not similar to the Project Area stratification, the existing inventory data must not be used. Apply the Measured Data pathway.
- Step 3: If the existing stratification is similar to the Project Area stratification, randomly establish 6 to 10 preliminary sample plots in each stratum as suggested for measuring the variance (Pearson et al., 2005) and measure the trees in the sample plots for their diameter at breast height and tree height. Guidance on the size and shape of the sample plots to be established is described in Section 7.1.2.2.
- Step 4: Estimate the carbon in the AGB for each forest product type (if specified) for each stratum using the biomass allometric method as presented in Section 3.2.1.2.
- Step 5: Calculate the carbon in the AGB per hectare and the 95 percent confidence interval for each stratum from the measured data. A 95 percent confidence interval is considered an appropriate measure for carbon stock (Pearson et al., 2005; Brown, 2002).
- Step 6: Estimate the carbon in the AGB per hectare for each forest product type (if specified) for each stratum from the existing data in the FIR or equivalent document, as stated in Section 3.2.1. The

specific method chosen in Section 3.2.1 depends on the type of data presented in the FIR or equivalent document.

Step 7: Compare the mean values between the measured and existing data:

If the mean value for the existing data is within the 95 percent confidence interval of the measured data, use the lower bound of the 95 percent confidence interval. This is a conservative estimate of carbon stock for the baseline scenario.

If the mean value for the existing data is greater than the 95 percent confidence interval, use the mean value of the measured data as a conservative estimate of carbon stock for the baseline scenario.

If the mean value for the existing data is lower than the 95 percent confidence interval, use the existing data, which is a conservative estimate of carbon stock for the baseline scenario.

If the FIR or an equivalent document contains inventory data collected more than five years ago, or the inventory data has been obtained without proper stratification and sampling of the Project Area, then the Project Proponent must take the Measured Data pathway for calculating carbon emissions for the baseline scenario.

3.2.1.2 Where a less detailed FIR is available

For a less detailed FIR, only stratified volume per hectare and area data is provided. Thus, the wood density and BCEF methods must be applied to calculate carbon in merchantable logs and growing stock.

3.2.1.2.1 Carbon in the merchantable logs using wood density method

A. Where inventory data does not distinguish between different forest product types

Step 1: Select a sustainable and commonly employed method of the host country to convert growing stock data to merchantable logs' volume to be removed. The Project Proponent must justify in the PD that the chosen method is appropriate, and demonstrate that the method is commonly employed in the host country.

Step 2: Apply the method selected and justified in Step 1 to strata level growing stock data ($\bar{V}_{gstock,j,t=0}$), to obtain merchantable logs' volume per hectare in stratum, j ($\bar{V}_{merch,j,t=0}$).

Step 3: Choose the most applicable wood density for a forest with corresponding climate region and ecological zone³ (see Appendix B, Section B.1, Table B-1).

Step 4: Choose the most applicable carbon fraction⁴ of wood from the following data sources:

- (i) National carbon fraction (e.g. from National GHG Inventory)
- (ii) Default IPCC carbon fraction of wood for a forest with corresponding climate domain (see Appendix B, Section B.2, Table B-2).

³ As defined in IPCC (2006) Table 4-1, p 4.46, for climate region and ecological zone of forests.

⁴ Throughout the Methodology, when determining carbon fraction, where merchantable logs are referred to, the carbon fraction of wood applies. Where aboveground biomass is referred to, the carbon fraction of aboveground biomass applies.

Step 5: Apply the following equation to convert the merchantable logs' volume to carbon per hectare in the stratum, j , using the wood density and carbon fraction:

Equation 3-3
$$\bar{C}_{merch,j,t=0} = D \times CF_{wood} \times \bar{V}_{merch,j,t=0}$$

Parameter	Description	Unit
$\bar{C}_{merch,j,t=0}$	Average carbon per hectare in merchantable logs in stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC ha ⁻¹
D	Wood density for the tropical forest with corresponding climate region and ecological zone (see Appendix B)	(t d.m.) m ⁻³
CF_{wood}	Carbon fraction of wood for the tropical forest (see Appendix B)	tC (t d.m.) ⁻¹
$\bar{V}_{merch,j,t=0}$	Average merchantable logs' volume per hectare in stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	m ³ ha ⁻¹

Step 6: Convert the strata level average carbon per hectare to the average carbon per hectare in the merchantable logs for the entire Project Area:

Equation 3-4
$$\bar{C}_{merch,t=0} = \frac{\sum_{j=1}^J (\bar{C}_{merch,j,t=0} \times A_{project,j,t=0})}{A_{project,t=0}}$$

Parameter	Description	Unit
$\bar{C}_{merch,t=0}$	Average carbon per hectare in merchantable logs in the Project Area determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC ha ⁻¹
$\bar{C}_{merch,j,t=0}$	Average carbon per hectare in merchantable logs in stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC ha ⁻¹
$A_{project,j,t=0}$	Project Area within each stratum j , (where $j=1,2,3 \dots J$ strata) where the IFM-LtPF project activity will be implemented; determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	ha

Parameter	Description	Unit
$A_{project,t=0}$	Project Area where the IFM-LtPF project activity will be implemented; determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	ha

B. Where inventory data distinguish between different forest product types

This Methodology accounts for three forest product types: sawlog, pulplog and commercially harvested fuelwood. These can be categorised into long-term and short-term harvested wood products as presented in Appendix B, Section B.9, Figure B-1. If the growing stock of different forest product types p , per hectare at the strata level j , is available, calculate the average carbon in the merchantable logs per hectare for these forest product types using the following procedure:

Step 1: Select a sustainable and commonly employed method of the host country to convert growing stock data to the merchantable logs' volume to be removed, for the different forest product types. The Project Proponent must justify that the chosen method is appropriate and demonstrate that the method is commonly employed in the host country.

Step 2: Apply the method selected and justified in Step 1, to strata level growing stock data ($\bar{V}_{gstock,p,j,t=0}$) to obtain the merchantable logs' volume per hectare at the strata level j ($\bar{V}_{merch,p,j,t=0}$).

Step 3: Choose the most applicable wood density for a forest with corresponding climate region and ecological zone⁴ (see Appendix B, section B.1, Table B-1).

Step 4: Choose the most applicable carbon fraction of wood from the data sources presented in Step 4 of Section 3.2.1.2.1,A.

Step 5: Apply the following equation to estimate average carbon per hectare of the forest product types at the strata level:

Equation 3-5
$$\bar{C}_{merch,p,j,t=0} = D \times CF_{wood} \times \bar{V}_{merch,p,j,t=0}$$

Parameter	Description	Unit
$\bar{C}_{merch,p,j,t=0}$	Average carbon per hectare in merchantable logs of forest product type p , in stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC ha ⁻¹
D	Wood density for the tropical forest with corresponding climate region and ecological zone (see Appendix B)	(t d.m.) m ⁻³
CF_{wood}	Carbon fraction of wood for the tropical forest (see Appendix B)	tC (t d.m.) ⁻¹

Parameter	Description	Unit
$\bar{V}_{merch,p,j,t=0}$	Average merchantable logs' volume per hectare of forest product type p , in stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	$m^3 ha^{-1}$

Step 6: The average carbon per hectare in the merchantable logs for different forest product types in the Project Area is calculated as follows:

Equation 3-6

$$\bar{C}_{merch,p,t=0} = \frac{\sum_{j=1}^J (\bar{C}_{merch,p,j,t=0} \times A_{project,j,t=0})}{A_{project,t=0}}$$

Parameter	Description	Unit
$\bar{C}_{merch,p,t=0}$	Average carbon per hectare in merchantable logs of forest product type p , in the Project Area determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC ha^{-1}
$\bar{C}_{merch,p,j,t=0}$	Average carbon per hectare in merchantable logs of forest product type p , in stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC ha^{-1}
$A_{project,j,t=0}$	Project Area within each stratum j , (where $j=1,2,3 \dots J$ strata) where the IFM-LtPF project activity will be implemented; determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	ha
$A_{project,t=0}$	Project Area where the IFM-LtPF project activity will be implemented; determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	ha

Step 7: The average carbon per hectare in all the merchantable logs including the different forest product types in the Project Area, is calculated as follows:

Equation 3-7

$$\bar{C}_{merch,t=0} = \sum_{p=1}^P \bar{C}_{merch,p,t=0}$$

Parameter	Description	Unit
$\bar{C}_{merch,t=0}$	Average carbon per hectare in merchantable logs in the Project Area determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC ha ⁻¹
$\bar{C}_{merch,p,t=0}$	Average carbon per hectare in merchantable logs of forest product type p , in the Project Area determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC ha ⁻¹

3.2.1.2.2 Carbon in the AGB of the growing stock using BCEF method

The carbon in the AGB of the growing stock can be calculated by applying the stratum-level BCEF method as presented in Figure 3-1 using the following procedure:

Step 1: Choose the most applicable BCEF value for each stratum in the Project Area from the following data sources:

- (i) National factor (e.g. from National GHG Inventory)
- (ii) Default IPCC BCEF for the corresponding forest, climate and growing stock in the Project Area (see IPCC, 2006, Chapter 4, Table 4.5, pp. 4.51-4.52).

If a factor for the forest type and the climatic region of the Project Area is not found, select the factor with the closest corresponding information.

Step 2: Choose the most applicable carbon fraction in the AGB of trees from the data sources presented in Step 4 of Section 3.2.1.2.1.A.

Step 3: Apply the following equation to convert the average growing stock per hectare to average carbon per hectare in the AGB in the growing stock in stratum j , using the values for BCEF and carbon fraction:

Equation 3-8
$$\bar{C}_{AGB_gstock,j,t=0} = BCEF_j \times CF_{AGB} \times \bar{V}_{gstock,j,t=0}$$

Parameter	Description	Unit
$\bar{C}_{AGB_gstock,j,t=0}$	Average carbon per hectare in the aboveground biomass of the growing stock in stratum j (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC ha ⁻¹
$BCEF_j$	Biomass conversion and expansion factor for converting average growing stock per hectare to carbon in the aboveground biomass for stratum j (where $j=1,2,3 \dots J$ strata)	(t d.m.) m ⁻³
CF_{AGB}	Carbon fraction in the aboveground biomass of trees for the tropical forest (see Appendix B)	tC (t d.m.) ⁻¹

Parameter	Description	Unit
$\bar{V}_{gstock,j,t=0}$	Average growing stock per hectare for stratum j (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	$m^3 ha^{-1}$

Step 4: Convert the average carbon per hectare in the AGB of the strata level to the whole forest level:

Equation 3-9

$$\bar{C}_{AGB_gstock,t=0} = \frac{\sum_{j=1}^J (\bar{C}_{AGB_gstock,j,t=0} \times A_{project,j,t=0})}{A_{project,t=0}}$$

Parameter	Description	Unit
$\bar{C}_{AGB_gstock,t=0}$	Average carbon per hectare in the aboveground biomass of the growing stock in the Project Area determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	$tC ha^{-1}$
$\bar{C}_{AGB_gstock,j,t=0}$	Average carbon per hectare in the aboveground biomass of the growing stock in stratum j (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	$tC ha^{-1}$
$A_{project,j,t=0}$	Project Area within each stratum j (where $j=1,2,3 \dots J$ strata) where the IFM-LtPF project activity will be implemented; determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	ha
$A_{project,t=0}$	Project Area where the IFM-LtPF project activity will be implemented; determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	ha

3.2.1.3 Where a detailed FIR is available

If the FIR or an equivalent document provides field inventory data on tree species, diameter at breast height (DBH) and tree height (H) of the growing stock at the sample plot level s , in each stratum j , the volume allometric method can be used for estimating the carbon in the merchantable logs and in the AGB of the growing stock.

3.2.1.3.1 Carbon in the merchantable logs using volume allometric method

Use the following steps to calculate the carbon in the merchantable logs applying the volume allometric method:

A. Where inventory data does not distinguish between different forest product types

Step 1: To estimate the growing stock per hectare using DBH and tree height for each tree, select species-specific or group of species-specific volume allometric equations from the following data sources:

- (i) Published peer reviewed studies for equations from tropical forests with corresponding climate region and ecological zone (e.g. Akindele and Lemay, 2006; Segura and Kanninen, 2005)
- (ii) Published documents from the relevant forestry authority of the host country, e.g. National Inventory Report.

If species-specific or group of species-specific volume allometric equations are not available, general volume equations can be obtained from the relevant forestry authority of the host country. For this case, the Project Proponent is required to verify the applicability of this equation in the first monitoring event (see Section 7.2.4.2). If the equation is not applicable, derive a Project Area-specific equation (see specifically Step 5 of Section 7.2.4.2 for guidelines).

Step 2: Apply the following equation to sum the volume of each tree n , of species i , in sample plot s , to obtain the average growing stock in all the sample plots within a stratum j .

Equation 3-10
$$\bar{V}_{gstock,j,t=0} = \frac{1}{A_{s,j,t=0}} \times \sum_{s=1}^S \left\{ \sum_{i=1}^I \left[\sum_{n=1}^{N_i} f_V(DBH_{n,i,s,j,t=0}, H_{n,i,s,j,t=0}) \right] \right\}$$

Parameter	Description	Unit
$\bar{V}_{gstock,j,t=0}$	Average growing stock per hectare for stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	m ³ ha ⁻¹
$f_V(DBH_{n,i,s,j,t=0}, H_{n,i,s,j,t=0})$	Volume allometric equation as a function of diameter at breast height and height; determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	dimensionless
$DBH_{n,i,s,j,t=0}$	Diameter at breast height for individual tree n (where $n=1,2,3 \dots N_i$ for species, i), of species i (where $i=1,2,3 \dots I$ species) in sample plot s (where $s=1,2,3 \dots S$ sample plot), of stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	cm
$H_{n,i,s,j,t=0}$	Height for individual tree n , (where $n=1,2,3 \dots N_i$ for species, i), of species i , (where $i=1,2,3 \dots I$ species) in sample plot s , (where $s=1,2,3 \dots S$ sample plot), of stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	m
$A_{s,j,t=0}$	Total area of sample plots s , (where $s=1,2,3 \dots S$ sample plot), in stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	ha

Step 3: After estimating growing stock per hectare at the strata level, to calculate the carbon in the merchantable logs ($\bar{C}_{merch,t=0}$), apply Steps 2 to 6 as shown in the Section 3.2.1.2.1,A.

B. Where inventory data distinguish between different forest product types

If the different forest product types are distinguished based on the DBH of the tree, and are to be accounted for in GHG emissions calculations, apply the following steps:

Step 1: To estimate growing stock per hectare for each product type using DBH and tree height for each tree, select species-specific or group of species-specific volume allometric equations from the data sources presented in Step 1 of Section 3.2.1.3.1,A.

If species-specific or group of species-specific volume allometric equations are not available, general volume equations can be obtained from the relevant forestry authority of the host country. For this case, the Project Proponent is required to verify the applicability of this equation in the first monitoring event (see Section 7.2.4.2). If the equation is not applicable, derive a Project Area-specific equation (see specifically Step 5 of Section 7.2.4.2 for guidelines).

Step 2: Apply the allometric equations to individual tree n , of species i , to determine the average growing stock per hectare of the forest product type p , in a stratum j , as follows:

Equation 3-11

$$\bar{V}_{gstock,p,j,t=0} = \frac{1}{A_{s,j,t=0}} \times \sum_{s=1}^S \left\{ \sum_{i=1}^I \left[\sum_{n=1}^{N_i} f_V(DBH_{n,i,s,j,t=0}, H_{n,i,s,j,t=0})_p \right] \right\}$$

Parameter	Description	Unit
$\bar{V}_{gstock,p,j,t=0}$	Average growing stock per hectare for product type p , in stratum j (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	$m^3 \text{ ha}^{-1}$
$f_V(DBH_{n,i,s,j,t=0}, H_{n,i,s,j,t=0})_p$	Volume allometric equation as a function of diameter at breast height and height for the forest product type p , determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	dimensionless
$DBH_{n,i,s,j,t=0}$	Diameter at breast height for individual tree n (where $n=1,2,3 \dots N_i$ for species, j), of species i (where $i=1,2,3 \dots I$ species) in sample plot s (where $s=1,2,3 \dots S$ sample plot), of stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	cm
$H_{n,i,s,j,t=0}$	Height for individual tree n , (where $n=1,2,3 \dots N_i$ for species, j), of species i (where $i=1,2,3 \dots I$ species) in sample plot s , (where $s=1,2,3 \dots S$ sample plot), of stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	m

Parameter	Description	Unit
$A_{s,j,t=0}$	Total area of sample plots s , (where $s=1,2,3 \dots S$ sample plot) in stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	ha

Step 3: After estimating growing stock per hectare for different forest product types p , at the strata level, to calculate the average carbon per hectare in the merchantable logs ($\bar{C}_{merch,t=0}$), apply Steps 1 to 7 as shown in the Section 3.2.1.2.1,B.

3.2.1.3.2 Carbon in the AGB of the growing stock using biomass allometric method

The carbon in the AGB of the growing stock per hectare can be calculated based on the carbon fraction and growing stock. The growing stock is calculated by applying biomass allometric equations as a function of DBH, tree height, and wood density. This method of carbon estimation in the AGB of the growing stock is also used for estimating the carbon in the growth foregone due to selective logging as described in Section 3.3.4. The data on DBH and tree height are obtained from measurement of trees in the PSPs using the method presented in Section 7.1.2.

Calculate the carbon in the AGB of the growing stock applying the biomass allometric method by using the following procedure:

Step 1: Select species-specific or group of species-specific biomass allometric equations for estimating the AGB using the tree measurements from the following data sources:

- (i) Published peer reviewed studies for equations from tropical forests with corresponding climate region and ecological zone (e.g. Pearson et al., 2005)
- (ii) Published documents from the relevant authority of the host country, e.g. National Inventory Report.

If species-specific or group of species-specific biomass allometric equations are not available, general biomass allometric equations can be obtained from literature such as the IPCC GPG LULUCF (2003), Chapter 4, Annex 4A.1, Table 4.A.1, p. 4.114, or Pearson et. al. (2005), Appendix C, p. 43. For this case, select the most applicable allometric equation for a tropical forest with corresponding climate region and ecological zone and verify the applicability of this equation in the first monitoring event (see Section 7.2.4.2). If the equation is not applicable, derive a Project Area-specific equation (see specifically Steps 1 and 5 in Section 7.2.4.2 for guidelines).

Step 2: Find species-specific or group of species-specific wood densities from the data sources presented in Appendix B, Section B.1.

Step 3: Apply the biomass allometric equation to convert the DBH, tree height and wood density to the AGB of all the individual trees n , of species i , in the sample plot s , of the stratum j .

Equation 3-12

$$\bar{B}_{AGB_gstock,j,t=0} = \frac{1}{A_{s,j,t=0}} \times \sum_{s=1}^S \left\{ \sum_{i=1}^I \left[\sum_{n=1}^{N_i} f_B(DBH_{n,i,s,j,t=0}, H_{n,s,i,j,t=0}, D_i) \right] \right\}$$

Parameter	Description	Unit
$\bar{B}_{AGB_gstock,j,t=0}$	Average aboveground biomass of the growing stock in stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	(t d.m.) ha ⁻¹
$f_B(DBH_{n,i,s,j,t=0}, H_{n,i,s,j,t=0}, D_i)$	Biomass allometric equation as a function of diameter at breast height, height determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year, and wood density	dimensionless
$DBH_{n,i,s,j,t=0}$	Diameter at breast height for individual tree n , (where $n=1,2,3 \dots N_i$ for species, i), of species i , (where $i=1,2,3 \dots I$ species) in sample plot s , (where $s=1,2,3 \dots S$ sample plot), of stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	cm
$H_{n,i,s,j,t=0}$	Height for individual tree n , (where $n=1,2,3 \dots N_i$ for species, i), of species i , (where $i=1,2,3 \dots I$ species) in sample plot s , (where $s=1,2,3 \dots S$ sample plot), of stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	m
D_i	Wood density for species, i if D_i is not available, the density (D) for the tropical forest with the corresponding climate region and ecological zone	(t d.m.) m ⁻³
$A_{s,j,t=0}$	Total area of sample plots s , (where $s=1,2,3 \dots S$ sample plot), in stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	ha

Note that for estimating the annual average growth foregone in the Project Area due to selective logging, the AGB of the merchantable trees ($\bar{B}_{AGB_merch,j,m_1}$ and $\bar{B}_{AGB_merch,j,m_2}$, see Equation 3-36a) are obtained by applying the DBH and height of the merchantable trees at the stratum level, $DBH_{n,i,s,j,t}$ and $H_{n,i,s,j,t}$, respectively, to Equation 3-12 in monitoring events m_1 and m_2 . Similarly, the annual average regrowth of the AGB in the naturally disturbed area ($\bar{B}_{AGB_nd,j,m_1}$ and $\bar{B}_{AGB_nd,j,m_2}$, see Equation 4-17b) is estimated by applying the DBH and height of all trees $DBH_{tree_nd,n,i,s_{nd},j,t}$ and $H_{tree_nd,n,i,s_{nd},j,t}$, respectively, in the sample plot established in the naturally disturbed areas, s_{nd} , to Equation 3-12 (see Step 5 of Section 4.4). Note that the subscript s in Equation 3-12 must consequently be replaced with the subscript s_{nd} .

Step 4: Convert the average AGB for the strata level to average AGB in the Project Area:

Equation 3-13

$$\bar{B}_{AGB_gstock,t=0} = \frac{\sum_{j=1}^J (\bar{B}_{AGB_gstock,j,t=0} \times A_{project,j,t=0})}{A_{project,t=0}}$$

Parameter	Description	Unit
$\bar{B}_{AGB_gstock,t=0}$	Average aboveground biomass per hectare of the growing stock in the Project Area determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	(t d.m.) ha ⁻¹
$\bar{B}_{AGB_gstock,j,t=0}$	Average aboveground biomass of the growing stock in stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	(t d.m.) ha ⁻¹
$A_{project,j,t=0}$	Project Area within each stratum j , (where $j=1,2,3 \dots J$ strata) where the IFM-LtPF project activity will be implemented; determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	ha
$A_{project,t=0}$	Project Area where the IFM-LtPF project activity will be implemented; determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	ha

Step 5: Choose the carbon fraction of the AGB from the data sources presented in Step 4 of Section 3.2.1.2.1,A.

Step 6: Convert the average AGB to average carbon per hectare in the AGB in the Project Area:

Equation 3-14

$$\bar{C}_{AGB_gstock,t=0} = \bar{B}_{AGB_gstock,t=0} \times CF_{AGB}$$

Parameter	Description	Unit
$\bar{C}_{AGB_gstock,t=0}$	Average carbon per hectare in the aboveground biomass of the growing stock in the Project Area determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC ha ⁻¹
$\bar{B}_{AGB_gstock,t=0}$	Average aboveground biomass per hectare of the growing stock in the Project Area determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	(t d.m.) ha ⁻¹
CF_{AGB}	Carbon fraction in the aboveground biomass of trees for the tropical forest (see Appendix B)	tC (t d.m.) ⁻¹

3.2.2 MEASURED DATA PATHWAY

If the Project Proponent has obtained approval for selective logging from the relevant authority of the host country and also provides evidence of preparation of the FIR or an equivalent document, then data collection through field measurements in PSPs (see Section 7.1.2) after proper stratification (see Section 2.2.1.1.1) is required. This approach has been termed as Measured Data pathway in this Methodology.

The data collected through the measurement in the PSPs provides the diameter at breast height (DBH) and tree height (H) of the growing stock at the sample plot level s , in each stratum j . The volume and biomass allometric methods described for the Existing Inventory Data pathway where the detailed FIR is available, Sections 3.2.1.3.1 and 3.2.1.3.2, respectively, will be used for calculating the carbon in the merchantable logs and the AGB in the growing stock.

3.2.3 DETERMINATION OF ANNUAL HARVEST VOLUME AND NET HARVEST AREA

3.2.3.1 Where the FIR or equivalent document contains a detailed harvesting plan

In the case where the FIR or equivalent document, contains a detailed harvesting plan, the plan must provide a detailed prescription for selective logging, clearly stating the total volume of wood to be harvested, the annual net harvest area, as well as a specification of the period when selective logging operations are to be carried out in the Project Area. Since the document is legally approved by the relevant authority in the host country, the total volume of wood to be harvested must be converted to an annual harvest volume ($V_{merch,t}$) based on the growing stock ($V_{gstock,t=0}$) and the area where the selective logging occurs, i.e., the annual net harvest area ($A_{NHA_annual,t}$). The latter, if known, is the summation of the annual net harvest area at the stratum level ($A_{NHA_annual,j,t}$). These parameters must be used in calculating the net GHG emission reductions for the baseline scenario.

3.2.3.2 Where a detailed harvesting plan is not available

In the case where the Project Proponent has obtained the approval for selective logging and also provides evidence of the preparation of the FIR or equivalent document, but this document is not complete, the annual harvest volume and annual net harvest area (at the stratum level) are calculated from the growing stock data obtained from field measurements in the PSPs (i.e. the Measured Data pathway) using the following procedure:

- Step 1: The Project Proponent must select a sustainable and commonly employed method in the host country and prepare a timber harvesting plan for the selective logging in the Project Area. The Project Proponent must demonstrate that the plan is realistic and sustainable and will be implemented for selective logging if the IFM-LtPF project does not go ahead. This timber harvesting plan must provide the annual harvest volume (volume of merchantable logs to be removed) ($V_{merch,t}$) and the annual net harvest area ($A_{NHA_annual,t}$), which, if known, is the summation of the annual net harvest area at the stratum level ($A_{NHA_annual,j,t}$), where the harvesting would occur.
- Step 2: Since the annual harvest volume and annual net harvest area are based on the measured data, the Project Proponent must obtain approval from the relevant authority in the host country and also provide the documentary evidence in the PD.

The Project Proponent must also provide evidence or documentation that the chosen method for preparing the timber harvesting plan is sustainable and a commonly employed method in the host country.

3.2.4 ANNUAL TOTAL CARBON IN THE MERCHANTABLE LOGS

The annual total carbon in the merchantable logs, $C_{merch,t}$ is calculated by multiplying the average carbon per hectare in the merchantable logs by the annual net harvest area applying Equation 3-15a. In the case where the annual net harvest area at the stratum level is known, apply Equation 3-15b:

Equation 3-15a
$$C_{merch,t} = \bar{C}_{merch,t=0} \times A_{NHA_annual,t}$$

Equation 3-15b
$$C_{merch,t} = \sum_{j=1}^J (\bar{C}_{merch,j,t=0} \times A_{NHA_annual,j,t})$$

Parameter	Description	Unit
$C_{merch,t}$	Annual total carbon in the merchantable logs harvested in the Project Area in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\bar{C}_{merch,t=0}$	Average carbon per hectare in the merchantable logs determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC ha ⁻¹
$\bar{C}_{merch,j,t=0}$	Average carbon per hectare in the merchantable logs in stratum j (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC ha ⁻¹
$A_{NHA_annual,t}$	Annual net harvest area for the Project Area in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	ha
$A_{NHA_annual,j,t}$	Annual net harvest area for the Project Area in stratum j (where $j=1,2,3 \dots J$ strata) in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	ha

3.2.5 ANNUAL TOTAL CARBON IN THE ABOVEGROUND BIOMASS OF THE GROWING STOCK

The annual total carbon in the AGB of the growing stock, $C_{AGB_gstock,t}$ is calculated by multiplying the average carbon per hectare in the AGB of the growing stock by the annual net harvest area applying Equation 3-16a. In the case where the annual net harvest area at the stratum level is known, apply Equation 3-16b.

Equation 3-16a
$$C_{AGB_gstock,t} = \bar{C}_{AGB_gstock,t=0} \times A_{NHA_annual,t}$$

Equation 3-16b

$$C_{AGB_gstock,t} = \sum_{j=1}^J (\bar{C}_{AGB_gstock,j,t=0} \times A_{NHA_annual,j,t})$$

Parameter	Description	Unit
$C_{AGB_gstock,t}$	Annual total carbon in the aboveground biomass of the growing stock harvested every year in the Project Area in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\bar{C}_{AGB_gstock,t=0}$	Average carbon per hectare in the aboveground biomass of the growing stock determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC ha ⁻¹
$\bar{C}_{AGB_gstock,j,t=0}$	Average carbon per hectare in the aboveground biomass of the growing stock in stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC ha ⁻¹
$A_{NHA_annual,t}$	Annual net harvest area for the Project Area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	ha
$A_{NHA_annual,j,t}$	Annual net harvest area for the Project Area in stratum j , (where $j=1,2,3 \dots J$ strata) in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	ha

3.3 CARBON CHANGES DUE TO DEGRADATION UNDER THE BASELINE SCENARIO

The primary parameters, annual net harvest area, $A_{NHA_annual,t}$, annual total carbon in the merchantable logs, $C_{merch,t}$, and annual total carbon in the AGB of the growing stock, $C_{AGB_gstock,t}$, calculated in Section 3.2, are then used to calculate each of the five parameters of Equation 3-2 to determine the annual carbon lost from degradation. The procedures for calculating these five parameters are provided in the subsequent sections.

3.3.1 NET CARBON FROM THE DEADWOOD POOL

The DW pool contains carbon in coarse woody debris, standing dead trees and other dead material not included in the litter or soil carbon pools (IPCC, 2006, p. 4.73). The Methodology assumes that the proportion of DW accumulation to the DW pool and rate of decay from the pool due to natural mortality and disturbance will be the same for both the baseline and with-project scenarios, and is therefore not included.

In this Methodology, for the baseline scenario of selective logging, the additions (also referred to as inputs) to the carbon in the DW pool are:

- (i) Carbon from residual stand damage
- (ii) Carbon in branches and trimmings.

The carbon from DW is a function of the (delayed) decay of the combined inputs to the DW pool and is given by:

$$C_{DW_{decay},t} = f(C_{DW_{in},t}, k_{decay})$$

Equation 3-17

Parameter	Description	Unit
$C_{DW_{decay},t}$	Annual carbon leaving the deadwood pool due to the decay of deadwood in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{DW_{in},t}$	Annual total carbon input to the deadwood pool in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
k_{decay}	Rate of decay of the deadwood pool	yr ⁻¹

The annual total carbon input into the deadwood pool is calculated as follows:

$$C_{DW_{in},t} = C_{RSD,t} + C_{branch_trim,t}$$

Equation 3-18

Parameter	Description	Unit
$C_{DW_{in},t}$	Annual total carbon input to the deadwood pool in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{RSD,t}$	Annual carbon in the residual stand damage in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{branch_trim,t}$	Annual carbon in branches and trimmings left over from harvesting in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC

3.3.1.1 Carbon from residual stand damage

In the process of selective logging, the residual stand is damaged either by being knocked down, snapped off, or due to limb breakage. The fraction of residual stand that is damaged over the amount of timber that is extracted, is denoted as the factor for residual stand damage, f_{RSD} . Data on carbon impact from damage to residual stand by selective logging is very limited (Brown et. al., 2005).

Brown et al. (2005) compiled the factor for residual stand damage (damage over extracted) for various countries and revealed a strong relationship with commercial log length (Brown et al., 2005, Figure 11, p. 16). The results have been summarised in this Methodology in Table B-3 in Appendix B. The carbon from residual stand damage can be determined using the following procedure:

Step 1: Determine the average commercial log length based on the inventory data for the Project Area.

Step 2: Select an appropriate and conservative value for f_{RSD} from publications specific to the host country. Table B-3, Section B.3, Appendix B is provided as a guidance to determine a suitable f_{RSD} .

Step 3: Apply the following equation to calculate carbon in the residual stand damage:

Equation 3-19
$$C_{RSD,t} = f_{RSD} \times C_{merch,t}$$

Parameter	Description	Unit
$C_{RSD,t}$	Annual carbon in the residual stand damage in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
f_{RSD}	Factor for residual stand damage, based on the fraction of quantity of carbon damaged in the residual stand to the quantity of carbon in total merchantable logs harvested	dimensionless
$C_{merch,t}$	Annual total carbon in merchantable logs harvested in the Project Area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity) (See Equations 3-15a and 3-15b)	tC

3.3.1.2 Carbon in branches and trimmings

In the baseline scenario where selective logging occurs, trees are trimmed of their branches, twigs, and foliage to produce the merchantable logs intended for processing. Branches and trimmings is the term used to denote the trimmed components such as large and small branches, whose carbon will be transferred into the DW pool. Biomass from the twigs, foliage and bark are transferred into the litter pool. In accordance with the VCS rules, litter is not included as a carbon pool in this Methodology. Hence, only the fraction of branches and trimmings are accounted for in the dead wood pool.

To determine this fraction, a branch-trim factor, f_{branch_trim} , is employed (see Appendix A). The factor can be derived from the ratio of biomass of large branches and coarse woody debris over the total AGB of trees.

Determine the carbon in branches and trimmings using the following procedure:

Step 1: Find a branch-trim factor, f_{branch_trim} , from published peer reviewed literature for tropical forests of corresponding climate domain and ecological zone. If the branch-trim factor obtained from the literature does not match the forest type and climatic region of the Project Area, it will require validation via the steps outlined in Section 7.2.4.3. If a branch-trim factor can be obtained from the literature, proceed to Step 2. If a branch-trim factor cannot be obtained from the literature, it is conservative to set $C_{branch_trim,t}$ equal to zero in Equation 3-20. Carbon from branches and trimmings is thus omitted in the ex ante calculation and must be included when a

project-specific f_{branch_trim} can be derived from measurements in the first monitoring event (Section 7.2.4.3).

Step 2: Apply f_{branch_trim} in the following equation to estimate the carbon from branches and trimmings transferring into the DW pool:

Equation 3-20
$$C_{branch_trim,t} = f_{branch_trim} \times C_{merch,t}$$

Parameter	Description	Unit
$C_{branch_trim,t}$	Annual carbon in branches and trimmings left over from harvesting in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{merch,t}$	Annual total carbon in the merchantable logs harvested in the Project Area in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
f_{branch_trim}	The fraction of branches and trimmings in the aboveground biomass remaining after trimming of the merchantable logs transferred to the DW pool	dimensionless

Note: it is conservative to use $C_{merch,t}$ to determine $C_{branch_trim,t}$.

3.3.1.3 Carbon emissions due to decay of the DW pool

Step 1: Select a value for the rate of decay, k_{decay} , from published peer reviewed literature, verified publication or equivalent document, for tropical forests of corresponding climate domain and ecological zone (e.g. Delaney et al., 1998) for the first year of the project.

If a range of values for the rate of decay is provided in the literature, the Project Proponent is required to provide justification for the choice of the rate of decay in the PD. If no justification can be derived, it is conservative to select the lower end of the range.

If in the subsequent monitoring events, the Project Proponent decides to estimate the project specific rate of decay, then refer to the guidance proposed by Scott and Brown (2008).

Step 2: Determine the annual fraction of DW that would remain in the DW pool in year t (years elapsed since the start of the IFM-LtPF project activity) by applying the rate of decay in the following equation:

Equation 3-21
$$F_{DW_remain,t} = e^{-k_{decay} \times t}$$

Parameter	Description	Unit
$F_{DW_remain,t}$	Annual fraction of carbon in the deadwood pool that would remain in the deadwood pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity) after applying the rate of decay	dimensionless
k_{decay}	Rate of decay of the deadwood pool	yr ⁻¹
t	1,2,3 ... t^* years elapsed since the start of the IFM-LtPF project activity	yr

Step 3: Calculate the carbon remaining and accumulating in the DW pool in year t (years elapsed since the start of the IFM-LtPF project activity). For a fixed annual harvest volume as set out in the harvesting plan or equivalent document, implying a fixed input into the deadwood pool, use Equation 3-22a. For a variable annual harvest volume as set out in the harvesting plan or equivalent document, implying a variable input into the deadwood pool, use Equation 3-22b. Where Equation 3-22b is to be used, see Appendix D for an example application.

Equation 3-22a

$$C_{DW_{pool},t} = \sum_{t=1}^{t^*} (F_{DW_remain,t} \times C_{DW_{in},t})$$

Equation 3-22b

$$C_{DW_{pool},t} = (F_{DW_remain,t-(t-1)} \times C_{DW_{in},t}) + (F_{DW_remain,t-(t-2)} \times C_{DW_{in},t-1}) + (F_{DW_remain,t-(t-3)} \times C_{DW_{in},t-2}) + K + (F_{DW_remain,t-(t-t^*)} \times C_{DW_{in},t-(t^*-1)})$$

Parameter	Description	Unit
$C_{DW_{pool},t}$	Cumulative carbon remaining in the deadwood pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$F_{DW_remain,t}$	Annual fraction of carbon in the deadwood pool that would remain in the deadwood pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity) after applying the rate of decay	dimensionless
$C_{DW_{in},t}$	Annual total carbon input to the deadwood pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC

Step 4: Calculate the cumulative overall emissions leaving the DW pool in year t (years elapsed since the start of the IFM-LtPF project activity):

Equation 3-23

$$C_{DW_{out},t} = \sum_{t=1}^{t^*} C_{DW_{in},t} - C_{DW_{pool},t}$$

Parameter	Description	Unit
$C_{DW_{out},t}$	Cumulative carbon leaving the deadwood pool and emitted into the atmosphere in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{DW_{in},t}$	Annual total carbon input to the deadwood pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{DW_{pool},t}$	Cumulative carbon remaining in the deadwood pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC

Step 5: To calculate the annual emissions leaving the DW pool in year t (years elapsed since the start of the IFM-LtPF project activity), apply the following equation:

Equation 3-24

$$C_{DW_{decay},t} = C_{DW_{out},t} - C_{DW_{out},t-1}$$

Parameter	Description	Unit
$C_{DW_{decay},t}$	Annual carbon leaving the deadwood pool due to the decay of deadwood in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{DW_{out},t}$	Cumulative carbon leaving the deadwood pool and emitted into the atmosphere in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{DW_{out},t-1}$	Cumulative carbon leaving the deadwood pool and emitted into the atmosphere in year $t-1$, (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC

Note that for the first year of the project where $t=1$, Equation 3-24 shows that the annual carbon leaving the DW pool ($C_{DW_{decay},t}$) is equal to the cumulative carbon leaving the DW pool ($C_{DW_{out},t}$).

3.3.2 Net Carbon from the Long-term HWP Pool

Long-term HWPs (ltHWPs) are those with a half-life of over 30 years while products with a half-life of two years or less are known as short-term HWPs (stHWPs) (IPCC, 2006, Chapter 12, Section 12.2.2, Table 12-2).

The approach taken for ltHWPs is for analysis to be based on domestic harvest volume and not domestic consumption volume. In the case where a FIR is not available (see Figure 3-1) and where no other information is available to determine what quantity or ratio of the total merchantable logs' volume

is classed as stHWPs, then it is conservative to assume that all the merchantable logs' volume goes into the ltHWP pool.

In order to apply this Methodology, data availability with respect to the following parameters must be determined:

- average carbon in the ltHWPs
- lumber recovery factor (see Appendix B)
- ltHWP rate of oxidation (see Appendix B).

The carbon from ltHWPs is a combination of the immediate oxidation of ltHWP residues as a result of timber processing, and the delayed oxidation of ltHWPs, as given by:

$$C_{ltHWP_{oxidation},t} = C_{ltHWP_{residues},t} + C_{ltHWP_{net_out},t}$$

Equation 3-25

Parameter	Description	Unit
$C_{ltHWP_{oxidation},t}$	Annual carbon due to the combined delayed oxidation of long-term harvested wood products and immediate oxidation of long-term harvested wood products residues in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{ltHWP_{residues},t}$	Annual carbon due to the immediate oxidation of long-term harvested wood products residues in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{ltHWP_{net_out},t}$	Annual net carbon due to the delayed oxidation of the long-term harvested wood products, leaving the long-term harvested wood products pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC

3.3.2.1 Carbon due to long-term HWP residues

Long-term HWP residues are known to be comprised of saw dust, trimmings and other conversion losses (Enters, 2001). The lumber recovery factor for merchantable log processing is required in order to determine the carbon associated with sawmill residues. This can be determined using the following procedure:

Step 1: Select a lumber recovery factor from the following data sources:

- National lumber recovery factors in relevant literature such as FAO (2004). Various country specific lumber recovery factors have been compiled in Table B-4 in Appendix B; or
- If a national lumber recovery factor is not available, select a lumber recovery factor from a country that uses similar timber processing technology to that of the project's host country.

If a range for the lumber recovery factor is provided, the Project Proponent is required to provide justification for the choice of lumber recovery factor in the PD. If no justification can be derived, it is conservative to select the upper end of the range.

Step 2: Assuming that the sawmill residues undergo oxidation in the same year of processing, annual emissions due to ltHWP residues are determined by:

Equation 3-26

$$C_{ltHWP_{residues},t} = \bar{C}_{merch,p,t=0} \times (1 - f_{lumber_recovery}) \times A_{NHA_annual,t}$$

Parameter	Description	Unit
$C_{ltHWP_{residues},t}$	Annual carbon due to the immediate oxidation of long-term harvested wood products residues in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\bar{C}_{merch,p,t=0}$	Average carbon per hectare in merchantable logs of forest product type p =sawlog, in the Project Area determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC ha ⁻¹
$f_{lumber_recovery}$	Lumber recovery factor for proportion of merchantable log converted to harvested wood product	dimensionless
$A_{NHA_annual,t}$	Annual net harvest area for the Project Area in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	ha

3.3.2.2 Carbon due to oxidation of the long-term HWP pool

Calculate the average amount of carbon expected to be transferred into the HWP pool each year and stored over the long term. Wood harvested for production of paper products (and the like) and commercially harvested fuelwood, such as from pulplog (see HWPs - stHWPs in Appendix A), are not considered to be ltHWPs and are therefore accounted in Section 3.3.3 instead.

According to the IPCC's default values for oxidation of specific ltHWPs (IPCC, 2006, Chapter 12) it is noteworthy that the oxidation factors among these ltHWPs do not vary greatly. Hence, it is assumed that a single oxidation factor applied to ltHWPs would suffice.

Step 1: Select a lumber recovery factor from the data sources presented in Step 1 of Section 3.3.2.1.

Step 2: Multiply the carbon in the forest product type such as sawlog that contributes to the ltHWPs (derived in Equation 3-6) by the lumber recovery factor and annual net harvest area, as given by:

Equation 3-27

$$C_{ltHWP_{in},t} = \bar{C}_{merch,p,t=0} \times f_{lumber_recovery} \times A_{NHA_annual,t}$$

Parameter	Description	Unit
$C_{ltHWP_m,t}$	Annual carbon input to the long-term harvested wood products pool from sawlog in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\bar{C}_{merch,p,t=0}$	Average carbon per hectare in merchantable logs of forest product type p , in the Project Area determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC
$f_{lumber_recovery}$	Lumber recovery factor for proportion of merchantable log converted to harvested wood product	dimensionless
$A_{NHA_annual,t}$	Annual net harvest area for the Project Area in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	ha

Step 3: Find an appropriate rate of carbon oxidation⁵ (k_{ltHWP_ox}) specific to ltHWPs. For example, if it is known that the ltHWPs are to be made into sawnwood furniture (i.e. solidwood products from sawlogs), then the rate of carbon oxidation should (closely) reflect solidwood products. The IPCC (2006) have default rates of oxidation for HWP categories (see Appendix B, Section B.5, Table B-5).

If a range of values for the carbon rate of oxidation is provided in the literature, the Project Proponent is required to provide justification for the rate of carbon oxidation in the PD. If no justification can be derived, it is conservative to select the lower end of the range.

Step 4: Determine the annual fraction of ltHWP that would remain in the ltHWP pool in year t (years elapsed since the start of the IFM-LtPF project activity) by applying the rate of oxidation in the following equation:

Equation 3-28
$$F_{ltHWP_remain,t} = e^{-k_{ltHWP_ox} \times t}$$

Parameter	Description	Unit
$F_{ltHWP_remain,t}$	Annual fraction of ltHWP that would remain in the ltHWP pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity) after applying the rate of oxidation	dimensionless

⁵ The term "rate of carbon oxidation" is synonymous with "rate of decay". However in this Methodology, for improved clarity to differentiate between the rates used between long-term harvested wood products (ltHWPs) and deadwood (DW), "rate of carbon oxidation" is used in association with ltHWPs while "rate of decay" is used in association with DW.

Parameter	Description	Unit
k_{ltHWP_ox}	Rate of oxidation for long-term harvested wood products	yr ⁻¹
t	1,2,3 ... t^* years elapsed since the start of the IFM-LtPF project activity	yr

Step 5: Calculate the carbon remaining and accumulating in the ltHWP pool in year t (years elapsed since the start of the IFM-LtPF project activity). For a fixed annual harvest volume as set out in the harvesting plan or equivalent document, implying a fixed input into the ltHWP pool, use Equation 3-29a. For a variable harvest volume as set out in the harvesting plan or equivalent document, implying a variable input into the ltHWP pool, use Equation 3-29b. Where Equation 3-29b is to be used, see Appendix D for an example application:

Equation 3-29a

$$C_{ltHWP_{pool},t} = \sum_{t=1}^{t^*} (F_{ltHWP_remain,t} \times C_{ltHWP_{in},t})$$

Equation 3-29b

$$C_{ltHWP_{pool},t} = (F_{ltHWP_remain,t-(t-1)} \times C_{ltHWP_{in},t}) + (F_{ltHWP_remain,t-(t-2)} \times C_{ltHWP_{in},t-1}) + (F_{ltHWP_remain,t-(t-3)} \times C_{ltHWP_{in},t-2}) + K + (F_{ltHWP_remain,t-(t-t^*)} \times C_{ltHWP_{in},t-(t^*-1)})$$

Parameter	Description	Unit
$C_{ltHWP_{pool},t}$	Cumulative carbon remaining in the ltHWP pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$F_{ltHWP_remain,t}$	Annual fraction of ltHWP that would remain in the ltHWP pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity) after applying the rate of oxidation	dimensionless
$C_{ltHWP_{in},t}$	Annual carbon input to the long-term harvested wood products pool from sawlog in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC

Step 6: Calculate the cumulative overall emissions leaving the ltHWP pool in year t (years elapsed since the start of the IFM-LtPF project activity):

Equation 3-30

$$C_{ltHWP_{out},t} = \sum_{t=1}^{t^*} C_{ltHWP_{in},t} - C_{ltHWP_{pool},t}$$

Parameter	Description	Unit
$C_{ltHWP_{out},t}$	Cumulative carbon leaving the ltHWP pool and emitted into the atmosphere from year $t=1$ to year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{ltHWP_{in},t}$	Annual carbon input to the long-term harvested wood products pool from sawlog in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{ltHWP_{pool},t}$	Cumulative carbon remaining in the ltHWP pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC

Step 7: To calculate the annual emissions leaving the ltHWP pool in t years elapsed since the start of the IFM-LtPF project activity, apply the following equation:

Equation 3-31
$$C_{ltHWP_{net_out},t} = C_{ltHWP_{out},t} - C_{ltHWP_{out},t-1}$$

Parameter	Description	Unit
$C_{ltHWP_{net_out},t}$	Annual net carbon due to the delayed oxidation of the long-term harvested wood products, leaving the long-term harvested wood products pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{ltHWP_{out},t}$	Cumulative carbon leaving the ltHWP pool and emitted into the atmosphere in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{ltHWP_{out},t-1}$	Cumulative carbon leaving the ltHWP pool and emitted into the atmosphere in year $t-1$, (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC

Note, that for the first year of the project where $t=1$, Equation 3-31 shows that the annual carbon leaving the ltHWP pool ($C_{ltHWP_{net_out},t}$) is equal to the cumulative carbon leaving the ltHWP pool ($C_{ltHWP_{out},t}$).

3.3.3 Net Carbon from the Short-term HWP Pool

Certain merchantable logs are destined to be converted into HWPs with a half-life of two years or less. These are referred to here as short-term HWPs (stHWPs; IPCC, 2006, Chapter 12, Section 12.2.2, p. 12.17). In this Methodology, the two forest product types contributing to stHWPs are pulplog (for paper products) and commercially harvested fuelwood. Similar to ltHWPs, the approach taken for stHWPs is for analysis to be based on domestic harvest volume and not domestic consumption volume.

In the case where inventory data do not distinguish the forest product types and/or where no other information is available to determine what quantity or ratio of the total merchantable logs' volume is classed into stHWPs, then it is conservative to assume that all the merchantable logs' volume goes into the ltHWP pool.

The annual carbon from stHWPs is a combination of the immediate oxidation of commercially harvested fuelwood and delayed oxidation of paper products, or solely from commercially harvested fuelwood, or solely from paper products. Equation 3-32 provides the annual carbon from the combination of both stHWPs. In the case where oxidation is attributable to only commercially harvested fuelwood, the parameter for paper products becomes zero - and vice versa.

$$C_{stHWP_{oxidation},t} = C_{stHWP_{comm_FW},t} + C_{stHWP_{net_out},t}$$

Equation 3-32

Parameter	Description	Unit
$C_{stHWP_{oxidation},t}$	Annual carbon due to immediate oxidation of short-term harvested wood products (commercially harvested fuelwood) and delayed oxidation of short-term harvested wood products (paper products) in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{stHWP_{comm_FW},t}$	Annual carbon due to immediate oxidation of short-term harvested wood products (commercially harvested fuelwood) leaving the project boundary in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{stHWP_{net_out},t}$	Annual net carbon due to the delayed oxidation of short-term harvested wood products (paper products), leaving the short-term harvested wood products pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC

3.3.3.1 Carbon in short-term HWPs - commercially harvested fuelwood

For commercially harvested fuelwood, immediate oxidation occurs upon consumption - and therefore is regarded as an (immediate) emission in the year of harvest. The carbon from stHWPs is thus considered from the immediate oxidation of the stHWP merchantable logs.

The annual carbon from stHWPs - commercially harvested fuelwood, is calculated by multiplying the average carbon per hectare in the commercially harvested fuelwood with the annual net harvest area:

$$C_{stHWP_{comm_FW},t} = \bar{C}_{merch,p,t=0} \times A_{NHA_annual,t}$$

Equation 3-33

Parameter	Description	Unit
$C_{stHWP_{comm_FW},t}$	Annual carbon due to immediate oxidation of short-term harvested wood products (commercially harvested fuelwood) leaving the project boundary in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\bar{C}_{merch,p,t=0}$	Average carbon per hectare in merchantable logs of forest product type p =pullog, in the Project Area determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC ha ⁻¹
$A_{NHA_annual,t}$	Annual net harvest area for the Project Area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	ha

3.3.3.2 Carbon in short-term HWP's - pulplog

Pulplogs are mainly used for paper products. For paper products, the IPCC (2006) suggests to use a (delayed) rate of oxidation (see Appendix B). Hence, use the following procedure:

Step 1: Find an appropriate rate of carbon oxidation (k_{stHWP_ox}) specific to paper products (or see Appendix B). If a range of values for the rate of oxidation is provided in the literature, the Project Proponent is required to provide justification for the choice of the rate of oxidation in the PD. If no justification can be derived, it is conservative to select the lower end of the range. Then apply the following equation:

Equation 3-34
$$F_{stHWP_remain,t} = e^{-k_{stHWP_ox} \times t}$$

Parameter	Description	Unit
$F_{stHWP_remain,t}$	Annual fraction of stHWP that would remain in the stHWP pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity) after applying the rate of oxidation	dimensionless
k_{stHWP_ox}	Rate of oxidation for short-term harvested wood products	yr ⁻¹
t	1,2,3 ... t^* years elapsed since the start of the IFM-LtPF project activity	yr

Step 2: Apply the following equation (note this is modified from Equation 3-33 because paper products are not immediately oxidized, these go into the stHWP pool).

Equation 3-35
$$C_{stHWP_{in},t} = \bar{C}_{merch,p,t=0} \times A_{NHA_annual,t}$$

Parameter	Description	Unit
$C_{stHWP_{in},t}$	Annual carbon input to the short-term harvested wood products pool, for the specific forest product type, pulplog, in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\bar{C}_{merch,p,t=0}$	Average carbon per hectare in merchantable logs of forest product type p =pulplog, in the Project Area determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	tC ha ⁻¹
$A_{NHA_annual,t}$	Annual net harvest area for the Project Area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	ha

Step 3: Apply Equations 3-29a or 3-29b through to 3-31, changing the descriptions (and subscripts) from ltHWP to stHWP to calculate the annual net carbon due to the delayed oxidation of the stHWPs, leaving the stHWP pool in t years elapsed since start of the IFM-LtPF project activity.

3.3.4 CARBON IN THE GROWTH FOREGONE DUE TO SELECTIVE LOGGING

The selective logging operations removes a prescribed amount of merchantable logs after harvesting of the selected merchantable trees from the Project Area under the baseline scenario. Consequently, the carbon that would have been stored from the annual growth in these trees is forfeited. The carbon lost here is denoted as carbon in the growth foregone, $C_{growth_foregone,t}$, and this can be calculated using the following procedure:

Step 1: Calculate the AGB in merchantable trees from height and diameter measurements of all the merchantable trees in the PSPs at the stratum level for two consecutive monitoring events using the method presented in Section 3.2.1.3.2. The procedure for establishing PSPs and tree measurements are described in Section 7.1.2.

The method to determine the annual growth parameter is based on the stock change method presented in the approved CDM Methodology for Afforestation and Reforestation project activities (CDM-EB, 2009, p. 8). The difference between the AGB of merchantable trees from data obtained in the PSP measurements at the stratum level for two consecutive monitoring events is determined and divided by the monitoring period, i.e. the time interval taken between the two monitoring events (in years), at the stratum level using Equation 3-36a and, for the Project Area using Equation 3-36b:

Equation 3-36a

$$\bar{G}_{growth_foregone,j,t} = \frac{\bar{B}_{AGB_merch,j,m_2} - \bar{B}_{AGB_merch,j,m_1}}{\Delta m}$$

Equation 3-36b

$$\bar{G}_{growth_foregone,t} = \frac{\sum_{j=1}^J (\bar{G}_{growth_foregone,j,t} \times A_{project,j,t=0})}{A_{project,t=0}}$$

Parameter	Description	Unit
$\bar{G}_{growth_foregone,j,t}$	Average growth per hectare per year in the aboveground biomass in the merchantable trees in the Project Area in stratum j , (where $j=1,2,3 \dots J$ strata) in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	(t d.m.) ha ⁻¹ yr ⁻¹
$\bar{B}_{AGB_merch,j,m_2}$	Average aboveground biomass of the merchantable trees in the Project Area in stratum j , (where $j=1,2,3 \dots J$ strata) in year, m_2 - when current monitoring event occurs	(t d.m.) ha ⁻¹
$\bar{B}_{AGB_merch,j,m_1}$	Average aboveground biomass of the merchantable trees in the Project Area in stratum j , (where $j=1,2,3 \dots J$ strata) in year, m_1 - when preceding monitoring event occurred	(t d.m.) ha ⁻¹

Parameter	Description	Unit
Δm	Monitoring period, the time taken between two consecutive monitoring events	yr
$\bar{G}_{growth_foregone,t}$	Average growth per hectare per year in the aboveground biomass in the merchantable trees in the Project Area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	(t d.m.) ha ⁻¹ yr ⁻¹
$A_{project,j,t=0}$	Project Area within each stratum j , (where $j=1,2,3 \dots J$ strata) where the IFM-LtPF project activity will be implemented; determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	ha
$A_{project,t=0}$	Project Area where the IFM-LtPF project activity will be implemented; determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	ha

Step 2: Calculate the carbon in the growth foregone due to the selective logging using Equation 3-37a. In the case where the annual net harvest area at the stratum level is known, apply Equation 3-37b:

Equation 3-37a
$$C_{growth_foregone,t} = CF_{AGB} \times \bar{G}_{growth_foregone,t} \times \sum_{t=1}^{t^*} A_{NHA_annual,t}$$

Equation 3-37b
$$C_{growth_foregone,t} = CF_{AGB} \times \sum_{j=1}^J \left(\bar{G}_{growth_foregone,j,t} \times \sum_{t=1}^{t^*} A_{NHA_annual,j,t} \right)$$

Parameter	Description	Unit
$C_{growth_foregone,t}$	Annual carbon lost due to growth foregone in the aboveground biomass in the Project Area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\bar{G}_{growth_foregone,t}$	Average growth per hectare per year in the aboveground biomass in the merchantable trees in the Project Area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	(t d.m.) ha ⁻¹ yr ⁻¹
$\bar{G}_{growth_foregone,j,t}$	Average growth per hectare per year in the aboveground biomass in the merchantable trees in the Project Area in stratum j , (where $j=1,2,3 \dots J$ strata) in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	(t d.m.) ha ⁻¹ yr ⁻¹

Parameter	Description	Unit
CF_{AGB}	Carbon fraction in the aboveground biomass of trees for the tropical forest (see Appendix B)	tC (t d.m.) ⁻¹
$A_{NHA_annual,j,t}$	Annual net harvest area for the Project Area in stratum j , (where $j=1,2,3 \dots J$ strata) in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	ha

3.3.5 Carbon in the Regrowth after Selective Logging

It is good practice to account for regrowth in the gaps following logging. The IPCC (2006) suggested approach for estimating the regrowth is based on the mean annual growth in the AGB after logging, denoted here as $\bar{G}_{regrowth,t}$.

This Methodology applies a conservative approach for area of regrowth by considering that the entire annual net harvest area would permit regrowth each year. The annual net harvest area for the selective logging is calculated in Section 3.2.3.

Determine carbon in the regrowth after selective logging using the following procedure:

Step 1: Select the factor for the annual average growth in the aboveground biomass after selective logging, or a time-dependent growth model from the following sources:

- (i) Local growth tables or growth models
- (ii) National growth tables or growth models (e.g., from National GHG Inventory)
- (iii) Published peer reviewed studies for growth models/tables from previous logged tropical forests with corresponding age, climate region and ecological zone.

Step 2: Determine the carbon due to regrowth by multiplying the annual net harvest area(s) and the average growth rate per hectare per year by the carbon fraction:

Equation 3-38

$$C_{regrowth,t} = (\bar{G}_{regrowth,t} \times CF_{AGB}) \times \sum_{t=1}^{t^*} A_{NHA_annual,t}$$

Parameter	Description	Unit
$C_{regrowth,t}$	Annual carbon increase in the biomass due to regrowth following logging in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\bar{G}_{regrowth,t}$	Average regrowth per hectare per year of the aboveground biomass after logging in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	(t d.m.) ha ⁻¹ yr ⁻¹

Parameter	Description	Unit
CF_{AGB}	Carbon fraction in the aboveground biomass of trees for the tropical forest (see Appendix B)	tC (t d.m.) ⁻¹
$A_{NHA_annual,t}$	Annual net harvest area for the Project Area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity) ^{^^}	ha

3.4 Baseline Activity Emissions

A number of emission sources are associated with the implementation of the baseline (selective logging) activity. Omission of any or all of the associated emission sources is conservative. The Project Proponent intending to calculate baseline activity emissions must select data required in the calculations on a conservative basis.

The stages of operation for timber harvesting include, but are not limited to, the process steps outlined in Figure 3-2 below:

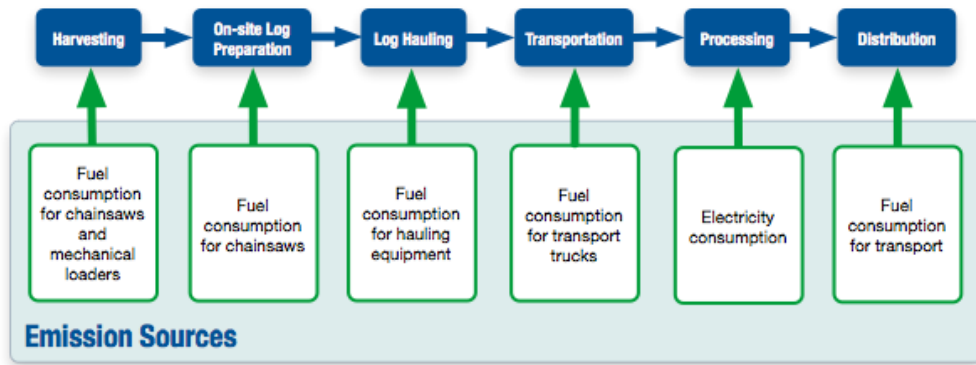


Figure 3-2. An outline of timber harvesting operations and the associated emission sources

The total emissions due to the baseline activity ($C'_{emissions,t}$) is determined from the summation of all the emission sources presented in Figure 3-2 using the following equation:

Equation 3-39
$$C'_{emissions,t} = E_{harvest,t} + E_{onsiteprep,t} + E_{hauling,t} + E_{transport,t} + E_{processing,t} + E_{distribution,t}$$

Parameter	Description	Unit
$C'_{emissions,t}$	Annual total carbon emissions associated with the baseline activity of selective logging operations in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$E_{harvest,t}$	Annual emissions due to harvesting operations such as felling and snigging in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

Parameter	Description	Unit
$E_{onsiteprep,t}$	Annual emissions due to on-site preparation such as trimming of tree heads, butts, branches and defective components in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$E_{hauling,t}$	Annual emissions due to log hauling in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$E_{transport,t}$	Annual emissions due to log transport from collection depot to processing plant in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$E_{processing,t}$	Annual emissions due to electricity consumption in sawmill in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$E_{distribution,t}$	Annual emissions due to transport of the sawn product from the mill to the wharf for export or to the depot for local usage in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

Ex ante baseline activity emissions $C'_{emissions,t}$ will be made using default emission factor data and activity data based on common practice in the area. The Project Proponent must ensure that the default values and activity data used are conservative.

3.4.1 Emissions Due to Harvesting Operations

Mechanical harvesting operations ($E_{harvest,t}$) of merchantable logs contributes to GHG emissions. This operation in countries containing tropical forests commonly employs chainsaws, while in other regions harvesters are commonly employed. Emissions from log harvesting are determined using the following procedure:

Step 1: For the common practice in the host country, select the typical equipment type, size and type of fuel consumed that is employed for harvesting. Information on harvesting practices can be found from reports on previous and existing harvesting practices in the host country.

Step 2: Find the fuel consumption (kL m⁻³) of the selected equipment from the following data sources:

Reports on common practice for harvesting in the host country and/or manufacturers specifications

Published peer reviewed studies on harvesting operations (e.g. Klvac and Skoupy (2009) in Table B-6, Appendix B).

If a range for fuel consumption is provided, the Project Proponent is required to provide justification for their choice of fuel consumption in the PD. If no justification can be derived, it is conservative to select the lower end of the range.

Step 3: Select the emission factor associated with the fuel employed for harvesting from IPCC default emission factors for CO₂, CH₄ and N₂O emissions for Off-Road Mobile Sources and Machinery for the Forestry Sector (IPCC, 2006, Volume 2, Chapter 3, Section 3.3.1, Table 3.3.1, p. 3.36). Guidance for converting the emission factors to a CO₂ equivalent (CO₂-e) emission factor (EF_{fuel}) is provided in Appendix C.2. Guidance for unit conversions of fuel emission factors is provided in Appendix C.3.

Step 4: Emissions from harvesting of $V_{merch,t}$ are then determined by applying the following equation:

Equation 3-40
$$E_{harvest,t} = FC_{harvest} \times EF_{fuel} \times V_{merch,t}$$

Parameter	Description	Unit
$E_{harvest,t}$	Annual emissions due to harvesting operations such as felling and snigging in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$FC_{harvest}$	Fuel consumption of equipment employed for felling and snigging per m ³ of merchantable log harvested	kL m ⁻³
EF_{fuel}	Fuel emission factor	tCO ₂ -e kL ⁻¹
$V_{merch,t}$	Annual volume of merchantable logs in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity) (see Section 3.2.3)	m ³

3.4.2 Emissions Due to On-site Preparation

Emissions due to on-site preparation ($E_{onsiteprep,t}$) such as trimming of tree heads, butts, branches and defective components are estimated using the following procedure:

Step 1: Select the typical equipment type, size and type of fuel consumed that is employed for trimming based on common practice for harvesting in the host country. For example, in tropical countries, chainsaws are commonly employed. Information on harvesting practices can be found from reports on previous and existing harvesting practices in the host country.

Step 2: Find the fuel consumption (KL m^{-3}) of the selected equipment from reports on common practice for harvesting in the host country and/or manufacturers specifications.

If a range for fuel consumption is provided, the Project Proponent is required to provide justification for their choice of fuel consumption factor in the PD. If no justification can be derived, it is conservative to select the lower end of the range.

Step 3: Select the emission factor associated with CO_2 , CH_4 and N_2O emissions for the fuel employed from IPCC default emission factors for Off-Road Mobile Sources and Machinery for the Forestry Sector (IPCC, 2006, Volume 2, Chapter 3, Section 3.3.1, Table 3.3.1, p 3.36). Guidance for converting the emission factors to a CO_2 equivalent ($\text{CO}_2\text{-e}$) emission factor (EF_{fuel}) is provided in Appendix C.2. Guidance for unit conversions of fuel emission factors is provided in Appendix C.3.

Step 4: Choose the most applicable wood density for a forest with corresponding climate region and ecological zone, and choose the most applicable carbon fraction of wood from the data sources presented in Step 4 of Section 3.2.1.2.1,A. Then determine the volume of trimmings and branches using the following equation:

Equation 3-41

$$V_{branch_trim,t} = \frac{C_{branch_trim,t}}{CF_{wood}} \times D$$

Parameter	Description	Unit
$V_{branch_trim,t}$	Annual volume of the trimmings and branches produced from harvesting in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	m^3
$C_{branch_trim,t}$	Annual carbon in branches and trimmings left over from harvesting in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
CF_{wood}	Carbon fraction of wood for the tropical forest (see Appendix B)	tC (t d.m.) ⁻¹
D	Wood density for the tropical forest with corresponding climate region and ecological zone (see Appendix B)	(t d.m.) m^{-3}

Step 5: Emissions from on-site preparation is then determined by applying the following equation:

Equation 3-42

$$E_{onsiteprep,t} = FC_{trim_equip} \times EF_{fuel} \times V_{branch_trim,t}$$

Parameter	Description	Unit
$E_{onsiteprep,t}$	Annual emissions due to on-site preparation such as trimming of tree heads, butts, branches and defective components in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
FC_{trim_equip}	Fuel consumption of equipment employed for trimming per m ³ of trimmed material	kL m ⁻³
EF_{fuel}	Fuel emission factor	tCO ₂ -e kL ⁻¹
$V_{branch_trim,t}$	Annual volume of the trimmings and branches produced from harvesting in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	m ³

3.4.3 Emissions Due to Log Hauling

Emissions associated with log hauling ($E_{hauling,t}$) to the collection depot are calculated using the following procedure:

Step 1: Select the typical equipment type, size and type of fuel consumed that is employed for hauling based on common practice for harvesting in the host country. Information on harvesting practices can be found from reports on previous and existing harvesting practices in the host country.

Step 2: Find the fuel consumption (kL m⁻³) of the selected equipment from reports on common practice for harvesting in the host country and/or manufacturers specifications.

If a range for fuel consumption is provided, the Project Proponent is required to provide justification for the choice of fuel consumption factor in the PD. If no justification can be derived, it is conservative to select the lower end of the range.

Step 3: Select the emission factor associated with CO₂, CH₄ and N₂O emissions for the fuel employed from IPCC default emission factors for Off-Road Mobile Sources and Machinery for the Forestry Sector (IPCC, 2006, Volume 2, Chapter 3, Section 3.3.1, Table 3.3.1, p. 3.36). Guidance for converting the emission factors to a CO₂ equivalent (CO₂-e) emission factor (EF_{fuel}) is provided in Appendix C.2. Guidance for unit conversions of fuel emission factors is provided in Appendix C.3.

Step 4: Emissions from log hauling are then determined by applying the following equation:

Equation 3-43
$$E_{hauling,t} = FC_{hauling} \times EF_{fuel} \times V_{merch,t}$$

Parameter	Description	Unit
$E_{hauling,t}$	Annual emissions due to log hauling in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$FC_{hauling}$	Fuel consumption of equipment for hauling one m ³ of merchantable log	kL m ⁻³
EF_{fuel}	Fuel emission factor	tCO ₂ -e kL ⁻¹
$V_{merch,t}$	Annual volume of merchantable logs in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity) (see Section 3.2.3)	m ³

3.4.4 Emissions Due to Log Transport

Emissions associated with log transport ($E_{transport,t}$) from the collection depot to the processing plant are calculated using the following procedure:

Step 1: Select the vehicle: type, load capacity (m³ truck⁻¹) and type of fuel consumed for log transport based on common practice for transport in the host country. Information on harvesting practices can be found from reports on previous and existing harvesting practices in the host country.

Step 2: Determine the number of trucks required using the following equation:

Equation 3-44

$$N_{trucks_transport,t} = \frac{V_{merch,t}}{Cap_{truck}}$$

Parameter	Description	Unit
$N_{trucks_transport,t}$	Number of truck trips required for log transport from collection depot to processing plant in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	truck
$V_{merch,t}$	Annual volume of merchantable logs in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity) (see Section 3.2.3)	m ³
Cap_{truck}	Truck load capacity	m ³ truck ⁻¹

Step 3: Find the fuel efficiency (km kL⁻¹) of the selected vehicle from the following data sources:

- (i) Manufacturers specifications
- (ii) Published peer reviewed studies on harvesting operations (e.g. Kinjo et al., 2005).

If a range for fuel efficiency is provided, the Project Proponent is required to provide justification for the choice of fuel efficiency factor in the PD. If no justification can be derived, it is conservative to select the upper end of the range.

Step 4: Select the emission factor associated with CO₂, CH₄ and N₂O emissions for the fuel employed from IPCC default emission factors for Road Transport (IPCC, 2006, Volume 2, Chapter 3, Section 3.2.1.2, Table 3.2.1-2, p. 3.16-3.21). Guidance for converting the emission factors to a CO₂ equivalent (CO₂-e) emission factor (EF_{fuel}) is provided in Appendix C.2. Guidance for unit conversions of fuel emission factors is provided in Appendix C.3.

Step 5: Estimate the transport distance from the collection depot to the processing plant ($KM_{transport,t}$) using the digital maps of the Project Area. If transport route(s) for the baseline scenario must be hypothesised, the Project Proponent is required to provide justification for their derivation of the transport route in the PD. In addition, to be conservative, the transport route proposed must be the minimum possible route.

The total log transport distance can be determined by the following:

Equation 3-45

$$KM_{transport_total,t} = KM_{transport,t} \times N_{trucks_transport,t} \times 2$$

Parameter	Description	Unit
$KM_{transport_total,t}$	Annual total log transport distance in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	km
$KM_{transport,t}$	Annual log transport distance from collection depot to processing plant in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	km truck ⁻¹
$N_{trucks_transport,t}$	Number of truck trips required for log transport from collection depot to processing plant in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	truck
2	Constant, indicating return trip	dimensionless

Step 6: Emissions due to log transport are then determined by applying the following equation:

Equation 3-46

$$E_{transport,t} = \frac{KM_{transport_total,t}}{Eff_{vehicle}} \times EF_{fuel}$$

Parameter	Description	Unit
$E_{transport,t}$	Annual emissions due to log transport haulage from felling location to the collection depot/ sawmill in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$KM_{transport_total,t}$	Annual total log transport distance in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	km
$Eff_{vehicle}$	Fuel efficiency for vehicle type	km kL ⁻¹
EF_{fuel}	Fuel emission factor	tCO ₂ -e kL ⁻¹

3.4.5 Emissions Due to Timber Processing

Emissions as a result of processing ($E_{processing,t}$), where the processing plant is situated in the Project Area, depend on the electricity source available at the processing site. Electricity can be supplied via the national grid, or where this is not available, supplied via on-site generators. In addition, mill residue/waste may also be used as an energy source during timber processing.

To avoid double accounting, if a processing plant utilises mill residue/waste as an electricity source, then the emissions from electricity generated by mill residue/waste must not be considered here - as these emissions have already been accounted for in Section 3.3.2.1.

3.4.5.1 Emissions from processing where grid electricity is available

If grid electricity is available, then emissions due to processing will be calculated using the following procedure:

Step 1: Select an electricity demand factor (e_{demand}) for the timber processing facility from the following data sources:

- (i) National electricity demand factors in relevant literature for timber processing. Various country specific electricity demand factors have been compiled in Table B-7 in Appendix B
- (ii) If a national electricity demand factor is not available, select an electricity demand factor from a country that uses similar timber processing technology to that of the project's host country.

If a range for the electricity demand factor is provided, the Project Proponent is required to provide justification for the choice of electricity demand factor in the PD. If no justification can be derived, it is conservative to select the lower end of the range.

Step 2: Using the volume of merchantable logs, determine the electricity consumption required for the processing mill:

Equation 3-47
$$Q_{processing,t} = V_{merch,t} \times e_{demand}$$

Parameter	Description	Unit
$Q_{processing,t}$	Annual quantity of electricity consumption for processing in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	kWh
$V_{merch,t}$	Annual volume of merchantable logs in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity) (see Section 3.2.3)	m ³
e_{demand}	Electricity demand for processing per volume processed	kWh m ⁻³

Step 3: Select an emission factor for electricity for the host country from IEA (2009), CO₂ emissions per kWh from electricity and heat generation, pp. 101-103.

Step 4: Apply the following equation to determine the emissions due to processing:

Equation 3-48
$$E_{processing,t} = Q_{processing,t} \times EF_{electricity}$$

Parameter	Description	Unit
$E_{processing,t}$	Annual emissions due to electricity consumption in sawmill in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$Q_{processing,t}$	Annual quantity of electricity consumption for processing in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	kWh
$EF_{electricity}$	Electricity emission factor for the host country	tCO ₂ -e kWh ⁻¹

3.4.5.2 Emissions from processing where a generator is required

If grid electricity is not available, and an on-site generator is required to power the processing facility, then calculation of processing emissions is determined from the emissions associated with fuel consumption of the generator. Apply Steps 1 and 2 of Section 3.4.5.1 to determine the electricity consumption involved for processing ($Q_{processing,t}$). Then use the following procedure:

Step 1: Select the typical load capacity (1/4, 1/2, full load), type of fuel consumed and operating time of the generator employed to power the processing facility. Such data can be obtained from the following data sources:

- (i) National reports from the relevant national forestry authority
- (ii) Published peer reviewed literature on timber processing for cases of equivalent volumes and processing modes and of a country that uses similar timber processing technology to that of the project's host country.

Step 2: Determine the power rating of the generator from the electricity required for processing and the operating hours required:

Equation 3-49

$$PR_{generator,t} = \frac{Q_{processing,t}}{t_{generator,t}}$$

Parameter	Description	Unit
$PR_{generator,t}$	Power rating of generator or generator size in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	kW
$Q_{processing,t}$	Annual quantity of electricity consumption for processing in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	kWh
$t_{generator,t}$	Annual operating time of generator in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	h

Step 3: Using a fuel consumption chart for the generator selected for the baseline case, apply the power rating (kW) and an appropriate load capacity (1/4, 1/2, full) to determine the fuel consumption

(kL h^{-1}) of the generator. Fuel consumption charts can be found from manufacturers of generators (see Table B-8, Appendix B for an example).

Step 4: Select an emissions factor for the fuel employed in the generator from IPCC (2006), Volume 2, Chapter 2 Stationary Combustion, Table 2.5, p. 2.22.

Step 5: Apply the following equation to determine emissions due to processing:

Equation 3-50
$$E_{\text{processing},t} = FC_{\text{generator}} \times t_{\text{generator},t} \times EF_{\text{fuel}}$$

Parameter	Description	Unit
$E_{\text{processing},t}$	Annual emissions due to electricity consumption in sawmill in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	$\text{tCO}_2\text{-e}$
$FC_{\text{generator}}$	Fuel consumption per hour of operation of generator	kL h^{-1}
$t_{\text{generator},t}$	Annual operating time of generator in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	h
EF_{fuel}	Fuel emission factor	$\text{tCO}_2\text{-e kL}^{-1}$

3.4.6 Emissions Due to Log Distribution

Emissions due to distributing logs ($E_{\text{distribution},t}$) are calculated using the following procedure:

Step 1: Select the vehicle type, load capacity (for e.g. $\text{m}^3 \text{ truck}^{-1}$) and type of fuel consumed, for log distribution based on common practice in the host country.

Step 2: Calculate the annual total volume of LtHWP to be distributed per year from the input of carbon in LtHWP pool as determined from Equation 3-27 in Section 3.3.2.2, and the most applicable density and carbon fraction of wood for a forest with corresponding climate region and ecological zone (see Section B-1 and B-2 of Appendix B).

Equation 3-51
$$V_{\text{LtHWP},t} = \frac{C_{\text{LtHWP}_{\text{in}},t}}{D \times CF_{\text{wood}}}$$

Parameter	Description	Unit
$V_{\text{LtHWP},t}$	Annual volume of long-term harvested wood products in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	m^3
$C_{\text{LtHWP}_{\text{in}},t}$	Annual carbon input to the long-term harvested wood products pool from sawlog in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC

Parameter	Description	Unit
CF_{wood}	Carbon fraction of wood for the tropical forest (see Appendix B)	tC (t d.m.) ⁻¹
D	Wood density for the tropical forest with corresponding climate region and ecological zone (see Appendix B)	(t d.m.) m ⁻³

Step 3: Determine the number of trucks required for distribution by dividing the long-term HWP volume by the load capacity of the vehicle:

Equation 3-52

$$N_{trucks_distrib,t} = \frac{V_{ltHWP,t}}{Cap_{truck}}$$

Parameter	Description	Unit
$N_{trucks_distrib,t}$	Annual number of truck trips required for log distribution from processing plant to distribution/export point in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	truck
$V_{ltHWP,t}$	Annual volume of long-term harvested wood products in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	m ³
Cap_{truck}	Truck load capacity	m ³ truck ⁻¹

Step 4: Find the fuel efficiency (km kL⁻¹) of the selected vehicle from the following data sources:

- (i) Manufacturers specifications
- (ii) Published peer reviewed studies on harvesting operations (e.g. Kinjo et al., 2005).

If a range for fuel efficiency is provided, the Project Proponent is required to provide justification for the choice of fuel efficiency factor in the PD. If no justification can be derived, it is conservative to select the upper end of the range.

Step 5: Select the emission factor associated with CO₂, CH₄ and N₂O emissions for the fuel employed from IPCC default emission factors for Road Transport (IPCC, 2006, Volume 2, Chapter 3, Section 3.2.1.2, Table 3.2.1-2, p 3.16-3.21). Guidance for converting the emission factors to a CO₂ equivalent (CO₂-e) emission factor (EF_{fuel}) is provided in Appendix C.2. Guidance for unit conversions of fuel emission factors is provided in Appendix C.3.

Step 6: Estimate the distance for distribution from the processing plant to the distribution/export point ($KM_{distrib,t}$) using digital maps. If distribution route(s) for the baseline scenario must be hypothesised, the Project Proponent is required to provide justification for the derivation of a distribution route in the PD. In addition, to be conservative, the distribution route proposed must be the minimum possible route.

The total distance for distribution can be determined by the following:

Equation 3-53

$$KM_{distribtotal,t} = KM_{distrib,t} \times N_{trucks_distrib,t} \times 2$$

Parameter	Description	Unit
$KM_{distribtotal,t}$	Annual total distribution distance in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	km
$KM_{distrib,t}$	Annual distance of transport from point of processing to distribution/export point in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	km truck ⁻¹
$N_{trucks_distrib,t}$	Annual number of truck trips required for log distribution from processing plant to distribution/export point in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	truck
2	Constant, indicating return trip	dimensionless

Step 7: Apply the following equation to determine the emissions due to log distribution:

Equation 3-54

$$E_{distribution,t} = \frac{KM_{distribtotal,t}}{Eff_{vehicle}} \times EF_{fuel}$$

Parameter	Description	Unit
$E_{distribution,t}$	Annual emissions due to transport of the sawn product from the mill to the wharf for export or to the depot for local usage in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$KM_{distribtotal,t}$	Annual total distribution distance in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	km
$Eff_{vehicle}$	Fuel efficiency for vehicle type	km kL ⁻¹
EF_{fuel}	Fuel emission factor	tCO ₂ -e kL ⁻¹

4 Actual Project Activity Emissions

The project activity is the introduction of forest protection without selective logging (i.e. no emissions from harvesting) - Improved Forest Management (Logged Forest to Protected Forest). However, implementing the IFM-LtPF project activity will incur quantifiable emissions. In addition, the forest in the Project Area is likely to be disturbed by natural causes such as fires, landslides, volcanoes etc. and may also be harvested illegally. The Project Proponent must take all possible measures to prevent carbon emissions from such causes and must keep detail and accurate records of emissions associated with the project activities including the monitoring of the Project Area for natural disturbances. Such records and data must be submitted along with the monitoring report to the Verifier.

Emissions associated with the actual project implementation is given by Equation 4-1, and include, but are not limited to:

- administration and planning (electricity and fuel consumption) ($E_{projplan,t}$)
- travel for design and set up (e.g. consultation and education) (fuel consumption) ($E_{design,t}$)
- travel for implementing monitoring plan (e.g. from on-the-ground forest surveillance) ($E_{monitoring,t}$)
- natural disturbances such as forest fires ($C_{natdisturb,t}$)
- illegal harvesting ($C_{illegal_harvest,t}$) .

Equation 4-1
$$C'_{actual,t} = E_{projplan,t} + E_{design,t} + E_{monitoring,t} + \left[(C_{natdisturb,t} + C_{illegal_harvest,t}) \times \frac{44}{12} \right]$$

Parameter	Description	Unit
$C'_{actual,t}$	Annual total carbon emissions associated with the project activity in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$E_{projplan,t}$	Annual emissions due to administration and project planning in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$E_{design,t}$	Annual emissions from travel for design and set up in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$E_{monitoring,t}$	Annual emissions due to monitoring for field work in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$C_{natdisturb,t}$	Annual carbon losses due to natural disturbance(s) in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC

Parameter	Description	Unit
$C_{illegal_harvest,t}$	Annual carbon losses due to illegal harvesting in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
44/12	The ratio of molecular weight of carbon dioxide to carbon, <i>see</i> Appendix C	tCO ₂ -e tC ⁻¹

The Project Proponent must use the guidelines established by The Greenhouse Gas Protocol (WBCSD and WRI, 2005) to determine the boundary for accounting of the activities of the above emission sources.

Omission of any of the above emissions sources must follow the significance guidelines outlined in Section 1.2.3.

Ex ante actual project activity emissions $C'_{actual,t}$ will be made using default emission factor data and projected activity data for planning, design and monitoring based on the project implementation requirements. The Project Proponent must ensure that the default values and activity data used are conservative.

4.1 Emissions Due to Project Planning

Project planning involves administrative activities and travel. Emissions associated with project planning are calculated as follows:

Equation 4-2
$$E_{projplan,t} = E_{admin,t} + E_{plan_travel,t}$$

Parameter	Description	Unit
$E_{projplan,t}$	Annual emissions due to administration and project planning in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$E_{admin,t}$	Annual emissions due to electricity consumption required for administration of the project activity in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$E_{plan_travel,t}$	Annual emissions due to travel for project planning in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

4.1.1 Emissions Due to Administration

Where administration and planning operations are conducted in the project's host country, emissions from these operations will be accounted for. Emissions from such activities including data processing are estimated using the following procedure:

Step 1: Determine annual electricity consumption from the following:

- (i) Electricity bills summated to determine the annual electricity consumption
- (ii) Derived from logged hours of operation for electrical equipment employed for administration and planning multiplied by the power rating of the equipment, as per the following equation:

Equation 4-3

$$Q_{admin,t} = \sum_{ee=1}^{EE} PR_{equip,ee,t} \times t_{op_equip,ee,t}$$

Parameter	Description	Unit
$Q_{admin,t}$	Annual electricity consumption due to administration of the project activity in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	kWh
$PR_{equip,ee,t}$	Power rating for electrical equipment, ee (where $ee=1,2,3 \dots EE$ pieces of equipment), in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	kW
$t_{op_equip,ee,t}$	Annual hours of operation per year of electrical equipment, ee (where $ee=1,2,3 \dots EE$ pieces of equipment) in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	h

If administration activities occur outside the host country, they are not considered in the project accounting.

Step 2: Select an emission factor for electricity for the project's host country from data sources such as IEA (2009), CO₂ emissions per kWh from electricity and heat generation, pp. 101-103.

Step 3: Determine the emissions due to electricity consumption required for administration and planning by the following equation:

Equation 4-4

$$E_{admin,t} = Q_{admin,t} \times EF_{electricity}$$

Parameter	Description	Unit
$E_{admin,t}$	Annual emissions due to electricity consumption required for administration of the project activity in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$Q_{admin,t}$	Annual electricity consumption due to administration of the project activity in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	kWh

Parameter	Description	Unit
$EF_{electricity}$	Electricity emission factor for the host country	tCO ₂ -e kWh ⁻¹

4.1.2 Emissions Due to Travel

Emissions as a result of travel depend on the mode of transport. For project planning this will typically and principally be either aircraft or ground transportation in the equation below.

Equation 4-5
$$E_{plan_travel,t} = E_{plan_flight,t} + E_{plan_ground,t}$$

Parameter	Description	Unit
$E_{plan_travel,t}$	Annual emissions due to travel for project planning in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$E_{plan_flight,t}$	Annual emissions due to flights in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$E_{plan_ground,t}$	Annual emissions due to ground transportation in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

Note: If transportation occurs outside the host country, it will not be considered in the project accounting.

4.1.2.1 Emissions due to flights

A flight log must be maintained throughout the project lifetime to provide the information required for the emission calculations. A flight log must have the following information:

- (i) Departure and destination location, or airport codes for arrival and departure
- (ii) Number of passengers per trip
- (iii) Type of aircraft
- (iv) Date of travel.

Emissions due to flights are calculated using the following procedure:

Step 1: From the travel log, determine in year t , for each trip y , the distance required for each trip and the number of passengers.

Step 2: Using distance and/or aircraft information, select an appropriate flight emission factor from the following data sources:

- (i) Published peer-reviewed studies for air transportation (e.g. Miyoshi and Mason, 2009; Babikian et al., 2002)
- (ii) Relevant studies from international industry experts (e.g. Ross, 2009; DEFRA, 2008).

Step 3: Calculate the total emissions due to flights for project planning using the emission factor (with units of tCO₂-e per passenger.km) by the following equation:

Equation 4-6

$$E_{plan_flight,t} = \sum_{y=1}^Y (KM_{plan_flight,y,t} \times N_{plan_flight,y,t} \times EF_{flight,y})$$

Parameter	Description	Unit
$E_{plan_flight,t}$	Annual emissions due to flights in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$N_{plan_flight,y,t}$	Annual number of passengers per trip y , (where $y=1,2,3 \dots Y$ trips) in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	passenger
$EF_{flight,y}$	Flight emission factor for trip, y (where $y=1,2,3 \dots Y$ trips)	tCO ₂ -e (passenger.km) ⁻¹
$KM_{plan_flight,y,t}$	Annual distance travelled per trip y , (where $y=1,2,3 \dots Y$ trips), in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	km

4.1.2.2 Emissions due to ground transport

A ground transport log must be maintained throughout the project lifetime to provide the information required for the emission calculations. A ground travel log must have the following information for each trip taken:

- (i) Departure and arrival points, or volume of fuel consumed
- (ii) Type of vehicle employed
- (iii) Type of fuel consumed
- (iv) Date of travel.

Emissions due to ground transport are calculated using the following procedure:

Step 1: If volume of fuel consumed per trip has been recorded in the travel log, go to Step 4. If the departure and arrival destinations have been provided, determine the distance travelled for each trip, y ($KM_{plan_ground,y,t}$).

Step 2: For the vehicle employed and fuel consumed, select the fuel efficiency of the vehicle from the following data sources:

- (i) Manufacturers' specifications for vehicle employed
- (ii) National database for vehicle specifications (e.g. Greenvehicle Guide, Australia, 2010)

Step 3: Calculate the volume of fuel consumed per trip by the following equation:

Equation 4-7

$$V_{fuel_plan_ground,y,t} = \frac{KM_{plan_ground,y,t}}{Eff_{vehicle}}$$

Parameter	Description	Unit
$V_{fuel_plan_ground,y,t}$	Annual volume of fuel consumed per trip y , (where $y=1,2,3 \dots Y$ trips), in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	kL
$Eff_{vehicle}$	Fuel efficiency for vehicle type	km kL ⁻¹
$KM_{plan_ground,y,t}$	Annual distance travelled per trip y , (where $y=1,2,3 \dots Y$ trips) in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	km

Step 4: From the type of fuel consumed, select the fuel emission factor(s) from the following sources:

- (i) National country-specific fuel emission factors
- (ii) Global fuel emission factors (e.g. IPCC (2006), Volume 2, Chapter 3, Table 3.2.1, p. 3.16 for CO₂ and Table 3.2.2, p. 3.21 for CH₄ and N₂O). Guidance for converting the emission factors to a CO₂ equivalent (CO₂-e) emission factor (EF_{fuel}) is provided in Appendix C.2. Guidance for unit conversions of fuel emission factors is provided in Appendix C.3.

Step 5: Calculate the total emissions due to ground transport for project planning by the following equations:

Equation 4-8

$$E_{plan_ground,t} = \sum_{y=1}^Y (V_{fuel_plan_ground,y,t} \times EF_{fuel})$$

Parameter	Description	Unit
$E_{plan_ground,t}$	Annual emissions due to ground transportation in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$V_{fuel_plan_ground,y,t}$	Annual volume of fuel consumed per trip y , (where $y=1,2,3 \dots Y$ trips), in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	kL
EF_{fuel}	Fuel emission factor	tCO ₂ -e kL ⁻¹

4.2 Emissions Due to Design

Project design involves travel to the host country for stakeholder consultation(s), thus emissions associated with project design are calculated from both ground and air travel emissions:

Equation 4-9
$$E_{design,t} = E_{design_ground,t} + E_{design_flight,t}$$

Parameter	Description	Unit
$E_{design,t}$	Annual emissions from travel for design and set up in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$E_{design_ground,t}$	Annual emissions due to ground travel for the design stage in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$E_{design_flight,t}$	Annual emissions due to air travel for the design stage in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

4.2.1 Emissions Due to Flights

Determine the distance, number of passengers and emission factor per trip for project design using Steps 1 to 3 in Section 4.1.2.1. Emissions from flights for project design and aerial surveillance for monitoring the Project Area can be determined as below:

Equation 4-10
$$E_{design_flight,t} = \sum_{y=1}^Y (KM_{design_flight,y,t} \times N_{design_flight,y,t} \times EF_{flight,y})$$

Parameter	Description	Unit
$E_{design_flight,t}$	Annual emissions due to air travel for the design stage in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$N_{design_flight,y,t}$	Annual number of passengers per trip y (where $y=1,2,3 \dots Y$ trips) in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	passenger
$EF_{flight,y}$	Flight emission factor for trip, y (where $y=1,2,3 \dots Y$ trips)	tCO ₂ -e (passenger.km) ⁻¹
$KM_{design_flight,y,t}$	Annual distance travelled per trip y (where $y=1,2,3 \dots Y$ trips) in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	km

4.2.2 Emissions Due to Ground Transport

Determine the distance, vehicle fuel efficiency and fuel emission factor per trip for project design using Steps 1 to 3 in Section 4.1.2.2. Emissions from ground transport for project design and ground patrolling for monitoring the Project Area can be determine as below:

Equation 4-11

$$E_{design_ground,t} = \sum_{y=1}^Y \left(\frac{KM_{design_ground,y,t}}{Eff_{vehicle}} \times EF_{fuel} \right)$$

Parameter	Description	Unit
$E_{design_ground,t}$	Annual emissions due to ground travel for the design stage in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$Eff_{vehicle}$	Fuel efficiency for vehicle type	km kL ⁻¹
EF_{fuel}	Fuel emission factor	tCO ₂ -e kL ⁻¹
$KM_{design_ground,y,t}$	Annual distance travelled per trip y , (where $y=1,2,3 \dots Y$ trips) in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	km

4.3 Emissions Due to Monitoring

Project monitoring requires travel to the Project Area for field work, thus emissions associated with project monitoring including the aerial surveillance and ground patrolling are deduced from travel emissions:

Equation 4-12

$$E_{monitoring,t} = KM_{monitoring_ground,t} + KM_{monitoring_flight,t}$$

Parameter	Description	Unit
$E_{monitoring,t}$	Annual emissions due to monitoring for field work in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$KM_{monitoring_ground,t}$	Annual emissions due to ground travel for monitoring team in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$KM_{monitoring_flight,t}$	Annual emissions due to air travel for monitoring team in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

4.3.1 Emissions Due to Flights

Determine the distance, number of passengers and emission factor per trip for monitoring using Steps 1 to 3 in Section 4.1.2.1. Emissions from flights including aerial surveillance for monitoring can be determined as below:

Equation 4-13

$$E_{monitoring_flight,t} = \sum_{y=1}^Y (KM_{monitoring_flight,y,t} \times N_{monitoring_flight,y,t} \times EF_{flight,y})$$

Parameter	Description	Unit
$E_{monitoring_flight,t}$	Annual emissions due to air travel for monitoring team in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$N_{monitoring_flight,y,t}$	Annual number of passengers per trip y , (where $y=1,2,3 \dots Y$ trips) in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	passenger
$EF_{flight,y}$	Flight emission factor for trip, y (where $y=1,2,3 \dots Y$ trips)	tCO ₂ -e (passenger.km) ⁻¹
$KM_{monitoring_flight,y,t}$	Annual distance travelled per trip y , (where $y=1,2,3 \dots Y$ trips) in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	km

4.3.2 Emissions Due to Ground Transport

Determine the distance, vehicle fuel efficiency and fuel emission factor per trip for monitoring using Steps 1 to 3 in Section 4.1.2.2. Emissions from ground transport including ground patrolling for monitoring can be determined as below:

Equation 4-14

$$E_{monitoring_ground,t} = \sum_{y=1}^Y \left(\frac{KM_{monitoring_ground,y,t}}{Eff_{vehicle}} \times EF_{fuel} \right)$$

Parameter	Description	Unit
$E_{monitoring_ground,t}$	Annual emissions due to ground travel for monitoring team in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$KM_{monitoring_ground,y,t}$	Annual distance travelled per trip y , (where $y=1,2,3 \dots Y$ trips) in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	km
EF_{fuel}	Fuel emission factor	tCO ₂ -e kL ⁻¹
$Eff_{vehicle}$	Fuel efficiency for vehicle type	km kL ⁻¹

4.4 Emissions Due to Natural Disturbances

Natural disturbances caused by fire (including human induced), windstorms, landslides or volcanoes can have impacts on both the extent of the forest area and the carbon. As a consequence, emissions from

such disturbance must be measured and factored out of the estimation of the ex post net anthropogenic GHG emission reductions. Events due to natural disturbances will be identified using the following procedures:

- Regular familiarisation with meteorological reports of the Project Area - the frequency of monitoring is dependent on the susceptibility of the region to natural disturbances
- Analysis of best available satellite imagery to monitor the Project Area, specifically to identify and estimate the areas that have been disturbed within the Project Area
- Aerial surveillance and/or ground patrolling followed by field checking to verify the extent (area) of the damage and quantification of the GHG emissions.

Once an event has been identified, emissions attributed to that naturally disturbed area must be monitored annually over the project crediting period and must be estimated using the following procedure:

Step 1: Identify the naturally disturbed areas, $A_{nd,t}$, using satellite data and locate them at the strata level. The satellite data analysis must follow a standard procedure for remote sensing, and must be verified by a field team (e.g. Fagan and DeFries, 2009).

Step 2: Calculate the area of natural disturbance in each stratum, $A_{nd,j,t}$, using satellite data and employ a field team to conduct on-ground verification of the damaged areas by measuring the area and extent of damage.

Step 3: Obtain the growing stock per hectare for the respective stratum, j , where the natural disturbance has occurred and also select appropriate BCEF at the stratum level (see Step 1 of Section 3.2.1.2.2) and carbon fraction (see Step 4 of Section 3.2.1.2.1,A) and then apply the following equation to obtain the annual carbon in the naturally disturbed area(s):

Equation 4-15
$$C_{AGB_nd,j,t} = \left[\bar{V}_{gstock,j,t=0} \times BCEF_j \times \left(\sum_{nd=1}^{ND} A_{nd,j,t} \right) \right] \times CF_{AGB}$$

Parameter	Description	Unit
$C_{AGB_nd,j,t}$	Annual carbon in the aboveground biomass of the growing stock in the naturally disturbed area in stratum j (where $j=1,2,3 \dots J$ strata) in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\bar{V}_{gstock,j,t=0}$	Average growing stock per hectare for stratum j (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	m ³ ha ⁻¹
$A_{nd,j,t}$	Annual area of natural disturbance nd (where $nd=1,2,3 \dots ND$ naturally disturbed areas) in stratum j (where $j=1,2,3 \dots J$ strata) in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	ha

Parameter	Description	Unit
$BCEF_j$	Biomass conversion and expansion factor for converting growing stock to carbon in the aboveground biomass for stratum j , (where $j=1,2,3 \dots J$ strata)	(t d.m.) m^{-3}
CF_{AGB}	Carbon fraction in the aboveground biomass of trees for the forest (see Appendix B)	tC (t d.m.) $^{-1}$

Step 4: Using direct observations of the area(s) damaged, the field team must provide an estimate of the fraction of the growing stock naturally disturbed based on a comparison with an adjacent representative undisturbed forest ($f_{natdisturb,j,t}$) within the stratum, then apply the following equation:

Equation 4-16

$$C_{AGB_nd,t} = \sum_{j=1}^J (C_{AGB_nd,j,t} \times f_{natdisturb,j,t})$$

Parameter	Description	Unit
$C_{AGB_nd,t}$	Annual carbon losses in the aboveground biomass of the growing stock due to natural disturbance(s) in the Project Area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{AGB_nd,j,t}$	Annual carbon in the aboveground biomass of the growing stock in the naturally disturbed area in stratum j , (where $j=1,2,3 \dots J$ strata) in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$f_{natdisturb,j,t}$	Fraction of the growing stock naturally damaged in stratum j (where $j = 1,2,3 \dots J$ strata) in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	dimensionless

Step 5: After a natural disturbance, regrowth is likely to occur in the naturally disturbed area and hence, acts as a carbon sink. Note that the rate of regrowth in the naturally disturbed area is not the same as the rate of regrowth in the gaps created after selective harvesting under the baseline scenario. The annual carbon increase due to the regrowth in the naturally disturbed area will be calculated via Equation 4-17a.

This Methodology has accounted for the carbon in the AGB of trees (see Table 2-3 in Section 2.2.3) which conservatively estimates the carbon due to regrowth. The annual average regrowth in the AGB in the naturally disturbed area is derived by direct measurement of all trees, i.e. trees with a minimum diameter at breast height as specified by the relevant authority in the

host country (see footnote ## under Table 2-3 in Section 2.2.3 of this Methodology), in the sample plots, s_{nd} , established in the naturally disturbed stratum during the monitoring event. The procedure for establishing the sample plots and the measurement techniques are described in Section 7.1.2. Number of sample plots within a stratum for regrowth measurement is determined in accordance with Pearson et al (2005) (see pp. 15-17).

The annual average regrowth post natural disturbance ($\bar{G}_{regrowth_nd,j,t}$) is calculated based on the AGB in the naturally disturbed area estimated using Step 3 of Section 3.2.1.3.2, as follow:

- (i) when the regrowth is measured for the first time in a monitoring event, the AGB at the stratum level obtained from the measurement is divided by the number of years of regrowth post natural disturbance
- (ii) in subsequent monitoring events, the difference in the AGB at the stratum level between the two consecutive monitoring events are divided by the monitoring period as presented in Equation 4-17b.

Equation 4-17a

$$C_{regrowth_nd,t} = CF_{AGB} \times \sum_{j=1}^J \left[\sum_{nd=1}^{ND} (A_{nd,j,t}) \times \bar{G}_{regrowth_nd,j,t} \right]$$

Equation 4-17b

$$\bar{G}_{regrowth_nd,j,t} = \frac{\bar{B}_{AGB_nd,j,m_2} - \bar{B}_{AGB_nd,j,m_1}}{\Delta m}$$

Parameter	Description	Unit
$C_{regrowth_nd,t}$	Annual carbon increase due to the regrowth in the naturally disturbed area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\bar{G}_{regrowth_nd,j,t}$	Average regrowth per hectare per year in the aboveground biomass after natural disturbance in stratum j , (where $j=1,2,3 \dots J$ strata) in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	(t d.m.) ha ⁻¹ yr ⁻¹
CF_{AGB}	Carbon fraction in the aboveground biomass of trees for the forest (see Appendix B)	tC (t d.m.) ⁻¹
$A_{nd,j,t}$	Annual area of natural disturbance nd , (where $nd=1,2,3 \dots ND$ naturally disturbed areas) in stratum j , (where $j=1,2,3 \dots J$ strata) in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity) ^{^^}	ha
$\bar{B}_{AGB_nd,j,m_2}$	Average aboveground biomass of the trees in the area of natural disturbance nd , (where $nd=1,2,3 \dots ND$ naturally disturbed areas) in stratum j , (where $j=1,2,3 \dots J$ strata) in year, m_2 - when current monitoring event occurs	(t d.m.) ha ⁻¹

Parameter	Description	Unit
$\bar{B}_{AGB_nd,j,m_1}$	Average aboveground biomass of the trees in the area of natural disturbance nd , (where $nd=1,2,3 \dots ND$ naturally disturbed areas) in stratum j , (where $j=1,2,3 \dots J$ strata) in year, m_1 - when preceding monitoring event occurred	(t d.m.) ha ⁻¹
Δm	Monitoring period, the time taken between two consecutive monitoring events	yr

Step 6: In the case of a natural disturbance such as fire, CH₄, N₂O and CO₂ gases are generated. The emissions of the non-CO₂ gases can be estimated based on the total carbon emitted, as suggested by the IPCC GPG for LULUCF (IPCC, 2003, Chapter 3, Equation 3.2.19, p. 3.49). Emission ratios for CH₄ and N₂O can be obtained from IPCC (2003) in Table 3A.1.15 and an appropriate ratio for nitrogen-to-carbon ($R_{N/C}$) for the material burnt must be used. If the specific $R_{N/C}$ cannot be found, a default value of 0.01 that applies to leaf litter can be used instead, noting that lower values would be appropriate for woody material (IPCC, 2003, Chapter 3, p. 3.50). The emissions of both CH₄ and N₂O are then estimated by applying the following equations:

Equation 4-18a
$$E_{CH_4,t} = C_{AGB_nd,t} \times R_{CH_4} \times \frac{16}{12}$$

Equation 4-18b
$$E_{N_2O,t} = C_{AGB_nd,t} \times R_{N/C} \times R_{N_2O} \times \frac{44}{28}$$

Parameter	Description	Unit
$E_{CH_4,t}$	Annual emissions due to CH ₄ in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCH ₄
$C_{AGB_nd,t}$	Annual carbon losses in the aboveground biomass of the growing stock due to natural disturbance(s) in the Project Area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
R_{CH_4}	Emission ratio for CH ₄	dimensionless
$E_{N_2O,t}$	Annual emissions due to N ₂ O in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tN ₂ O
$R_{N/C}$	Ratio of nitrogen to carbon	tN tC ⁻¹
R_{N_2O}	Emission ratio for N ₂ O	dimensionless

Parameter	Description	Unit
16 / 12	The ratio of molecular weight of CH ₄ to carbon	tCH ₄ tC ⁻¹
44 / 28	The ratio of molecular weight of N ₂ O to N	tN ₂ O tN ⁻¹

Step 7: Select the most current Global Warming Potentials (GWPs) for CH₄ and N₂O from UNFCCC documents and apply the GWPs in the following equation to aggregate the carbon emissions from the non-CO₂ gases. See also Section 2.2.4.1 Table 2.4 for more information. Then apply the following equation:

Equation 4-19
$$C_{CH_4N_2O,t} = \left(E_{CH_4,t} \times GWP_{CH_4} + E_{N_2O,t} \times GWP_{N_2O} \right) \times \frac{12}{44}$$

Parameter	Description	Unit
$C_{CH_4N_2O,t}$	Annual carbon from CH ₄ and N ₂ O emissions in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$E_{CH_4,t}$	Annual emissions due to CH ₄ in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCH ₄
GWP_{CH_4}	Global warming potential of CH ₄	tCO ₂ -e tCH ₄ ⁻¹
$E_{N_2O,t}$	Annual emissions due to N ₂ O in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tN ₂ O
GWP_{N_2O}	Global warming potential of N ₂ O	tCO ₂ -e tN ₂ O ⁻¹
12 / 44	The ratio of molecular weight of carbon to carbon carbon dioxide, see Appendix C	tC tCO ₂ -e ⁻¹

Step 8: Calculate the total carbon emissions due to natural disturbances by using the outputs from Equations 4-16, 4-17a and 4-19. If natural disturbance is caused by an event other than fire, non-CO₂ gases (CH₄ and N₂O) would not be produced. Hence the terms from Equations 4-17a and 4-19 would be zero. The regrowth in the naturally disturbed area is subtracted because it is considered as a carbon sink in this equation.

Equation 4-20
$$C_{natdisturb,t} = \sum_{t=1}^{t^*} \left(C_{AGB_nd,t} - C_{regrowth_nd,t} + C_{CH_4N_2O,t} \right)$$

Parameter	Description	Unit
$C_{natdisturb,t}$	Annual total carbon losses due to natural disturbance(s) in the Project Area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{AGB_nd,t}$	Annual carbon losses in the aboveground biomass of the growing stock due to natural disturbance(s) in the Project Area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{regrowth_nd,t}$	Annual carbon increase due to the regrowth in the naturally disturbed area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$C_{CH_4N_2O,t}$	Annual carbon from CH_4 and N_2O emissions in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC

Ex ante estimations for natural disturbances will be made based on the likely scenario in the Project Area. The Project Proponent must justify the likely scenario of natural disturbances using the historical data from satellite imagery or regional/local documentation from the relevant authority.

4.5 Emissions Due to Illegal Harvesting

Under the IFM-LtPF project scenario, the Project Area is completely protected and the Project Proponent must stop selective harvesting operation as well as harvesting by other agents of deforestation. All harvesting operations in the Project Area during the Project Crediting Period are considered as illegal harvesting. Ex ante estimations for illegal harvesting will be made based on the likely scenario in the Project Area. The Project Proponent must justify the likely scenario of illegal harvesting using the historical data from satellite imagery or regional/local documentation from the relevant authority. Ex post calculation for emissions due to illegal harvesting is described below for two different scenarios depending upon the method of data collected: the field inventory method or using satellite data must be used.

4.5.1 Field Inventory Method

Where field teams are able to collect data on the quantity of wood illegally harvested from the Project Area through observation or interviews based on Participatory Rural Appraisal techniques (e.g. see Surhone et al., 2010) carbon losses are quantified using the following procedure:

Step 1: Quantify the total volume of wood illegally harvested in the Project Area through field surveys ($V_{illegal_harvest,t}$) by measuring the stump diameter and developing species specific model to predict tree volumes based on the relevant peer reviewed literature (e.g. Corral-Rival et al., 2007)

Step 2: Choose the most applicable biomass expansion factor (BEF) value from the following data sources:

- (i) National factor (e.g. from National GHG Inventory)
- (ii) Default IPCC factor (see IPCC, 2003, Table 3A.1.10; and IPCC, 2006, Table 4.5)

It is noteworthy that the BEF values in IPCC literature and national GHG inventory are usually applicable to closed canopy forest. If applied to individual trees growing in an open field, the selected BEF value be increased by a further 30 percent (CDM-EB, 2009, p. 24).

Step 3: Choose the most applicable wood density for a forest with corresponding climate region and ecological zone (see Appendix B, Section B.1, Table B-1).

Step 4: Find the appropriate factor for residual stand damage (f_{RSD}), and carbon fraction (Step 4 of Section 3.2.1.2.1,A) and apply the following equation (which has been based on IPCC (2003), Equation 3.2.7, p. 3.27) to obtain the total carbon due to illegal harvesting in the Project Area:

Equation 4-21
$$C_{illegal_harvest,t} = V_{illegal_harvest,t} \times (1 + f_{RSD}) \times BEF \times D \times CF_{AGB}$$

Parameter	Description	Unit
$C_{illegal_harvest,t}$	Annual carbon losses due to illegal harvesting in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$V_{illegal_harvest,t}$	Annual volume of wood sold as determined from field surveys in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	m ³
BEF	Biomass expansion factor for converting volume of extracted roundwood to total aboveground biomass (including bark)	(t d.m.) m ⁻³
D	Wood density for the tropical forest with corresponding climate region and ecological zone (see Appendix B)	(t d.m.) m ⁻³
CF_{AGB}	Carbon fraction in the aboveground biomass of trees for the tropical forest (see Appendix B)	tC (t d.m.) ⁻¹
f_{RSD}	Factor for residual stand damage, based on the fraction of quantity of carbon damaged in the residual stand to the quantity of carbon in the total merchantable logs harvested	dimensionless

4.5.2 Using Satellite Data

Where high resolution satellite data are used, carbon losses are quantified by:

Step 1: Identify and calculate the illegally harvested areas ($A_{illegal_harvest,t}$) using satellite data and locate them at the strata level. The satellite data analysis must follow a standard procedure (e.g. Skole et al., 1998), and must be verified by a field team or teams.

Step 2: Verify the areas by sending a field team or teams.

Step 3: Obtain the growing stock per hectare for the respective stratum, j , where the illegal harvesting has occurred and also select appropriate BCEF (Step 1 of Section 3.2.1.2.2) and carbon fraction (Step 4 of Section 3.2.1.2.1,A) and then apply the following equation to obtain the total carbon per year due to illegal harvesting:

Equation 4-22

$$C_{illegal_harvest,t} = \sum_{j=1}^J \left[\bar{V}_{gstock,j,t=0} \times BCEF_j \times \left(\sum_{il=1}^{IL} A_{illegal_harvest,j,t} \right) \right] \times CF_{AGB}$$

Parameter	Description	Unit
$C_{illegal_harvest,t}$	Annual carbon losses due to illegal harvesting in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\bar{V}_{gstock,j,t=0}$	Average growing stock per hectare for stratum j , (where $j=1,2,3 \dots J$ strata) determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	m ³ ha ⁻¹
$A_{illegal_harvest,j,t}$	Annual area of illegal harvest il (where $il=1,2,3 \dots IL$ illegal harvested areas) in stratum j , (where $j=1,2,3 \dots J$ strata) in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	ha
$BCEF_j$	Biomass conversion and expansion factor for converting growing stock to the aboveground biomass for stratum j , (where $j=1,2,3 \dots J$ strata)	(t d.m.) m ⁻³
CF_{AGB}	Carbon fraction in the aboveground biomass of trees for the tropical forest (see Appendix B)	tC (t d.m.) ⁻¹

5 Leakage Assessment and Management

5.1 Identifying Leakage

The objective of leakage assessment and management is to provide an ex post estimate of the actual decrease in carbon and increase in GHG emissions (other than carbon changes) that arise as a result of the implementation of the project activity. Where an increase in GHG emissions occurs outside a project's boundary but is measurable and attributable to the project activity, this is known as project leakage and the associated emissions in tonnes of CO₂-e per year are termed $C'_{leakage,t}$.

For an IFM-LtPF project activity, there are two sources of leakage that need to be considered and addressed in this Methodology:

- (i) Carbon from degradation due to shifting of the baseline activity, $CL_{activityshifting,t}$ i.e. removal of harvested wood products including sawlog, pulplog and commercially harvested fuel wood and emissions from the associated activities outside the Project Area, $CL'_{emissions,t}$
- (ii) Carbon from market leakage $CL_{market,t}$, due to shifts in supply and demand of the products and services affected by the project activity, which in this case is the supply and demand of timber.

On this basis, project leakage is the combined total of the above leakage parameters:

Equation 5-1
$$C'_{leakage,t} = (CL_{activityshifting,t} + CL_{market,t}) \times \frac{44}{12} + CL'_{emissions,t}$$

Parameter	Description	Unit
$C'_{leakage,t}$	Annual total carbon emissions associated with leakage in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$CL_{activityshifting,t}$	Annual carbon losses due to baseline activity shifting in other lands managed or operated by the Project Proponent in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$CL_{market,t}$	Annual carbon due to market leakage effects in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$CL'_{emissions,t}$	Annual emissions due to implementation of the shifted baseline activity in other lands managed or operated by the Project Proponent in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
44/12	The ratio of molecular weight of carbon dioxide to carbon, see Appendix C	tCO ₂ -e tC ⁻¹

The emissions determined in $C'_{leakage,t}$ are subsequently accounted for in the net anthropogenic GHG emissions for the IFM-LtPF project (Equation 1-1).

5.2 Leakage Due to Activity Shifting

Any activity shifting due to selective logging which includes removal of harvested wood projects comprising sawlog, pulplog and commercially harvested fuel wood by the Project Proponent must be assessed as a component of leakage.

Activity shifting leakage situations become evident where the Project Proponent has:

- (i) Intensified operations for selective logging, i.e. has legal authorisation for selective logging and increase logging operations in other owned and/or operated lands to recover the harvesting loss due to the IFM-LtPF project; or
- (ii) Shifted operations for selective logging from the Project Area to another forest area within the host country.

The Project Proponent must provide documentation for the potential leakage areas due to activity shifting i.e. other lands owned and/or operated by the Project Proponent, including geo-referenced or digital maps illustrating the physical location(s) and their boundaries, existing land uses and management plans at each verification period. When this documentation has been provided, the carbon losses from activity shifting, $CL_{activityshifting,t}$, can be assessed as described in the following sections.

Equation 5-2
$$CL_{activityshifting,t} = CL_{IH_activityshifting,t} + CL_{SH_activityshifting,t}$$

Parameter	Description	Unit
$CL_{activityshifting,t}$	Annual carbon losses due to baseline activity shifting in other lands managed or operated by the Project Proponent in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$CL_{IH_activityshifting,t}$	Annual carbon losses from activity shifting due to intensification of harvest volumes in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$CL_{SH_activityshifting,t}$	Annual carbon losses from activity shifting due to shifting of harvest to new area in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC

5.2.1 Intensification of Logging Operations

Activity shifting leakage through intensification of logging operations including removal of sawlog, pulplog and commercially harvested fuel wood is considered where the actual merchantable logs' volume in lands currently owned and/or operated by the Project Proponent (but not part of the Project Area) since the commencement date of the IFM-LtPF project activity, exceeds the common practice volumes. To identify and determine the carbon due to the intensification of logging operations, use the following procedure:

Step 1: Assign a historical reference period of five years (Refer to Section 2.2.2.3) where the harvesting operations have occurred over five years. In the case where harvesting operations have been conducted for less than five years, the harvest volume data from the forest with comparable situations and conditions at the local or regional or national level (with conservative value) shall be used. The Project Proponent shall provide the data source and collection method for the common harvest volume from the forest comparable situations and conditions to the Verifier along with the monitoring report for verification.

Step 2: Obtain the annual volume of harvest for each land l , currently owned and/or operated by the Project Proponent or the forest land with comparable situations and conditions at the local or regional or national level (with conservative value), for each historical reference year k , over the historical reference period K .

Step 3: Determine the average volume of harvest for each land l , currently owned and/or operated by the Project Proponent using the following equation:

Equation 5-3

$$V_{\text{historical_harvest},l,t=0} = \frac{\sum_{k=1}^K V'_{\text{historical_harvest},l,k}}{K}$$

Parameter	Description	Unit
$V_{\text{historical_harvest},l,t=0}$	Average volume of harvest for land l , (where $l=1,2,3 \dots L$ lands), that is owned and/or operated by the Project Proponent or the forest with comparable situations and conditions at local or regional or national level over the historical reference period K , determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	$\text{m}^3 \text{ yr}^{-1}$
$V'_{\text{historical_harvest},l,k}$	Total volume of harvest for land l , (where $l=1,2,3 \dots L$ lands), that is owned and/or operated by the Project Proponent during the historical reference year k , (where $k=1,2,3 \dots K$ historical reference years)	m^3
K	Historical reference period (see Section 2.2.2.3) or actual number of years of harvesting operation	yr

Step 4: Compare the annual actual volume of harvest in year t (years elapsed since the start of the IFM-LtPF project activity) with the average volume of harvest over the historical reference period determined ex ante, for each land l .

One of the following three situations would occur:

Equation 5-4 $V_{\text{historical_harvest},l,t=0} > V_{\text{actual_harvest},l,t} \Rightarrow \text{No Leakage}$

Equation 5-5 $V_{\text{historical_harvest},l,t=0} = V_{\text{actual_harvest},l,t} \Rightarrow \text{No Leakage}$

Equation 5-6 $V_{\text{historical_harvest},l,t=0} < V_{\text{actual_harvest},l,t} \Rightarrow \text{Leakage}$

Parameter	Description	Unit
$V_{historical_harvest,l,t=0}$	Average volume of harvest for land l , (where $l=1,2,3 \dots L$ lands), that is owned and/or operated by the Project Proponent over the historical reference period K , determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	$m^3 \text{ yr}^{-1}$
$V_{actual_harvest,l,t}$	Annual actual volume of harvest for land l , (where $l=1,2,3 \dots L$ lands), that is owned and/or operated by the Project Proponent in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	$m^3 \text{ yr}^{-1}$

Step 5: If there is leakage (i.e. Equation 5-6 is true), the total leakage in terms of the merchantable logs' volume is calculated as follows:

Equation 5-7

$$V_{IH_activityshifting,t} = \sum_{l=1}^L (V_{actual_harvest,l,t} - V_{historical_harvest,l,t=0})$$

Parameter	Description	Unit
$V_{IH_activityshifting,t}$	Annual total intensification of harvest volume per year in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	m^3
$V_{actual_harvest,l,t}$	Annual actual volume of harvest for land l , (where $l=1,2,3 \dots L$ lands), that is owned and/or operated by the Project Proponent in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	$m^3 \text{ yr}^{-1}$
$V_{historical_harvest,l,t=0}$	Average volume of harvest for land l , (where $l=1,2,3 \dots L$ lands), that is owned and/or operated by the Project Proponent over the historical reference period K , determined ex ante - before the start of the IFM-LtPF project activity, hence $t=0$ year	$m^3 \text{ yr}^{-1}$

Step 6: Choose the most applicable BEF value from the data sources presented in Step 2 of Section 4.5.1

Step 7: Choose the most applicable wood density for a forest with corresponding climate region and ecological zone (see Appendix B, Section B.1, Table B-1).

Step 8: Select an appropriate factor for residual stand damage (f_{RSD}) (use Table B-3, section B.3, Appendix B) and carbon fraction (see Step 4 of Section 3.2.1.2.1,A) and apply the following equation to get the carbon loss due to intensification of harvest volumes:

Equation 5-8

$$CL_{IH_activityshifting,t} = V_{IH_activityshifting,t} \times (1 + f_{RSD}) \times BEF \times D \times CF_{AGB}$$

Parameter	Description	Unit
$CL_{IH_activityshifting,t}$	Annual carbon losses from activity shifting due to intensification of harvest volumes in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$V_{IH_activityshifting,t}$	Annual total intensification of harvest volume per year in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	m^3
f_{RSD}	Factor for residual stand damage, based on the fraction of quantity of carbon damaged in the residual stand to the quantity of carbon in the total merchantable logs harvested	dimensionless
BEF	Biomass expansion factor for converting volume of extracted roundwood to total aboveground biomass (including bark)	dimensionless

Parameter	Description	Unit
D	Wood density for the tropical forest with corresponding climate region and ecological zone (see Appendix B)	(t d.m.) m ⁻³
CF_{AGB}	Carbon fraction in the aboveground biomass of trees for the tropical forest (see Appendix B)	tC (t d.m.) ⁻¹

5.2.2 Shifted Logging Operations

Activity shifting leakage due to shifted harvesting to a new forest area can be confirmed where documentation of the management plans and/or land-use designations reveal that the Project Proponent has commenced harvesting operation in any new land logged during the crediting period after the start date of the IFM-LtPF project. When this is the case, the carbon losses due to harvest shifting can be determined using the following procedure:

Step 1: Obtain the volume of harvest in the new area from management plans ($V_{SH_activityshifting,t}$)

Step 2: Choose the most applicable BEF value from the data sources presented in Step 2 of Section 4.5.1

Step 3: Choose the most applicable wood density for a forest with corresponding climate region and ecological zone (see Appendix B, Section B.1, Table B-1)

Step 4: Select an appropriate factor for residual stand damage (f_{RSD}) (use Table B-3, Section B.3, Appendix B) and carbon fraction (see Step 4 of Section 3.2.1.2.1,A) and apply the following equation to get the carbon loss due to shifted logging operations:

Equation 5-9
$$CL_{SH_activityshifting,t} = V_{SH_activityshifting,t} \times (1 + f_{RSD}) \times BEF \times D \times CF_{AGB}$$

Parameter	Description	Unit
$CL_{SH_activityshifting,t}$	Annual carbon losses from activity shifting due to shifting of harvest to new area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$V_{SH_activityshifting,t}$	Annual shifted harvest volume for new area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	m ³
f_{RSD}	Factor for residual stand damage, based on the fraction of quantity of carbon damaged in the residual stand to the quantity of carbon in the total merchantable logs harvested	dimensionless
BEF	Biomass expansion factor for converting volume of extracted roundwood to total aboveground biomass (including bark)	dimensionless
D	Wood density for the tropical forest with corresponding climate region and ecological zone (see Appendix B)	(t d.m.) m ⁻³
CF_{AGB}	Carbon fraction in the aboveground biomass of trees for the tropical forest (see Appendix B)	tC (t d.m.) ⁻¹

5.2.3 Shifted Baseline Activity Emissions

Following Section 5.2.1.1, if Equation 5-6 is true, implying that harvesting has intensified, and/or shifted operations for logging have been confirmed in Section 5.2.1.2, it implies leakage due to activity shifting of the baseline scenario has occurred. Consequently, emissions associated with implementation of the shifted baseline activity must be accounted for and determined using the same approach as presented in Section 3.4 for determining baseline activity emissions. In this case, the parameters E are replaced with L in Equation 3-54 to distinguish between the baseline and leakage scenarios:

Equation 5-10
$$CL'_{emissions,t} = L_{harvesting,t} + L_{onsiteprep,t} + L_{hauling,t} + L_{transport,t} + L_{processing,t} + L_{distribution,t}$$

Parameter	Description	Unit
$CL'_{emissions,t}$	Annual emissions due to implementation of the shifted baseline activity in other lands managed or operated by the Project Proponent in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$L_{harvesting,t}$	Annual shifted emissions due to harvesting operations such as felling and snigging in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$L_{onsiteprep,t}$	Annual shifted emissions due to on-site preparation such as trimming of tree heads, butts, branches and defective components in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$L_{transport,t}$	Annual shifted emissions due to log transport from collection depot to processing plant in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$L_{hauling,t}$	Annual shifted emissions due to log hauling in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$L_{processing,t}$	Annual shifted emissions due to electricity consumption in sawmill in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$L_{distribution,t}$	Annual shifted emissions due to transport of the sawn product from the mill to the wharf for export or to the depot for local usage in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

5.3 Market Leakage

Market effects due to the presence of an IFM-LtPF project could occur in two main ways:

- (i) Intensification of existing harvest practices
- (ii) Formation of new enterprises and hence new (or modified existing) FIRs for sanctioned selective logging.

The Project Proponent must demonstrate how market leakage has been accounted for in accordance with the most recent version of applicable VCS rules.

6 Uncertainty Assessment

The assessment of uncertainty is a required component of quality management procedures for the management of data and information (see Section 7.3). It is considered necessary that the Project Proponent identify the parameters that significantly affect the accuracy of the calculated emission reductions in order to reduce uncertainties related to the quantification of GHG emission reductions or removal enhancements. A procedure for the estimation of uncertainty of the GHG emission reductions arising from an IFM-LtPF project is presented in this section.

6.1 Identifying Uncertainty

In general, estimations of emission reductions that are a result of emissions and removals from IFM activities have various sources of uncertainties. Typically these uncertainties are associated with sample data, such as height and diameter measurements from PSPs, biomass growth and rates of decay, activity data, emission factors, and other coefficients (e.g. lumber recovery factor).

The values of the uncertainties associated with these parameters are derived from a number of different sources:

- (i) IPCC (2006) default data and guidelines
- (ii) Statistical sampling
- (iii) Expert judgement with justification.

Where the uncertainty of a parameter is not known, the value of the parameter will be assigned a value that is indisputably conservative, eliminating the corresponding uncertainty.

6.1.1 Calculating Uncertainty

In accordance with IPCC best practice, the methodology requires the use of 95% confidence interval for quantification of random errors (IPCC, 2006, Volume 1, Chapter 3, p. 3.6 and 3.9).

A simple approach to determine the uncertainty value of the overall emission reductions from the uncertainties of the individual components, is based on the basic error propagation rule set outlined below:

Rule 1

The uncertainty or absolute error for a function, $z = x + y$, where σ_x and σ_y represent the absolute error for x and y , respectively, is determined by the following:

Equation 6-1

$$\sigma_z = \sqrt{\sigma_x^2 + \sigma_y^2}$$

Rule 2

The uncertainty or absolute error for a function, $fn = x \times y \times z$, where σ_x , σ_y and σ_z represent the absolute error for x , y and z , respectively, is determined by the following:

Equation 6-2

$$\sigma_{fn} = \sqrt{\left[\frac{\partial f}{\partial x} \sigma_x\right]^2 + \left[\frac{\partial f}{\partial y} \sigma_y\right]^2 + \left[\frac{\partial f}{\partial z} \sigma_z\right]^2}$$

The relative error, or percentage error (U) is determined, for function, fn , as an example, via the following:

Equation 6-3

$$U_{fn} = \frac{\sigma_{fn}}{fn} \times 100$$

Project Proponents seeking more detailed guidance for combining uncertainties will find it provided in IPCC (2006), Volume 1, Chapter 3, Section 3.2.3, p. 3.27. The above method has been presented for the parameters specific to the IFM-LtPF project in the subsequent sections.

6.2 Overall IFM-LtPF Project Uncertainty

The net anthropogenic GHG emission reductions of the IFM-LtPF project activity, is deduced from the subtraction of the actual project emissions ($C'_{actual,t}$) and the leakage emissions ($C'_{leakage,t}$) from the proposed baseline emissions ($C'_{baseline,t}$), as presented in Equation 1-1 in Section 1.2.1.

An estimation of the overall uncertainty for the IFM-LtPF project is deduced by error propagation using the following equation:

Equation 6-4

$$\sigma_{IFM-LtPF,t} = \sqrt{(\sigma_{baseline,t})^2 + (\sigma_{actual,t})^2 + (\sigma_{leakage,t})^2}$$

Parameter	Description	Unit
$\sigma_{IFM-LtPF,t}$	Annual uncertainty (absolute error) for the overall IFM-LtPF project in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$\sigma_{baseline,t}$	Annual uncertainty in the annual total GHG emissions as a result of the baseline scenario in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$\sigma_{actual,t}$	Annual uncertainty in the actual (project) activity in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$\sigma_{leakage,t}$	Annual uncertainty in leakage in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

To track the changes in uncertainty associated with the IFM-LtPF activity, it is appropriate to use the relative uncertainty instead of the absolute. The relative error, or percentage error is expressed using the following equation:

Equation 6-5

$$U_{IFM-LtPF,t} = \frac{\sigma_{IFM-LtPF,t}}{C'_{IFM-LtPF,t}} \times 100$$

Parameter	Description	Unit
$U_{IFM-LtPF,t}$	Annual uncertainty (relative error) for the overall IFM-LtPF project in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	%
$\sigma_{IFM-LtPF,t}$	Annual uncertainty (absolute error) for the overall IFM-LtPF project in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$C'_{IFM-LtPF,t}$	Annual net anthropogenic GHG emission reductions in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

6.3 Uncertainty in Baseline Accounting

The total uncertainty related to the baseline scenario (see Equation 3-1) is a propagation of error associated with the uncertainty of the degradation plus the uncertainty in the estimation of emission sources due to the implementation of the baseline case, and is expressed using the following equation:

Equation 6-6

$$\sigma_{baseline,t} = \sqrt{(\sigma_{degradation,t})^2 + (\sigma_{emissions,t})^2}$$

Parameter	Description	Unit
$\sigma_{baseline,t}$	Annual uncertainty in the annual total GHG emissions as a result of the baseline scenario in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$\sigma_{degradation,t}$	Annual uncertainty in the annual emissions produced from degradation due to the baseline activity: selective logging in the Project Area in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$\sigma_{emissions,t}$	Annual uncertainty in the annual emissions from the baseline activity: selective logging operations in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

6.3.1 Degradation Uncertainty

The uncertainty associated with degradation in the baseline scenario (see Equation 3-2) is a propagation of errors of the individual components contributing to degradation and is expressed using the following equation:

Equation 6-7

$$\sigma_{degradation,t} = \left[\left(\sqrt{(\sigma_{DW_{decay},t})^2 + (\sigma_{ltHWP_{oxidation},t})^2 + (\sigma_{stHWP_{oxidation},t})^2 + (\sigma_{growth_foregone,t})^2 + (\sigma_{regrowth,t})^2} \right) \times \frac{44}{12} \right]$$

Parameter	Description	Unit
$\sigma_{degradation,t}$	Annual uncertainty in the annual emissions produced from degradation due to the baseline activity: selective logging in the Project Area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$\sigma_{DW_{decay},t}$	Annual uncertainty in carbon from the deadwood (DW) pool due to decay in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\sigma_{ltHWP_{oxidation},t}$	Annual uncertainty in carbon from the long-term harvested wood products (ltHWP) pool due to oxidation in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\sigma_{stHWP_{oxidation},t}$	Annual uncertainty in carbon from the short-term harvested wood products (stHWP) pool due to oxidation in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\sigma_{growth_foregone,t}$	Annual uncertainty in the carbon lost due to growth foregone in the annual net harvest area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\sigma_{regrowth,t}$	Annual uncertainty in the carbon increase in the biomass due to regrowth following logging in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
44/12	The ratio of molecular weight of carbon dioxide to carbon, <i>see</i> Appendix C	tCO ₂ -e tC ⁻¹

6.3.2 Baseline Emission Sources Uncertainty

The uncertainty due to emissions from the implementation of the baseline case is determined as follows:

Equation 6-8

$$\sigma_{emissions,t} = \sqrt{\sum_{g=1}^G \sigma_{baseline_source,g,t}^2}$$

Parameter	Description	Unit
$\sigma_{emissions,t}$	Annual uncertainty in the annual emissions from the baseline activity: selective logging operations in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$\sigma_{baseline_source,g,t}$	Annual uncertainty in the greenhouse gas emissions sources g , (where $g=1,2,3 \dots G$ emission sources) associated with the baseline activity in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

6.4 Uncertainty in Actual Project Accounting

The uncertainty associated with the emissions due to the implementation of the actual project activity can be tracked through error propagation related to the individual project activity emission sources as outlined by the following:

Equation 6-9

$$\sigma_{actual,t} = \sqrt{\sum_{g=1}^G \sigma_{proj_source,g,t}^2}$$

Parameter	Description	Unit
$\sigma_{actual,t}$	Annual uncertainty in the actual project activity in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$\sigma_{proj_source,g,t}$	Annual uncertainty in the greenhouse gas emissions sources g , (where $g=1,2,3 \dots G$ emission sources) associated with the actual project activity in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

6.5 Uncertainty in Leakage Accounting

The uncertainty associated with leakage takes into account uncertainty in the estimations of emissions due to activity shifting ($CL_{activityshifting,t}$), illegal harvesting ($CL_{illegal_harvest,t}$), natural disturbances ($CL_{natural_disturb,t}$) implementation of the shifted baseline activity ($CL'_{emissions,t}$), and market effects ($CL_{market,t}$).

Equation 6-10

$$\sigma_{leakage,t} = \left(\sqrt{(\sigma_{activityshifting,t})^2 + (\sigma_{market,t})^2 + (\sigma_{illegal_harvest,t})^2 + (\sigma_{natural_disturb,t})^2 + \left(\sigma_{CL'_{emissions,t}} \times \frac{12}{44} \right)^2} \right) \times \frac{44}{12}$$

Parameter	Description	Unit
$\sigma_{leakage,t}$	Annual uncertainty in leakage in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$\sigma_{activityshifting,t}$	Annual uncertainty associated with carbon due to activity shifting in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\sigma_{market,t}$	Annual uncertainty associated with carbon due to market effects in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\sigma_{illegal_harvest,t}$	Annual uncertainty associated with carbon due to illegal harvesting in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\sigma_{natural_disturb,t}$	Annual uncertainty associated with carbon due to natural disturbances in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$\sigma_{CL'_{emissions,t}}$	Annual uncertainty associated with emissions due to implementation of the shifted baseline activity in other lands managed or operated by the Project Proponent in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity); this parameter has been multiplied by 12/44 for conversion into tC	tCO ₂ -e
44/12	The ratio of molecular weight of carbon dioxide to carbon, <i>see</i> Appendix C	tCO ₂ -e tC ⁻¹

6.6 Uncertainty Deduction

Upon establishing the uncertainty percentage using Equation 6-5, apply the following procedure:

Step 1: If $U_{IFM-LtPF,t} \leq 10\%$, then no deduction will be applied for uncertainty. The Project Proponent shall proceed to Section 6.7 to further reduce uncertainty associated with parameters.

Step 2: If $U_{IFM-LtPF,t} > 10\%$, then apply the following equation:

Equation 6-11

$$CC_{IFM-LtPF,t} = \frac{100 - (U_{IFM-LtPF,t} - 10)}{100} \times C'_{IFM-LtPF,t}$$

Parameter	Description	Unit
$CC_{IFM-LtPF,t}$	Annual carbon credits post uncertainty deduction in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$U_{IFM-LtPF,t}$	Annual uncertainty (relative error) for the overall IFM-LtPF project in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	%
$C'_{IFM-LtPF,t}$	Annual net anthropogenic GHG emission reductions in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

After applying Equation 6-11, the Project Proponent proceeds to Section 6.7 to further reduce uncertainty associated with parameters, so as to maximise the estimates of carbon credits.

6.7 Reducing Uncertainty

Reducing the uncertainty associated with the net anthropogenic GHG emission reductions of the IFM-LtPF project activity can be achieved by identifying the parameters contributing the greatest to the uncertainty and conducting appropriate procedures to reduce the uncertainty of these parameters. This will be achieved using the following procedure:

Step 1: For each parameter used to calculate the net anthropogenic GHG emission reductions, e.g.

$C'_{baseline,t}$, $C'_{actual,t}$, and $C'_{leakage,t}$, derive the ratio of the uncertainty of each contributing parameter over the main parameter. For example, to determine the ratio of the uncertainty of the individual components of the baseline scenario: degradation and baseline activity emissions, the following ratios must be derived:

Equation 6-12

$$R_{degradation,t} = \frac{\sigma_{degradation,t}}{\sigma_{baseline,t}}$$

Equation 6-13

$$R_{emissions,t} = \frac{\sigma_{emissions,t}}{\sigma_{baseline,t}}$$

Parameter	Description	Unit
$R_{degradation,t}$	Annual ratio of uncertainty of emissions due to degradation over uncertainty of the annual total GHG emissions of the baseline scenario in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	dimensionless
$\sigma_{degradation,t}$	Annual uncertainty in the annual emissions produced from degradation due to the baseline activity: selective logging in the Project Area in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

Parameter	Description	Unit
$\sigma_{baseline,t}$	Annual uncertainty in the annual total GHG emissions as a result of the baseline scenario in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e
$R_{emissions,t}$	Annual ratio of uncertainty of emissions due to implementation of baseline activity over uncertainty of the annual total GHG emissions of the baseline scenario in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	dimensionless
$\sigma_{emissions,t}$	Annual uncertainty in the annual emissions from the baseline activity: selective logging operations in year t (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tCO ₂ -e

Step 2: Identify the parameters with the highest ratios and using the same procedure as in Step 1, investigate the parameters contributing to the identified parameter. Continue this process for all calculated/derived parameters until only sourced/measured parameters remain.

Step 3: For the sourced parameters identified as the greatest contributors to uncertainty from Step 2, analyse the source of the parameter value and the associated uncertainty.

If the parameter is either a default IPCC value or sourced from literature:

- (i) Can the accuracy of the parameter and hence the range of the uncertainty be reduced by selecting another appropriate data source? If yes, provide justification and new data source
- (ii) If no, can a project-specific parameter be measured? If yes, provide a description of the procedure and incorporate into the Monitoring Plan
- (iii) If no, summarise justification on why the uncertainty of the parameter cannot be improved upon.

If the parameter is a project-specific measured value:

- (i) Can the accuracy of the parameter and hence the range of the uncertainty be reduced by increasing the statistical sampling? If yes, provide statistical requirements and incorporate into Monitoring Plan
- (ii) If no, summarise justification on why the uncertainty of the parameter cannot be improved upon.

Step 4: Once uncertainty revisions have been conducted, re-calculate the net anthropogenic emission reductions, $C'_{IFM-LtPF}$, and follow Sections 6.2 to 6.5 to calculate the revised uncertainty for the net anthropogenic GHG emission reductions of the IFM-LtPF project activity using Equations 6-4 and 6-5. Compare revised percentage with the previous.

Step 5: Follow Steps 1 and 2 in Section 6.6 and where required, account for uncertainty deduction using Equation 6-11.

7 Monitoring

7.1 Monitoring Methodology

This section provides the methodology for monitoring the parameters employed to calculate carbon changes due to forest degradation, as well as emissions due to implementation of the project and baseline activities, plus emissions as a result of leakage.

In addition, the procedures for establishing Permanent Sample Plots (PSPs) and data collection presented herein also apply to the Measured Data pathway, i.e., the case where the preparation of FIR or equivalent document, is in progress (see Section 3.2.2).

After monitoring has been implemented, the results are consequently applied to revise the net anthropogenic GHG emission reductions for the subsequent reporting period.

7.1.1 Monitoring Plan

The Project Proponent is required to submit a detailed monitoring plan in accordance with the most recent version of applicable VCS rules.

Standard Operating Procedures (SOPs), training, and capacity building will be provided by the Project Proponent to the identified responsible entity as specified in the PD. This is to ensure that accurate data is collected as required. The data, its source, the entity responsible, the frequency of collection and the equation it applies to, is detailed in the monitoring tables provided in the following sections (Tables 7-1 to 7-3).

7.1.2 Procedure for Establishing the Permanent Sample Plots (PSPs) and Measurement

Forest inventory obtained in the PSPs that are established in the Project Area is required for:

- (i) The Measured Data pathway, ex ante estimation for the baseline degradation calculations (see Section 3.2.2)
- (ii) Monitoring of the carbon change throughout the crediting period to obtain ex post estimations of growth foregone (see Section 3.3.4) and emissions due to natural disturbances (see Section 4.4).

Sample plots are either permanent or temporary in nature. For measuring the carbon change under the IFM-LtPF project, Permanent Sample Plots (PSPs) are more suitable than Temporary Sample Plots, as they are statistically more accurate and efficient in estimating changes in forest carbon. The measurements collected within the same plots under the PSPs approach throughout the crediting period allows the monitoring of the growth of individual trees, survivors, mortality, and growth of new trees at specific time intervals. In addition, the PSPs also permit the verification body to find, measure at random, and to verify in quantitative terms, the design and implementation of the carbon monitoring plan. The following sections describe the procedures for establishing PSPs in the Project Area.

7.1.2.1 Stratification of the Project Area

To facilitate the field work and increase the accuracy and precision of the parameters that are to be measured, the Project Area is to be divided into sub-populations or “strata” that form relatively homogeneous units. This makes monitoring more cost effective because it decreases the sampling and monitoring efforts, whilst maintaining the same level of confidence. Stratification of the Project Area is presented in more detail Section 2.2.1.1.1.

7.1.2.2 Shape and size of plots

The measurement of tree parameters in a PSP requires the determination of the size and shape of the PSPs. Circular or rectangular plots with varying sizes can be used for measuring different diameter class trees. Nesting of plots is possible - a smaller size plot is established inside a larger plot for measuring small diameter class trees. Figure 7-1 shows an example of nested circular sample plots for measuring different diameter classes in a specified circular plot. The sample plot has three circular plots: a small plot with 4 m radius to measure trees with DBH between 5-20 cm, an intermediate plot with 14 m radius to measure trees with DBH between 20-50 cm, and a large plot with 20 m radius for measuring trees with DBH above 50 cm.

The Project Proponent must decide on the appropriate size and shape of the PSP based on the common practices employed for forest inventory in the host country.

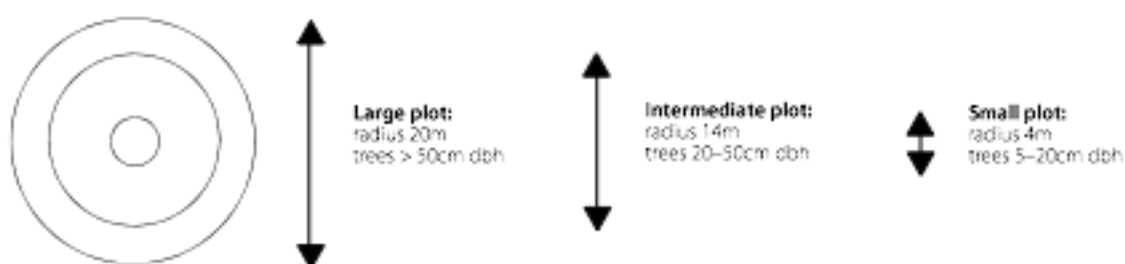


Figure 7-1 Circular three-nest sample plot sizes (Pearson et al. 2005)

7.1.2.3 Determining the number of PSPs

The number of PSPs in each stratum is determined by taking measurements in preliminary sample plots randomly laid in each stratum to achieve the desired level of precision using the following procedure:

- Step 1: Select the number of preliminary sample plots (sp) to be established (between 6 and 10 for each stratum).
- Step 2: Set the geographic location of the preliminary sample plots in each stratum by employing, for example, a random function available in the Geographic Information System (GIS) platform (e.g. ArcGIS).
- Step 3: Locate the preliminary sample plot on the ground using GPS and maps.
- Step 4: Establish the sample plot of specific size and shape as discussed in the Section 7.1.2.2.
- Step 5: Measure the DBH and tree height of all trees as defined by the relevant authority of the host country. Use the standard forestry techniques and equipment such as DBH tape and clinometer for measuring DBH and tree height, respectively. See Pearson et al. (2005) for details.

Step 6: Use field data to calculate the carbon in the AGB in each preliminary sample plot and stratum as per the calculation in Section 3.2.1.2, and employ the following equation to estimate the standard deviation for each stratum (sd_j):

Equation 7-1

$$sd_j = \sqrt{\frac{\sum_{sp=1}^{SP} (\bar{C}_{AGB_gstock,sp,j} - \bar{C}_{AGB_gstock,j})^2}{N_j - 1}}$$

Parameter	Description	Unit
sd_j	Standard deviation for carbon in the aboveground biomass in the stratum j (where $j=1,2,3, \dots J$ strata)	tC ha ⁻¹
$\bar{C}_{AGB_gstock,sp,j}$	Average carbon per hectare in the aboveground biomass of the growing stock in the preliminary sample plot sp (where $sp=1,2,3 \dots SP$ preliminary sample plots), of the stratum j (where $j=1,2,3, \dots J$ strata)	tC ha ⁻¹
$\bar{C}_{AGB_gstock,j}$	Average carbon per hectare in the aboveground biomass of the growing stock in the stratum j (where $j=1,2,3, \dots J$ strata)	tC ha ⁻¹
N_j	Total number of sampling units in the stratum j (where $j=1,2,3, \dots J$ strata)	dimensionless

Step 7: Apply the precision level of 10% of the true value of the mean at the 95% confidence interval for accurate estimation of net change in the carbon stock (Pearson et al., 2005) in the project area.

Step 8: Apply the equation below to estimate the total number of PSPs, N , in all strata for the entire Project Area (Pearson et al., 2005):

Equation 7-2

$$N = \frac{\left(\sum_{j=1}^J N_j \times sd_j \right)^2}{\left(\frac{N_j^2 \times err^2}{t_{stat}^2} \right) + \left(\sum_{j=1}^J N_j \times sd_j^2 \right)}$$

Parameter	Description	Unit
N	Total number of Permanent Sample Plots in the Project Area	Number
N_j	Number of sampling units in stratum j (where $j = 1,2,3 \dots J$ strata)	Number
sd_j	Standard deviation of carbon density for the stratum j , (where $j=1,2,3, \dots J$ strata)	tC ha ⁻¹

Parameter	Description	Unit
t_{stat}	Sample statistic from the t-distribution for the 95% confidence level	dimensionless
err	Allowable error (calculated by mean carbon density by desired precision)	dimensionless

Step 9: The following equation is then used to calculate the number of PSPs in each individual stratum (Pearson et al. 2005):

Equation 7-3

$$n_j = N \times \frac{N_j \times sd_j}{\sum_{j=1}^J N_j \times sd_j}$$

Parameter	Description	Unit
n_j	Number of Permanent Sample Plots in stratum j , (where $j=1,2,3, \dots J$ strata)	Number
N	Total number of Permanent Sample Plots in the Project Area	Number
N_j	Total number of sampling units in the stratum j , (where $j=1,2,3, \dots J$ strata)	Number
sd_j	Standard deviation of carbon density for the stratum j , (where $j=1,2,3, \dots J$ strata)	tC ha ⁻¹

Equation 7-3 determines the number of PSPs for each stratum based on the variability in the stratum. A stratum with higher variability will require a higher number of sample plots than a stratum with less variability.

In the case of the natural disturbances, the sample plots, s_{nd} , are established in the naturally disturbed stratum during the monitoring event for estimating the carbon increased in the AGB pool due to regrowth in the naturally disturbed areas (see Step 5 of Section 4.4).

7.1.2.4 Sampling design for the PSPs

Different sampling design approaches are available for laying out the ground location of the PSPs in each stratum. Some of the commonly used sampling designs in forestry are random sampling, stratified sampling, stratified random sampling, stratified systematic sampling and cluster sampling. A suitable sampling design for establishing the PSPs is required and is important to avoid bias, such as locating PSPs along a road for example, in order to maintain the statistical rigour and validity of the calculations.

7.1.2.5 Parameter measurement in the PSPs

After determining the number of PSPs in each stratum, and the sampling design for on the ground location of the PSPs, field measurements will be conducted. The field team should have sound

knowledge and experience in forest inventory, measurement procedures and equipment. Before the field work commences, the field team must prepare the following:

- A plan of the data collection procedures including the specification of merchantable tree in terms of species, minimum DBH and tree form, rules for borderline trees, locating trees during the following monitoring period, and measurement techniques for the DBH and tree height
- A plan of the data recording procedures and resources, such as paper forms or handheld PDAs
- Documentation of quality control and quality assurance procedures
- Documentation of data archiving methods
- Details of the responsibilities of the parties involved.

The Project Proponent will develop Standard Operating Procedures (SOPs) for each of the above steps and include these in the PD.

After establishing a PSP on the ground, all trees as defined by the relevant authority in the host country will be measured for their DBH and tree height (H) using appropriate equipment. The equipment employed must have been calibrated before taking to the field for measurements.

This Methodology has provided guidance on establishment of the PSPs and the measurement in the Section 7.1.2. However, the Project Proponent will be required to use methods that adhere to good practice internationally as well as locally in the host country. A detail of the chosen methods for the Project Area stratification, sampling design, intensity and the actual tree measurement must be provided in the PD. Some of the specific resource materials for monitoring the net GHG emissions are provided by the following literature: Pearson et al. (2005); Pearson et al. (2007); IPCC (2003); Hoover (2008).

7.1.3 Monitoring Frequency

Under this methodology, the Project Proponent should undertake measurement in the PSPs at intervals not exceeding five years. Parameters associated with the Project Area and leakage must be monitored annually.

7.2 Monitoring Implementation

The Project Proponent must prepare a monitoring plan for estimating carbon changes under the baseline scenario, project implementation and leakage. This section provides guidance on the specific parameters to be monitored under these scenarios, their reference in this Methodology, their sources, monitoring frequency and the relevant equation where the new value is to be applied for ex post calculations of net anthropogenic GHG emission reductions.

The data and the parameters associated with the calculation of the annual change in carbon due to degradation from selective logging ($C'_{degradation,t}$ in Equation 3-2) have been categorised into three groups:

- (i) parameters obtained from the literature or reports (not monitored)
- (ii) parameters measured once (not monitored)
- (iii) parameters that require monitoring.

Tables 7-1 through 7-3 summarise the specific parameters, the source(s) of each data or parameter, and the equation(s) in which each of the parameters are applied.

7.2.1 Parameters Obtained from the Literature/Reports to be Reviewed/Verified (Not Monitored)

Table 7-1 provides a list of parameters to be monitored during the project lifetime and used for updating the net anthropogenic GHG emissions calculation. Section 7.3.2 provides QA/QC procedures for selecting the most appropriate value. The Project Proponent must select the value which does not lead to an over-estimation of the net anthropogenic GHG emission reductions.

Table 7-1 Parameters obtained from the literature or reports to be reviewed or verified

Parameter	Summary of Description	Unit	Source of the parameter	Review frequency or validation	Application of the Parameter in Equation(s)
$BCEF_j$	Biomass conversion and expansion factor in stratum j	(t d.m.) m ⁻³	literature value or derive a local value	Review the literature or where required, verify or derive local value	3-8; 4-15; 4-22
BEF	Biomass expansion factor for converting volume of extracted roundwood to total aboveground biomass (including bark)	dimensionless	literature value or derive a local value	Review the literature or where required, verify or derive local value	4-21; 5-8; 5-9
CF_{wood}	Carbon fraction of wood for the tropical forest	tC (t d.m.) ⁻¹	literature value	Review the literature for appropriate value during the monitoring event	3-3; 3-5; 3-41; 3-51;
CF_{AGB}	Carbon fraction in the aboveground biomass of trees for the tropical forest	tC (t d.m.) ⁻¹	literature value	Review the literature for appropriate value during the monitoring event	3-8; 3-14; 3-37a; 3-37b; 3-38; 4-15; 4-17a; 4-21; 4-22; 5-8; 5-9
D	Wood density for the tropical forest with corresponding climate region and ecological zone	(t d.m.) m ⁻³	literature value or derive a local value	Review the literature or where required, verify or derive local value	3-3; 3-5; 3-41; 3-51; 4-21; 5-8; 5-9
D_i	Species-specific density of wood,	(t d.m.) m ⁻³	literature value	Review the literature or where required, verify or derive local value	3-12

Parameter	Summary of Description	Unit	Source of the parameter	Review frequency or validation	Application of the Parameter in Equation(s)
$f_V(DBH_{n,j,t,j,t=0}, H_{n,j,t,j,t=0})$	Volume allometric equation as a function of diameter at breast height and height; $t=0$ year	dimensionless	obtained from literature	Review the literature or where required, verify or derive local value	3-10; 3-11
$f_B(DBH_{n,j,t,j,t=0}, H_{n,j,t,j,t=0}, D_1)$	Biomass allometric equation as a function of diameter at breast height and height; $t=0$ year	dimensionless	obtained from literature	Review the literature or where required, verify or derive local value	3-12
k_{decay}	Rate of decay of the deadwood pool	yr ⁻¹	literature value	Review the literature for appropriate value during the monitoring event	3-17, 3-21
f_{RSD}	Factor for residual stand damage	dimensionless	literature value	Review the literature for appropriate value during the monitoring event	3-19; 4-21; 5-8, 5-9
f_{branch_trim}	Branch-trim factor	dimensionless	literature value or derive a local value	Review the literature or where required, verify or derive local value	3-20
$f_{lumber_recovery}$	Lumber recovery factor	dimensionless	literature value	Review the literature for appropriate value during the monitoring event	3-26, 3-27
k_{ltHWP_ox}	Rate of oxidation for long-term harvested wood products	yr ⁻¹	literature value	Review the literature for appropriate value during the monitoring event	3-28
k_{stHWP_ox}	Rate of oxidation for short-term harvested wood products	yr ⁻¹	literature value	Review the literature for appropriate value during the monitoring event	3-34
$\bar{G}_{regrowth,t}$	Average regrowth per hectare per year of the aboveground biomass after logging in year, t	(t d.m.) ha ⁻¹ yr ⁻¹	literature value	Review the literature for appropriate value during the monitoring event	3-38

Parameter	Summary of Description	Unit	Source of the parameter	Review frequency or validation	Application of the Parameter in Equation(s)
EF_{fuel}	Fuel emission factor	tCO ₂ -e kL ⁻¹	literature value	Review the literature for appropriate value during the monitoring event	3-40; 3-42; 3-43; 3-46; 3-50; 3-54; 4-8; 4-11; 4-14
$FC_{harvest}$	Fuel consumption of equipment employed for felling and snigging per m ³ of merchantable log harvested	kL m ⁻³	Manufacturer's specification/literature value	Review the literature for appropriate value during the monitoring event	3-40
FC_{trim_equip}	Fuel consumption of equipment employed for trimming per m ³ of trimmed material	kL m ⁻³	Manufacturer's specification	Review the literature for appropriate value during the monitoring event	3-42
$FC_{hauling}$	Fuel consumption of equipment for hauling one m ³ of merchantable log	kL m ⁻³	Manufacturer's specification	Review the literature for appropriate value during the monitoring event	3-43
Cap_{truck}	Truck load capacity	m ³ truck ⁻¹	literature value	Review the literature for appropriate value during the monitoring event	3-44; 3-52
$Eff_{vehicle}$	Fuel efficiency for vehicle type	km kL ⁻¹	Manufacturer's specification/National Database	Review the literature for appropriate value during the monitoring event	3-46; 3-52; 3-54; 4-7; 4-11; 4-14
e_{demand}	Electricity demand for processing per volume processed	kWh m ⁻³	literature value	Review the literature for appropriate value during the monitoring event	3-47
$EF_{electricity}$	Emission factor for electricity in the host country	tCO ₂ -e kWh ⁻¹	Country-specific data from International Energy Agency	Annually revised for the specific value	3-48; 4-4
$t_{generator,t}$	Total operating time of generator in year t	h	National reports on harvesting practice	Review the literature for appropriate value during the monitoring event	3-49; 3-50

Parameter	Summary of Description	Unit	Source of the parameter	Review frequency or validation	Application of the Parameter in Equation(s)
$FC_{generator}$	Fuel consumption per hour of operation of generator	kL h ⁻¹	Manufacturer's fuel consumption chart	Review the literature for appropriate value during the monitoring event	3-50
$PR_{equip,ee,t}$	Power rating for electrical equipment, <i>ee</i> , in year <i>t</i>	kW	Manufacturer's power rating for the equipment	Review the value during the monitoring event	4-3
$EF_{flight,y}$	Flight emission factor for trip <i>y</i>	tCO ₂ -e (passenger.km) ⁻¹	literature value	Annually revised for the specific value	4-6, 4-10; 4-13
R_{CH_4}	Emission ratio for CH ₄	dimensionless	literature value	Review the literature for appropriate value during the monitoring event	4-18a
R_{N_2O}	Emission ratio for N ₂ O	dimensionless	literature value	Review the literature for appropriate value during the monitoring event	4-18b
$R_{N/C}$	Ratio of nitrogen to carbon	dimensionless	literature value	Review the literature for appropriate value during the monitoring event	4-18b
GWP_{CH_4}	Global warming potential of CH ₄	tCO ₂ -e tCH ₄ ⁻¹	literature value	Review the literature for appropriate value during the monitoring event	4-19
GWP_{N_2O}	Global warming potential of N ₂ O	tCO ₂ -e tN ₂ O ⁻¹	literature value	Review the literature for appropriate value during the monitoring event	4-19

7.2.2 Parameters to be Measured Once (Not Monitored)

Table 7-2 provides a list of parameters measured ex ante and used in the calculation for the net anthropogenic GHG emission reductions. These parameters and data are validated or verified but not monitored.

Table 7-2 Parameters to be measured once but not monitored

Parameter	Summary of Description	Unit	Source of the parameter	Measurement Frequency	Application of the Parameter in Equation(s)
$A_{project,t=0}$	Project Area at time, $t=0$	ha	GPS data, GIS maps and satellite data	Validated/Verified before the IFM-LtPF project start date	2-1; 3-4; 3-6; 3-9; 3-13; 3-36b;
$A_{project,j,t=0}$	Project Area within each stratum, j , at time, $t=0$	ha	GPS data, GIS maps and satellite data	Validated/Verified before the IFM-LtPF project start date	2-1; 3-4; 3-6; 3-9; 3-13
$A_{s,j,t=0}$	Total area of sample plots, s , in stratum, j , $t=0$ year	ha	Ex ante field measurement in the sample plots	Validated/Verified before the IFM-LtPF project start date	3-10; 3-11; 3-12
$DBH_{n,i,s,j,t=0}$	Diameter at breast height $t=0$ year	cm	Ex ante field measurement in the sample plots	Validated/Verified before the IFM-LtPF project start date	3-10; 3-11; 3-12
$H_{n,i,s,j,t=0}$	Height for individual tree, $t=0$ year	m	Ex ante field measurement in the sample plots	Validated/Verified before the IFM-LtPF project start date	3-10; 3-11; 3-12
$A_{NHA_annual,t}$	Annual net harvest area for the Project Area in year, t	ha	Ex ante, obtained from the harvesting plan	Validated/Verified before the IFM-LtPF project start date	3-15a; 3-16a; 3-26; 3-27; 3-33; 3-35; 3-37a; 3-38
$A_{NHA_annual,j,t}$	Annual net harvest area at the stratum level in year, t	ha	Ex ante, obtained from the harvesting plan	Validated/Verified before the IFM-LtPF project start date	3-15b; 3-16b; 3-37b
$KM_{transport,t}$	Annual log transport distance from collection depot to processing plant	km truck ⁻¹	Digital maps	Validated/Verified before the IFM-LtPF project start date	3-45
$KM_{distrib,t}$	Annual distance of transport from point of processing to distribution/export point	km truck ⁻¹	Digital maps	Validated/Verified before the IFM-LtPF project start date	3-53
$V'_{historical_harvest,l,k}$	Total volume of harvest for land l that is owned and/or operated by the Project Proponent over the historical reference period	m ³	Project Proponent records	Validated/Verified before the IFM-LtPF project start date	5-3

7.2.3 Parameters to be Monitored

Table 7-3 provides a list of parameters to be monitored during the project lifetime which are then employed to update the net anthropogenic GHG emission reductions calculation.

Table 7-3 Parameters to be monitored in the Project Area

Parameter	Summary of Description	Unit	Source of the parameter	Measurement Frequency	Application of the Parameter in Equation(s)
$DBH_{n,i,s,j,t}$	Diameter at breast height for individual tree n , of species i , in sample plot s , of stratum j , in year t	cm	Measured using DBH tape	At intervals not exceeding five years after the first monitoring event	3-12
$DBH_{tree_nd,n,i,s_{nd},j,t}$	Diameter at breast height for individual tree n , of species i , in sample plot in the naturally disturbed area s_{nd} of stratum j , in year t	cm	Measured using DBH tape	At intervals not exceeding five years after the first monitoring event	3-12
$H_{n,i,s,j,t}$	Height for individual tree n , of species i , in sample plot s , of stratum j , in year t	m	Measured using tree height measurement equipment	At intervals not exceeding five years after the first monitoring event	3-12
$H_{tree_nd,n,i,s_{nd},j,t}$	Height for individual tree n , of species i , in sample plot in the naturally disturbed area s_{nd} of stratum j , in year t	m	Measured using tree height measurement equipment	At intervals not exceeding five years after the first monitoring event	3-12
$t_{op_equip,ee,t}$	Hours of operation of electrical equipment ee , in year t	h	Electrical equipment time log book	Annual	4-3
$KM_{plan_flight,y,t}$	Distance travelled per trip y , for a total of Y trips in year t	km	Flight travel log	Annual	4-6
$N_{plan_flight,y,t}$	Number of passengers per trip y in year t	Number	Flight travel log	Annual	4-6
$KM_{plan_ground,y,t}$	Distance travelled per trip y , for a total of Y trips in year t	km	Vehicle travel log	Annual	4-7

Parameter	Summary of Description	Unit	Source of the parameter	Measurement Frequency	Application of the Parameter in Equation(s)
$V_{fuel_plan_ground,y,t}$	Annual volume of fuel consumed per trip y in year t	kL	Vehicle travel log	Annual	4-8
$KM_{design_flight,y,t}$	Distance travelled per trip y , for a total of Y trips in year t	km	Flight travel log	Annual	4-10
$N_{design_flight,y,t}$	Number of passengers per trip y in year t	Number	Flight travel log	Annual	4-10
$KM_{design_ground,y,t}$	Distance travelled per trip y , for a total of Y trips in year t	km	Vehicle travel log	Annual	4-11
$KM_{monitoring_flight,y,t}$	Distance travelled per trip y , for a total of Y trips in year t	km	Flight travel log	Annual	4-13
$N_{monitoring_flight,y,t}$	Number of passengers per trip y , in year t	passenger	Flight travel log	Annual	4-13
$KM_{monitoring_ground,y,t}$	Distance travelled per trip y , for a total of Y trips in year t	km	Vehicle travel log	Annual	4-14
$A_{nd,j,t}$	Area of natural disturbance nd , in stratum j in year t	ha	Satellite imagery and field measurement	Annual	4-15; 4-17a
$f_{natdisturb,j,t}$	Fraction of the forest naturally damaged in stratum j in year t	dimensionless	Field survey	Annual	4-16
$V_{illegal_harvest,t}$	Volume of wood sold as determined from field surveys in year t	m ³	Field survey	Annual	4-21
$A_{illegal_harvest,j,t}$	Area of illegal harvest in stratum j in year t	ha	Satellite data	Annual	4-22

Parameter	Summary of Description	Unit	Source of the parameter	Measurement Frequency	Application of the Parameter in Equation(s)
$V_{actual_harvest,l,t}$	Annual actual volume of harvest for land /that is owned and/or operated by the Project Proponent or the forest with comparable situations and condition in local or regional or nation level in year t	$m^3 yr^{-1}$	Project Proponent records	Annual	5-4 to 5-7

7.2.4 Validating or Deriving the Parameters

To calculate the annual change in carbon due to degradation, the carbon in merchantable logs and AGB must be calculated. Good practice (IPCC, 2003) suggests the use of locally derived species-specific or group of species-specific default values for the net anthropogenic GHG emission reductions calculation. However, the application of default parameters available in the IPCC literature, national inventory reports or published peer-reviewed studies is acceptable if the parameter is applicable to a forest type and climatic region similar to the Project Area (CDM-EB, 2009, p. 26). Where the default parameter does not match the forest type and climatic region of the Project Area, it is required to validate the parameters using the destructive sampling approach described in the following section.

7.2.4.1 Validating or deriving the wood density

Where the default value for wood density is not obtained from the forest type and climatic region similar to the Project Area, validate or derive the wood density using the following procedures based on Pearson et al. (2005) and CDM-EB (2009):

Step 1: Select 20-30 trees⁶ representing all DBH classes found in the Project Area during the first monitoring event (within five years from the project start date).

Step 2: Measure the DBH and height of all trees.

Step 3: Fell the trees and separate each tree into the following components: merchantable log, large branches, small branches (less than 10 cm diameter), and foliage.

Step 4: Cut the merchantable logs into at least five sections, and the large branches into at least three sections.

Step 5: To determine the total volume of the merchantable logs and large branches, measure the length and the diameter at both ends of each section of the logs/branches; and calculate the total volume as in Pearson et al. (2005), Appendix B, Method 1, step 6a, p. 40.

⁶ According to the CDM at least five trees representing all DBH classes, is sufficient for the destructive sampling method for verifying default values or equations. However, this Methodology suggests using at least 20 trees, which provides a greater representation of the trees in the Project Area.

Step 6: Obtain a complete cross-sectional sample from each log and large branch and measure the fresh weight and calculate the volume of each sample. Oven-dry the samples at 70 degrees for 24 hours and measure the dry weight. Estimate the wood density (D) for the forest types as well as species-specific wood density (D_s) by dividing the dry weight by the volume.

Step 7: Select the wood density.

If the wood density for the forest types and the species-specific wood density in the Project Area derived from the direct measurement are within $\pm 10\%$ (CDM-EB, 2009, p. 26) of the corresponding default wood densities or lower than the measured value, it is conservative to use the selected default values for wood densities.

If the default wood densities are greater than the measured wood densities by 10% or more, it is conservative to use the wood densities derived from the samples.

If the default wood densities are lower than the measured wood densities by 10% or more, it is conservative to use the default wood densities.

7.2.4.2 Validating or deriving the volume and biomass allometric equations

Where the default volume and biomass allometric equations do not match the forest type and climatic region of the Project Area, it is required to validate or derive the allometric equations using the following destructive sampling procedure:

Step 1: Follow Steps 1 to 5 in Section 7.2.4.1 to calculate the total volume of the merchantable logs and large branches.

Step 2: Measure the fresh weight and oven dry weight of the small branches and foliage of all the felled trees.

Step 3: Estimate the biomass of the logs and large branches using the total volume and wood density (see Section 7.2.4.1) (biomass = volume x wood density). Derive the total AGB by adding the oven dry biomass of the leaves and small branches to the biomass of the large branches and logs.

Step 4: Apply the volume and biomass allometric equations to the measured data for 20-30 trees and obtain the mean volume and AGB for these trees.

Step 5: Select the equations.

If the volumes and AGB of the harvested trees derived from the direct measurement are within $\pm 10\%$ of the mean values predicted by the default allometric equations, or the prediction is lower than the measured value, it is conservative to use the selected default allometric equation.

If the default allometric equations overestimate the prediction compared with the measured volume or AGB, develop a project-specific volume and/or AGB allometric equation by applying regression to tree volume and/or AGB as an independent variable to DBH and height. The wood density is also used as an independent variable for the biomass allometric equation.

7.2.4.3 Validating or deriving the branch-trim factor

If a branch-trim factor, f_{branch_trim} , cannot be obtained from the literature, or the branch-trim factor obtained from the literature does not match the forest type and climatic region of the Project Area, it is required to derive or validate, respectively, a project-specific branch-trim factor in the first monitoring event using the following the procedure:

Step 1: Follow Steps 1 to 3 in Section 7.2.4.2 and obtain the biomass of large and small branches and the total AGB.

Step 2: Calculate the branch-trim factor as the ratio of biomass of large branches to the AGB of the tree via the following equation:

Equation 7-4

$$f_{branch_trim} = \frac{B_{branch_trim, m_1}}{B_{AGB_trees, m_1}}$$

Parameter	Description	Unit
f_{branch_trim}	The fraction of branches and trimmings in the aboveground biomass remaining after trimming of the merchantable logs transferred to the DW pool	dimensionless
B_{branch_trim, m_1}	Biomass in large and small branches in year m_1 of first monitoring event	t d.m.
B_{AGB_trees, m_1}	Biomass in the aboveground biomass of sample trees in year m_1 of first monitoring event	t d.m.

Step 3: If a branch-trim factor was employed in ex ante estimations, compare the project-specific branch-trim factor and the estimated branch-trim factor. If the estimated branch-trim factor is within $\pm 10\%$ of the branch-trim factor value, or the value is lower than the estimated value, it is conservative to use the selected branch-trim factor.

Step 4: If the branch-trim factor value is higher than the estimated value, use the estimated value for the project.

7.2.5 Non-Permanence Risk Assessment

7.2.5.1 Buffer determination

The amount of credits a project must allocate for buffer reserves in the AFOLU Pooled Buffer Account, $CC_{NPbuffer, t}$, is based on the project's potential for future carbon loss. The method to calculate $CC_{NPbuffer, t}$ has been presented in Section 1.2.4, Equation 1-2.

To obtain the project's non-permanence buffer withholding percentage, $NP_{buffer, t} \%$, a risk rating in relation to its potential for reversal of the sequestered/protected carbon will be determined using the most recent version of VCS tool for AFOLU Non-Permanence Risk Analysis and Buffer Determination. The risk assessment must be clearly documented and substantiated in the PD and/or monitoring report.

Once the amount of credits to be deposited in the buffer account is determined, the tradeable carbon credits, VCU_t , can be calculated using Equation 1-3, Section 1.2.4.

7.3 Quality Assurance and Quality Control

Quality assurance (QA) and quality control (QC) are important for verifying and certifying carbon changes throughout the crediting period. The Project Proponent must have QA and QC systems in place at the project development stage in order to provide enough confidence to all the stakeholders on the reliability, accuracy and precision of the measurement. Accurate and precise data collection and analysis requires a trained field team, appropriate equipment, standard data collection procedures and recording, analysis and storage of data.

The PD must describe the QA and QC approach for the IFM-LtPF project and must contain Standard Operating Procedures (SOPs) for:

- (i) Conducting field measurement
- (ii) Selecting literature values
- (iii) Data entry, maintenance and archiving
- (iv) Contract procurement.

To ensure good practice in monitoring and inventory a comprehensive QA and QC system will be implemented. The procedures will be based on ISO 9001 and encompass the elements as defined in the relevant sections of the IPCC 2006 Guidelines for National Greenhouse Gas Inventories. Applying the framework proposed by ISO 9001 will ensure that all procedures and systems will be subjected to internal audits as well as being independently verified.

The QA/QC and verification system will adopt a continuous improvement model with a particular emphasis on the key parameters within the Methodology that will be monitored.

7.3.1 QA/QC for Conducting Field Measurements

The capacity and skill set of field teams employed to establish and collect measurement data in PSPs associated with carbon inventory will be assessed by qualified personnel using a combination of theoretical and practical assessment. All field teams will receive extensive training so that they are fully cognisant of all procedures and understand the importance of collecting accurate data. Appropriate training will be developed encompassing a continuous improvement system. The QA/QC system will document and allow traceability of the skill sets, training and continuous improvement of all personnel associated with the project.

If the particular required skill sets are deemed unsatisfactory then the applicable personnel will be required to undergo a formal training program conducted by suitably qualified personnel. This will be recorded in the appropriate QA module.

Standard Operating Procedures (SOPs) will be developed to encompass all the steps required to establish the carbon inventory associated with the major pools, namely AGB, DW and where applicable HWP. The SOPs will be sufficiently robust to ensure that any new person sent into the field should be able to repeat the previous measurements.

SOPs associated with the number and placement of PSPs will also be developed to ensure their appropriate implementation. SOPs associated with the use and accuracy of all equipment used to take measurements will be developed and implemented.

All PSPs and measurements will in the first instance be established under the guidance of appropriately qualified personnel. The qualifications and expertise of these authorised staff will be identified in the appropriate QA module.

An evaluation of all field teams will be conducted to identify errors in field techniques, verify the measurement process and correct any identified problems before any measurements are undertaken. All PSPs and associated personnel who established and obtained the associated measurements will be subjected to a random audit program to ensure the accuracy and precision of data collected. The audit program will initially consist of 10% of PSPs being re-measured by suitably qualified personnel. The audit program will be fluid in nature so that it targets those plots and personnel that show a significant standard deviation. SOPs will be written to encompass this as well as a system that ensures that all PSPs and or specific measurements are re-taken and re-calculated. SOPs will be written to ensure that any field data collected over time is compared to the original data and discrepancies are verified again.

7.3.2 QA/QC for Selecting Literature Values

The selection of appropriate and current data from the literature is required in order to reduce uncertainty of the net anthropogenic emission reductions of the IFM-LtPF project. Types of data required from literature for the current Methodology are:

Emission factors

Rate of oxidation/decay

Forestry parameters and conversion factors, i.e. biomass expansion factor, biomass conversion and expansion factors, wood density and carbon fraction

Forestry process factors, i.e. lumber recovery factor, residual stand damage

Growth models

Allometric equations.

The Project Proponent is required to choose values/models from the literature from project(s) and area(s) with features and characteristics similar to the IFM-LtPF Project Area, for example a tropical forest with corresponding climate region and ecological zone. In addition when selecting data or parameters from the literature, the Project Proponent must adopt a conservative approach in regards to the net anthropogenic GHG emission reductions. This implies that if two or more values are available for the forest types and condition in the Project Area, the value which does not lead to an over-estimation of the net GHG emission reductions must always be chosen.

Literature values must be obtained from the following sources in the order of the most preferred data source:

- (i) Peer-reviewed literature values providing project-specific data
- (ii) Country-specific data

- (iii) Global IPCC default values.

When recording literature values, the Project Proponent is required to provide the following information in order for the data to be verified:

- (1) Source of data, date, table/figure, page number, from which the data is derived
- (2) Corresponding features that match with the Project Area, i.e. forest type, climate zone, etc.
- (3) Corresponding uncertainty (absolute or relative)
- (4) An explanation of the choice of parameter from the range, if a range is provided. The Project Proponent must choose a value that presents the net anthropogenic emission reductions as conservative.

7.3.3 QA/QC for Data Entry and Archiving

According to the guidelines published by the IPCC (2006) it is a prerequisite that all calculations leading to emission or removal estimates should be fully reproducible. As such, QC procedures must be created to address potential errors associated with input data, the conversion of algorithms of a calculation and the output. In addition, because of the relatively long term nature of the IFM-LtPF project activity, data archiving (maintenance and storage), QA procedures and SOPs will be established to ensure the traceability of data analysis and the documentation of calculations over the lifetime of the project. As such the following activities will be developed into formal SOPs.

- (i) Numbers entered into spreadsheets will reference the data sources and cells containing derived data as "results" will be clearly marked
- (ii) All calculations will be presented in the form of formulas so that auditing tools can be used to track back from a result to the data source, and calculations can be evaluated by analysing the formulae
- (iii) All databases, spreadsheets and recording sheets will be clearly referenced by name, version, author, updates, intended use and checking procedures, so that it can be used as a data source of the derived results and referenced further as required in the inventory process
- (iv) When using databases, the source of data tables must be referenced to the data source using a referencing column
- (v) Wherever possible, queries are to be used when using databases as a means to track back to the data table
- (vi) Where queries are not practical and new tables of data need to be generated, scripts or macros wherever used, are to be recorded and referenced in a referencing column
- (vii) All data are to be archived electronically so that these can be stored and backed up and kept for at least two years after the end of the crediting period. SOPs will be developed to ensure that this occurs in a timely manner
- (viii) Before any calculation is implemented it must be peer reviewed for accuracy and relevance (including unit conversion factors) by two personnel with the relevant skills and authority
- (ix) All calculations must be checked by at least one appropriately skilled independent person to ensure they are accurate and consistent with applicable time series

- (x) Wherever possible, estimates and calculations are to be compared with literature values. If they are considered significantly different (5%) they are to be checked for accuracy.

7.3.4 QA/QC for Contract Procurement

SOPs will be developed for the procurement of suppliers or contractors. This is to ensure that any analysis undertaken meets internationally recognised standards and that contractors if used, adopt best practice when undertaking specific tasks.

7.4 Ex post Calculation of Net Anthropogenic GHG Emission Reductions

After the completion of the monitoring plan as outlined in the previous sections, the baseline degradation, project activity, and leakage emissions and their uncertainties, must be revised and adjusted at the end of each monitoring period. The net anthropogenic GHG emission reductions on the basis of the ex post measured parameters can be recalculated via the following steps:

- Step 1: Calculate annual total carbon emissions due to degradation of the baseline activity, $C'_{degradation,t}$, using procedures outlined in Sections 3.2 and 3.3.
- Step 2: Calculate annual total carbon emissions due to implementation of baseline activity, $C'_{emissions,t}$, using procedures outlined in Section 3.4.
- Step 3: Calculate annual total carbon emissions associated with the baseline scenario, $C'_{baseline,t}$, using Equation 3-1, Section 3.1.
- Step 4: Calculate annual total carbon emissions associated with the project activity, $C'_{actual,t}$, using procedures outlined in Section 4.
- Step 5: Calculate the annual total carbon emissions associated with leakage, $C'_{leakage,t}$, using procedures outlined in Section 5.
- Step 6: Calculate the net anthropogenic GHG emission reductions, $C'_{IFM-LtPF,t}$, using Equation 1-1, Section 1.2.1.
- Step 7: Determine the absolute uncertainties for all parameters employed in the calculation of the net anthropogenic GHG emission reductions using the process outlined in Sections 6.2 to 6.5. Using Equations 6-4 and 6-5, calculate the absolute and relative uncertainty, respectively, associated with the net anthropogenic GHG emission reductions. Follow the procedure (Steps 1 and 2) in Section 6.6 to ascertain if an uncertainty deduction must be applied. Proceed to Section 6.7 to reduce the uncertainty of the net anthropogenic GHG emission reductions. Revise the calculation of the net anthropogenic GHG emission reductions.
- Step 8: Determine the non-permanence buffer withholding percentage, $NP_{buffer,t}\%$, from the Non-Permanence Risk Assessment (Section 7.2.4) and calculate the carbon credits to be deposited in the VCS buffer withholding pool, $CC_{NPbuffer,t}$, using Equation 1-2, Section 1.2.4.

Step 9: In the case where $C'_{actual,t} < C'_{baseline,t}$, calculate the carbon credits that can be traded after non-permanence buffer credits have been considered, VCU_t , using Equation 1-3, Section 1.2.4.

Step 10: In the case where $C'_{actual,t} > C'_{baseline,t}$, and carbon is lost from the project, the project will be subjected to procedures for cancellation of buffer credits in accordance with VCS rules.

The ex post calculation of the net GHG anthropogenic emission reductions presented in the monitoring report must be verified.

8 References

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Appendices

Appendix A: Definitions and Acronyms

Some of the important definitions of terms used throughout the proposed Methodology have been adapted from the GPG LULUCF published by the IPCC (2003). The definition of terms are presented in Appendix A.1.

The acronyms used throughout the Methodology represent internationally defined terms used widely by the Verified Carbon Standard (see VCS), the Intergovernmental Panel on Climate Change (see IPCC), the United Nations Framework Convention on Climate Change (see UNFCCC), and in the disciplines of agriculture, forestry and geography.

A.1 Definition of Terms

Aboveground Biomass, AGB	Living biomass of the trees above the ground that includes stems, stumps, branches, bark, seeds and foliage.
Wood Density, D	Ratio between oven dry biomass and fresh stem-wood volume without bark; units of (tonne of dry matter) m^{-3} ; used as a parameter in equations.
Branches and trimmings	The trimmed components such as large and small branches, whose carbon is transferred into the deadwood pool after harvesting.
Branch-trim factor	Factor accounting only for the carbon in the large and small branches that remain on the forest floor after the trimming process and whose carbon is transferred into the deadwood pool.
Biomass Conversion and Expansion Factor, BCEF	Factor converting growing stock to aboveground biomass, units of (tonne of dry matter) m^{-3} ; used as a parameter in equations.
Biomass Expansion Factor, BEF	Biomass expansion factor for converting volume of extracted roundwood to total AGB (including bark).
Carbon Fraction, CF	Ratio between quantity of carbon and dry biomass, units of (tonnes of carbon) (tonne of dry matter) $^{-1}$; used as a parameter in equations.
Carbon Pool	Reservoir that has the potential to accumulate (or lose) carbon over a period of time. In IFM-LtPF project activity, the pools to be considered are aboveground biomass, deadwood and wood products.

Commercial Log Length	The length of merchantable logs above the stump height to the minimum top diameter.
Crediting Period	Synonymous with project crediting period as defined in VCS rules
Deadwood, DW	Non-living biomass of woody vegetation not contained in the litter, either standing or lying above the soil level, and stumps larger or equal to 10 cm in diameter, or a nationally-specified diameter; used as a subscript in equations.
Dry Matter	Biomass that has been dried to an oven-dry condition, where the temperature of the oven is usually held at 70 °C.
Ex ante	Before an activity; the estimation of the net GHG emission reductions, before the start of the IFM-LtPF project activity, for the project crediting period.
Ex post	After an activity; the calculation of the actual net GHG emission reductions for the years elapsed since the start of the IFM-LtPF project activity.
Forest	Land with biomass that is defined by a minimum area, level of tree crown cover and tree height. Under the Kyoto Protocol, ranges for the three parameters are: 0.05 - 1.0 hectare with crown cover (or equivalent stocking level) of more than 10 - 30% comprising trees, with the potential of the trees to reach a height of 2 - 5 metres at maturity <i>in situ</i> .
Forest Degradation	Biomass that is lost through a measurable decrease in canopy cover, but which is not sufficient to reduce canopy cover to below the percent range specified for forest in the relevant country (Penman et al., 2003).
Forest Inventory Report, FIR	A legally approved or sanctioned document by the relevant authority of the host country that includes, but is not limited to, inventory data, forest management plan, and other forest inventory findings obtainable from the host country's forest inventory database.
Forest Product Type	Forest product type includes sawlog, pulplog and commercially harvested fuelwood. The definition of these products are obtained from the relevant authority of the host country.

Fuelwood, FW	Biomass collected for energy production purposes that includes commercial fellings and trees damaged by natural causes, but does not include wood that is produced as a by-product or residual matter from industrial processing of merchantable logs.
Growing Stock	The total volume (over bark) of all living trees with DBH larger than the minimum DBH as specified by the relevant authority in the host country.
Harvested Wood Products, HWPs	Merchantable logs that will be crafted into products, either classed as long-term or short-term products. An example of merchantable logs that will be made into long-term HWPs (ltHWPs) is sawlogs. Examples of merchantable logs that will be made into short-term HWPs (stHWPs) are pulplogs and logs commercially harvested for fuelwood.
Lumber Recovery Factor	Ratio of the volume of sawnwood products (ltHWPs) recovered from harvested sawlog volumes. Lumber recovery factor subtracted from unity results in sawmill residues from the processing of ltHWPs.
Merchantable Logs	Marketable portion of the merchantable trees as defined by the minimum diameter at breast height and tree top, as specified by the relevant authority in the host country.
Merchantable Trees	Trees with a minimum DBH and tree form as defined by the relevant authority in the host country.
Minimum DBH	The diameter at breast height used for defining a merchantable tree as specified by the relevant authority in the host country.
Monitoring event	The implementation of the monitoring activity for measuring the parameters for net GHG emission reductions after the commencement of the IFM-LtPF project.
Monitoring period	The time taken between monitoring events for collecting measurements in the PSPs in the Project Area, and for reviewing non-monitored parameters.
Non-tree	For the purpose of this Methodology, non-tree has been broadly defined as all the vegetation except the trees with the minimum DBH as specified by the relevant authority in the host country (see footnote ## on page 18). It comprises of ground vegetation (seedlings, saplings, herbs and shrubs), hanging vines and lianas and also woody climbers.

Project Area	Geographic area in which the IFM-LtPF project activity will be implemented that reduce emissions from forest degradation.
Project Boundary	Encompasses (i) geographical boundary, (ii) crediting period, (iii) sources and sinks and associated GHGs, and (iv) carbon pools. In this Methodology, on the basis that the overarching component is the geographical boundary, the Project Boundary is therefore referred to as the geographical area in which the project actions and activities will be implemented.
Residual Stand Damage, RSD	Damage to non-harvested trees due to harvesting operations. The residual stand are damaged either by being knocked down, snapped off or due to limb breakage.
Tropical Forests	Forests growing in the tropical region and comprised of the following four broad forest classes: Evergreen Tropical Rainforest, Moist Deciduous Tropical Forest, Dry Tropical Forest and Upland Tropical Forest (FAO, 1998).

A.2 List of Acronyms Used

Abbreviation	Term
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A.3 General Notation of Parameters

Parameter symbol type	Description of parameter function and unit type
$\bar{X}$	a per hectare value
X	a value in any particular year
X'	a tCO ₂ -e value in any particular year
X''	an absolute value

Reference Notes

FAO (1998). Guidelines for the Management of Tropical Forests 1. The Production of Wood, FAO Forestry Paper 135.

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Appendix B: Guidance for Parameter Selection

B.1 Wood Density

Guidance on selecting suitable wood density depends on the type of data available and its application:

Existing Inventory Data Pathway - BCF Method: Where Less Detailed FIR is Available

Where less detailed FIR information is available and species types are not known, a mean tree density (t d.m.) m^{-3} for the tropical region most appropriate to the Project Area should be employed.

The arithmetic mean and most common wood density values (t d.m.) m^{-3} for tropical tree species in tropical Africa, America and Asia presented in Table B-1, is based on a study by Reyes et al. (1992) using approximately 1280 tree species.

Table B-1 Arithmetic mean for tropical tree species by tropical region (Reyes et al. 1992)

Tropical Region	Arithmetic Mean, (t d.m.) m^{-3}
Africa	0.50
America	0.60
Asia	0.57

Existing Inventory Data Pathway - Allometric Method: Where Detailed FIR is Available

Where species are known, species-specific or group of species-specific wood density should be selected from the following sources:

- (i) National species-specific or group of species-specific densities (e.g., from National GHG inventory)
- (ii) Species-specific or group of species-specific densities from neighbouring countries of a similar climate region and ecological zone
- (iii) Globally species-specific densities (e.g. IPCC, 2006a, Chapter 4, Table 4.13, p. 4.64).

Such data types should be applied at either the species level, for example, in biomass allometric equations that employ species density, along with DBH and H, $f_i(\text{DBH}_i, H_i, D_i)$, or applied to determine a weighted average density, based on the dominant species measured in each volume.

A Project Area specific average density can be employed, see Section 7.2.4.1 on how to derive this value.

Measured Data Pathway - Allometric Method: Where FIR Data is not Available

Guidance on density selection is the same as for Existing Inventory Data Pathway - Allometric Method.

B.2 Carbon Fraction

Where species-specific carbon fraction data is not available, the carbon fraction should be obtained from IPCC (2006a), Chapter 4, Table 4.3, p. 4.48, presented here for the tropical and subtropical climate domain in Table B-2:

Table B-2 Carbon fraction for all of tree and wood component for tropical forests (adapted from IPCC, 2006, Chapter 4, Table 4.3, p. 4.49)

Part of Tree	Parameter	Carbon Fraction, tC (t d.m.) ⁻¹	References
Wood	CF_{wood}	0.49	Feldpausch et al., in IPCC (2006a)
Wood, tree DBH < 10 cm	CF_{wood}	0.46	Hughes et al., in IPCC (2006a)
Wood, tree DBH ≥ 10 cm	CF_{wood}	0.49	
All of tree (AGB)	CF_{AGB}	0.47	Andreae and Merlet, Chambers et al., Lasco and Pulhin in IPCC (2006a)
Default IPCC value, All of tree	CF_{AGB}	0.47	McGroddy et al., in IPCC (2006a)

B.3 Residual Stand Damage

Brown et al. (2005) compiled the factor for residual stand damage (damage over extracted) for various countries and revealed a strong relationship with average commercial log length. The results have been summarised in this Methodology in Table B-3 below.

Table B-3 Factor for residual stand damage (compiled from Brown et. al., 2005, Figure 11, p. 16)

f_{RSD} (tCdamaged tCextracted ⁻¹)	Commerical Log Length (m)	Country	Reference
1.74	22	Congo	Brown et al. (2005)
2.30	17	Malaysia	Pinard and Putz (1996)
2.78	10.8	Belize	Brown et al. (2005)

f_{RSD} (tCdamaged tCextracted ⁻¹)	Commerical Log Length (m)	Country	Reference
3.10	9.8	Bolivia	Pearson et al. (2005)

B.4 Lumber Recovery Factor

The lumber recovery factor for sawlog processing is required in order to calculate the volume of the long-term HWP carbon pool.

This factor vary from country to country (see Table B-4), and are dependent on a number of variables that include skilled labour, equipment used, and sawmill facilities (FAO, 2004; Pulkki, 1997).

Table B-4. Country-specific lumber recovery factors

Country (and Sawmill details)	$f_{lumber_recovery}$, (Range, Where Provided)	Reference
Conventional logging sawmills(<i>for comparative purposes</i>)	0.35	FAO (2004)
Cameroon, Malaysia - Sarawak for general use in the absence of country-specific factor	0.36-0.57, dependent on lumber market	FAO (2004)
Ghana	average of 0.40, maximum of 0.50	Loehnertz et al. (1996)
Indonesia (general)	0.45	Silitonga (1987)
Brazil, Old Amazon	0.47	Verissimo et al. (1992)
Indonesia, East Kalimantan	0.50	Muladi (1996)
Malaysia (general)	0.50	Bhargava and Kugan (1988)
Papua New Guinea	average of 0.52 (0.44-0.56)	Kilkki (1992)
Brazil, Maranhao State	0.55	Loehnertz et al. (1996)
Philippines	average value not provided (0.56- 0.68)	Niedermaier (1984)

B.5 HWP Rate of Oxidation

The rate of oxidation refers to the proportion of the HWP pool(s) which is oxidized, thereby releasing carbon to the atmosphere. Three approaches can be used to establish the product decay profile for HWP (Ford-Robertson, 2003):

- (i) linear decay over the lifetime - so there is nothing left at the end of that period
- (ii) exponential decay with a given half-life - in order to generate an equivalent “tonne-year” impact, the half life is equivalent to half of the lifetime used for linear decay
- (iii) instant decay of all emissions at the end of the product life - as with the exponential decay, the life used is half of that used for linear decay.

The IPCC default approach (IPCC, 2006b, Chapter 12, Section 12.2.2, p. 12.17) assumes that the carbon in woody material follows a first-order oxidation. It is based on the assumption that HWPs are discarded from use. Table B-5 presents the IPCC default rates of oxidation for HWP categories (ltHWPs and stHWPs).

Table B-5. Rate of oxidation for Harvested Wood Products (adapted from IPCC (2006b), Chapter 12, Table 12.2, p. 12.17)

Harvested Wood Products Category	k_{ltHWP_ox} Rate of oxidation
ltHWPs, e.g. solid wood	0.023
stHWPs, e.g. paper products	0.347

B.6 Harvester Fuel Consumption Factor

Klvac and Skoupy (2009) derived fuel consumption emission factors for equipment required for harvesting operations, as listed in Table B-6.

Table B-6. Equipment types and fuel consumption for harvesting operations (Klvac and Skoupy, 2009)

Operation	Equipment	Factor	Data	Unit
Felling	harvester	Fuel consumption, $FC_{harvest}$	1.28 - 1.73	L m ⁻³

B.7 Electricity Demand of Sawmill Processes

The electricity demand factor can be employed to determine the electricity consumption (kWh) required for a particular volume of merchantable logs. Table B-7 presents data from three international sawmill processes and shows that an approximate range for the electricity demand (e_{demand}) is 20-40 kWh m⁻³.

Where the country specific value for the electricity demand is available, it should be used. If this is not the case, then a factor from a country that uses similar timber processing technology to that of the project's host country should be used.

Table B-7. Electricity demand for sawmill processing in various countries

Country	Electricity (kWh day ⁻¹)	Harvest Volume (m ³ day ⁻¹)	e _{demand} (kWh m ⁻³)	Data Source
Indonesia	600	20	30	Budiono (unknown)
Brazil	2756	136	20	Poole and Pinheiro (2003)
New Zealand	-	-	26-41	Li et al. (2006)

B.8 Fuel Consumption Charts

The fuel consumption rate of the generator can be derived from fuel consumption charts for generators. An example is shown in Table B-8 using the generator size (kW) and the anticipated load.

Table B-8 Fuel consumption chart for a diesel generator (Diesel Service and Supply, 2009)

Generator Size (kW)	1/4 Load (kL h ⁻¹)	Full Load (kL h ⁻¹)	Generator Size (kW)	1/4 Load (kL h ⁻¹)	Full Load (kL h ⁻¹)
20	0.002	0.006	500	0.042	0.135
40	0.006	0.015	750	0.062	0.202
60	0.007	0.018	1000	0.082	0.269
100	0.01	0.028	1250	0.102	0.336
125	0.012	0.034	1500	0.122	0.403
150	0.014	0.041	1750	0.142	0.47
200	0.018	0.055	2000	0.162	0.537
300	0.026	0.081	2250	0.182	0.604

B.9 Forest Product Type

Classification of forest product types into short-term and long-term harvested wood products is presented in Figure B-1 below:

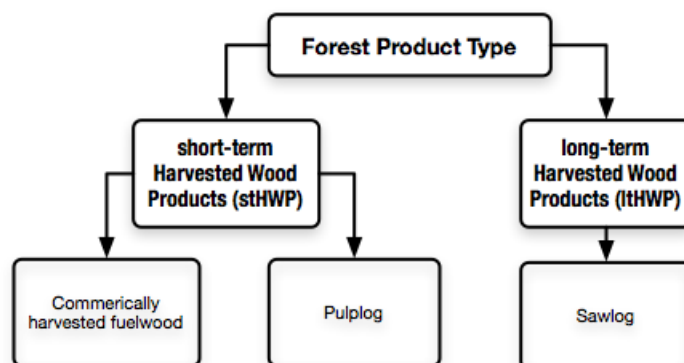


Figure B-1. Classification of forest product types into short-term and long-term harvested wood products

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Appendix C: Unit Conversion and GWP Calculations

C.1 Converting Mass of Carbon (tC) to Carbon Dioxide (tCO₂)

Mass of carbon, which is the typical unit expressed in AFOLU projects, is converted to mass of carbon dioxide gas using the following relationship:

Equation C-1
$$m_{CO_2} = m_{carbon} \times \frac{44}{12}$$

Parameter	Description	Unit
m_{CO_2}	Mass of carbon dioxide	tCO ₂
m_{carbon}	Mass of carbon	tC
44/12	Molecular ratio of carbon dioxide to carbon	tCO ₂ tC ⁻¹

C.2 Converting Other GHGs to Carbon Dioxide Equivalents (CO₂-e)

Other GHGs considered in this Methodology subject to significance as determined in Section 1.2.3, are methane (CH₄) and nitrous oxide (N₂O).

Non-CO₂ gas emission factors must be expressed in carbon dioxide equivalents by multiplying their emission factors by their corresponding global warming potentials (GWPs) and summing the results together to obtain an overall emission factor in terms of carbon dioxide equivalents:

Equation C-2
$$EF = (EF_{CH_4} \times GWP_{CH_4}) + (EF_{N_2O} \times GWP_{N_2O})$$

Parameter	Description	Unit
EF _{CH₄}	Emission factor for methane gas, activity data-specific	tCH ₄ (activity data unit [#]) ⁻¹
EF _{N₂O}	Emission factor for nitrous oxide gas, activity data-specific	tN ₂ O (activity data unit) ⁻¹
GWP _{CH₄}	Global warming potential of methane (<i>see</i> Table 2-4)	tCO ₂ -e tCH ₄ ⁻¹
GWP _{N₂O}	Global warming potential of nitrous oxide (<i>see</i> Table 2-4)	tCO ₂ -e tN ₂ O ⁻¹

activity data unit is the unit of the activity data of either the carbon source or sink, e.g. methane emissions, tCH₄ (kL of fuel)⁻¹

Where CH₄ and N₂O are determined to be insignificant (determined from Section 1.2.3) to the overall calculations, and HFCs, PFCs and SF₆ are not included, then CO₂ becomes the sole contributor.

Throughout this Methodology, even if CO₂ becomes the sole contributor toward GHG emissions, it is expressed as CO₂-e.

Global Warming Potentials (GWPs) for CH₄ and N₂O must be selected from the most recent UNFCCC publications (e.g. UNFCCC, 2004).

C.3 Converting Units for Fuel Emission Factor

IPCC (2006) emission factors for GHGs (CO₂, CH₄ and N₂O) are recorded in units of kg of GHG (TJ of fuel)⁻¹. As such, for each GHG, the emission factor must be multiplied by the fuel's heating value (in TJ kg⁻¹) and density (kg kL⁻¹) to obtain an emission factor in tGHG kL⁻¹. (e.g. tCO₂-e kL⁻¹, tCH₄ kL⁻¹ and tN₂O kL⁻¹).

Equation C-3

$$E_{fuel_GHG} = \frac{EF_{IPCCfuel_GHG} \times HV_{fuel} \times \rho_{fuel}}{1000}$$

Parameter	Description	Unit
EF_{fuel_GHG}	Fuel emission factor for the greenhouse gas (CO ₂ , CH ₄ or N ₂ O)	tGHG kL ⁻¹
$EF_{IPCCfuel_GHG}$	IPCC fuel emission factor for greenhouse gas (CO ₂ , CH ₄ or N ₂ O)	kgGHG TJ ⁻¹
HV_{fuel}	Heating value of fuel	TJ kg ⁻¹
ρ_{fuel}	Density of fuel	kg kL ⁻¹

Heating value and density of fuels can be found from the IPCC Emission Factor Database (2009).

A CO₂ equivalent (CO₂-e) fuel emission factor (EF_{fuel}) can be derived using the global warming potential (GWP) of each gas as outlined in Section C.2.

Reference Notes

IPCC (2006). 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Prepared by the National Greenhouse Gas Inventories Programme, Eggleston H.S., Buendia L., Miwa K., Ngara T. and Tanabe K. (eds). Institute for Global Environmental Strategies, Japan.

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UNFCCC (2004). Information on Global Warming Potentials: Technical Paper, FCCC/TP/2004/3, pp. 12.

Appendix D: Example Solution to Equations 3-22a and 3-22b

Equation 3-22a is straight-forward and therefore will not be represented here. However, the mathematical expression of Equation 3-22b is complex and therefore warrants an order of example solution of its application. This example applies also to Equation 3-29b as the procedure to the order of solution is similar. For applying this example to Equation 3-29b, replace the subscript "DW" with "ltHWP" or/and "stHWP" as required.

Equation 3-22b

$$C_{DW_{pool,t}} = (F_{DW_remain,t-(t-1)} \times C_{DW_{in,t}}) + (F_{DW_remain,t-(t-2)} \times C_{DW_{in,t-1}}) + (F_{DW_remain,t-(t-3)} \times C_{DW_{in,t-2}}) + K + (F_{DW_remain,t-(t-t^*)} \times C_{DW_{in,t-(t^*-1)}})$$

Parameter	Description	Unit
$C_{DW_{pool,t}}$	Cumulative carbon remaining in the deadwood pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC
$F_{DW_remain,t}$	Annual fraction of carbon in the deadwood pool that would remain in the deadwood pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity) after applying the rate of decay	dimensionless
$C_{DW_{in,t}}$	Annual total carbon input to the deadwood pool in year t , (where $t=1,2,3 \dots t^*$ years elapsed since the start of the IFM-LtPF project activity)	tC

This example assumes that a specific project has a lifetime of 5 years. Hence, the application of Equation 3-22b becomes:

$$\text{Year 1} \quad C_{DW_{pool,1}} = (F_{DW_remain,1} \times C_{DW_{in,1}})$$

$$\text{Year 2} \quad C_{DW_{pool,2}} = (F_{DW_remain,1} \times C_{DW_{in,2}}) + (F_{DW_remain,2} \times C_{DW_{in,1}})$$

$$\text{Year 3} \quad C_{DW_{pool,3}} = (F_{DW_remain,1} \times C_{DW_{in,3}}) + (F_{DW_remain,2} \times C_{DW_{in,2}}) + (F_{DW_remain,3} \times C_{DW_{in,1}})$$

$$\text{Year 4} \quad C_{DW_{pool,4}} = (F_{DW_remain,1} \times C_{DW_{in,4}}) + (F_{ltHWP_remain,2} \times C_{ltHWP_{in,3}}) + (F_{DW_remain,3} \times C_{DW_{in,2}}) + (F_{DW_remain,4} \times C_{DW_{in,1}})$$

$$\text{Year 5} \quad C_{DW_{pool,5}} = (F_{DW_remain,1} \times C_{DW_{in,5}}) + (F_{DW_remain,2} \times C_{DW_{in,4}}) + (F_{DW_remain,3} \times C_{DW_{in,3}}) + (F_{DW_remain,4} \times C_{DW_{in,2}}) + (F_{DW_remain,5} \times C_{DW_{in,1}})$$

Tables D-1 and D-2 present the application of Equations 3-21 through to 3-23, for a fixed and variable input to the deadwood pool, applying Equations 3-22a and 3-22b, respectively. For parameter descriptions, please refer to the description tables for relevant equations used in Section 3.3.1 (for deadwood pool) and Section 3.3.2 (for long-term HWP pool).

Table D-1. Numerical example for a fixed input into the deadwood pool (applying Equation 3-22a)

Assume that $C_{DW_{in},t}$ has been calculated to have a fixed annual value of 54,000 tC and k_{decay} determined to be 0.03 yr ⁻¹				
Project Lifetime	Equation 3-18	Equation 3-21	Equation 3-22a	Equation 3-23
t (yr)	$C_{DW_{in},t}$ (tC)	$F_{DW_remain,t}$ (tC)	$C_{DW_{pool},t}$ (dimensionless)	$C_{DW_{out},t}$ (tC)
1	54,000	0.970	52,404	1,596
2	54,000	0.942	103,259	4,741
3	54,000	0.914	152,612	9,388
4	54,000	0.887	200,505	15,495
5	54,000	0.861	246,984	23,016
* Please note that due to numerical rounding, the values presented in this table may not exactly reflect your calculated values.				

Table D-2. Numerical example for a variable input into the deadwood pool (applying Equation 3-22b)

Assume that $C_{DW_{in},t}$ has been calculated to have a variable annual value (see Column 2) and k_{decay} determined to be 0.03 yr ⁻¹				
Project Lifetime	Equation 3-18	Equation 3-21	Equation 3-22b	Equation 3-23
t (yr)	$C_{DW_{in},t}$ (tC)	$F_{DW_remain,t}$ (tC)	$C_{DW_{pool},t}$ (dimensionless)	$C_{DW_{out},t}$ (tC)

1	54,000	0.970	52,404	1,596
2	26,000	0.942	76,087	3,913
3	30,000	0.914	102,952	7,048
4	12,000	0.887	111,554	10,446
5	13,000	0.861	120,873	14,127
* Please note that due to numerical rounding, the values presented in this table may not exactly reflect your calculated values.				